

Rigorous spectral structure of first-passage times to a point target on the reflecting hypercubic lattice, $d \geq 2$ poles, interlacing, residues, and explicit bounds on the principal killed eigenvalue

Xiaoxiao Zhouyi
Companion note R2

1 October 2026

Note. This document is the companion note R2 of the article “Mean and most probable first-passage times of lazy random walks in reflecting hypercubic lattices” (`paper.pdf`, Section 7; Supplementary Material, Section S8), in the repository <https://github.com/zhouyi-xiaoxiao/master-dissertation>. In the repository, its scripts are in `code/msc_rigorous_spectral/` and its data in `data/msc_rigorous_spectral/`; paths written `scripts/...` and `data/...` below refer to these folders; the paths in the `% src` comments of the $\text{\LaTeX}$ source are relative to the root of the repository, and the run logs they cite are in `data/msc_rigorous_*/logs/`. Other run logs, the code of the internal checks and working notes mentioned below are not part of the repository; `notes/VERIFICATION.md` summarises how this note was checked.

Abstract

For the lazy nearest-neighbour walk on $\{1, \dots, N\}^d$ with reflecting (cancelled-move) walls and a single absorbing site, we replace the formal pole analysis of the article by theorems. Part I (any finite symmetric chain): the first-passage law is a finite signed mixture of exponentials (geometrics) whose rates are exactly the eigenvalues of the killed generator in the cyclic subspace of the target; they strictly interlace the eigenvalues of the reflecting walk that are visible from the target; degeneracies and eigenfunctions vanishing at the target are treated completely; the residues sum to one, the principal residue is positive, its quasi-stationary average is one, there are at least $\text{dist}(x_0, a) - 1$ sign changes, and for a start–target pair exchanged by a symmetry the sign of every residue is characterised (in particular $r_1 < 0$ for opposite corners, all N , $d \geq 2$; and for opposite corners the residues alternate in sign if and only if $d = 1$ or $N = 2$). Part II: explicit two-sided bounds on the principal eigenvalue σ_0 , the second pole σ_1 , the gap, and the residues r_0, r_1 in terms of three functionals of the lattice Green’s function at the target. Part III: for opposite corners, $r_0 > 1$ and $\sigma_0 \cdot \text{MFPT} > 1$ in every dimension and for every N (monotone coupling), and the functionals are evaluated: in $d = 2$ with an explicit $O(1)$ remainder (logarithmic growth) and the limiting constant term, in $d = 3$ with the Watson-type constant $C_3 = \sum_{y \in \{0,1\}^3} u(y)$, the second-order constant, an explicit $O(1)$ remainder for every N and the limiting constant term, and in every dimension up to explicit constants. Part IV: a bound on the contribution of all neglected poles, certified tables, and a list of what remains unproved (notably: any statement about the mode itself). Every lemma is accompanied by a numerical check; constants are certified by ball arithmetic.

Summary of results and their status

Unit-rate convention: $L = I - P_1$; for activity q the continuous-time rates are $q\sigma_j$ and the discrete-time multipliers $\nu_j = 1 - q\sigma_j$. τ is the mean hitting time of the target from a uniform start, m the MFPT from x_0 , Λ_1 the smallest non-zero eigenvalue of the reflecting walk visible from the target, $\bar{X} = \Lambda_1 \tau$.

Statement	where	status
$f(t) = \sum_j r_j \sigma_j e^{-\sigma_j t}$, $f(t) = \sum_j r_j (1 - \nu_j) \nu_j^{t-1}$; the rates σ_j are the zeros of $G(s) = \langle e_a, (L - s)^{-1} e_a \rangle =$ eigenvalues of L_a on K_0 (σ_j is absent from the law started at x iff $G_x(\sigma_j) = 0$); $\det(L_a - s) = G(s) \det(L - s)$	Thms 2.2, 3.1	proved
strict interlacing $\Lambda_j < \sigma_j < \Lambda_{j+1}$ in the visible/symmetric sector; multiplicities; equality cases	Thm 4.1, Cor. 4.2	proved
$\sum r_j = 1$; $r_0 > 0$; $\geq D-1$ sign changes; full alternation in $d = 1$; quasi-stationary averages $\sum_x \nu(x) r_j(x) = \delta_{j0}$, $\sum_x \nu(x) m(x) = 1/\sigma_0$; sign rule for symmetric pairs; $r_1 < 0$ (CC, $d \geq 2$, all N); stationary start: all weights > 0 ; $\sum_j r_j \leq \sqrt{n-1}$	Thm 5.1, 5.6, Props. 5.4, 5.5	proved
$\tau \leq 1/\sigma_0 \leq \tau + 1/\Lambda_1$; $\sigma_0 \tau = 1 - \kappa + O(\kappa^2)$ two-sided; bounds for $\sigma_1, \sigma_1 - \sigma_0, r_0, r_1$ (general chain)	Thms 6.2–6.8	proved
sign monotonicity across eigenvalues of pure parity; CC: the residues alternate <i>iff</i> $d = 1$ or $N = 2$ (the conjecture “alternation in $d \geq 2$ ” is false)	Prop. 5.9, Thm 5.10	proved
CC, any $d \geq 1$, any N : $r_0 > 1$, $1 < \sigma_0 m$, $\tau + \frac{1}{\Lambda_1} \leq m$; $\frac{1}{3}d(N^2 - 1) \leq m - \tau \leq \frac{1+\ln W}{\Lambda_1}$	Lem. 7.2, 7.8, Thm 7.9	proved
CC, any $d \geq 2$: $1 + \frac{\mu_- - 1}{X+1} \leq \sigma_0 m < 1 + \frac{1+\ln 2d}{X}$; all bounds explicit in $\bar{X}$	Thm 7.10	proved (uses (7.7) if $Z = Z_d$)
Z_2, Z_3, Z'_2, Z'_3 (Epstein-type sums), 16 certified decimals	Lem. 7.6	computer-assisted
$d = 2$: $\tau_2 = \frac{8}{\pi} N^2 \ln N + c_\tau N^2 + R$, $-2.14 \leq R \leq 3.53$; $m_2 - \tau_2 = \frac{4 \ln 2}{\pi} N^2 + R'$	Thm 9.1	computer-assisted (constants)
$d = 2$: $R \rightarrow c_\tau^{(0)} = 0.8830352\dots$, $R' \rightarrow c_{m\tau}^{(0)} = -0.3508943\dots$ (constant term $C_2^{(0)} = 0.5321408\dots$ of the MFPT)	Thm 9.8	proved (limit); values computer-assisted
$d = 3$: $\tau_3 = C_3 N^3 + c_{\tau 3} N^2 + R_3$, $-17.4 \leq R_3 \leq 19.4$, $m_3 - \tau_3 = c_{m\tau 3} N^2 + R'_3$, $ R'_3 + 1 \leq 81.4$, all $N \geq 2$; $C_3 = 4.320460169373\dots$	Thm 10.1(a)–(c)	computer-assisted (constants)
$d = 3$: $R_3 \rightarrow 1.7573985\dots$, $R'_3 \rightarrow -0.2989172\dots$ (constant term $C'_3 = 1.4584812\dots$ of the MFPT)	Thm 10.1(d)	proved (limit); values computer-assisted
N -explicit bounds on $\bar{X}$, $\sigma_0, \sigma_1, r_0, r_1$ for $d = 2, 3$	Cor. 11.1, 11.2	computer-assisted
these bounds with truncated decimal constants are false for $N \geq 8872$ ($d = 2$) and $N \geq 182\,420$ ($d = 3$)	Prop. 11.5	computer-assisted (refutation)
first-order pole structure (the formal relations of the computations of Sections 5–6 of the article)	Thm 11.3	proved (asymptotic)
tail of all neglected poles, continuous and discrete time	Thm 12.1, Cor. 12.2	proved
finite- N enclosures of $\sigma_0, \sigma_1, r_0, r_1, \tau, m$	Section 13	computer-assisted
certified sign patterns of all residues for 20 pairs (d, N)	Prop. 14.1	computer-assisted
sign statistics for larger N ; monotonicity of the remainders in N and their rate of convergence	Obs. 14.2, 9.7, Rem. 10.12	numerical observation
location of the mode; unimodality	Section 14	not proved here

Convention for decimals. A decimal expansion followed by dots (...) is *truncated*: every digit shown is a digit of the exact value (the list of all such strings is checked by script 21). Decimals in inequalities are exact rational numbers.

Contents

Summary of results and their status

1

1 Setting, conventions, standing assumptions

4

1.1	An abstract symmetric chain	4
1.2	The reflecting box	4
2	Structure of the killed generator	5
3	Spectral representation of the first-passage law	6
4	Interlacing	7
5	Residues: normalisation, moments, signs	8
5.1	Reflection-symmetric pairs: the sign of r_1	10
5.2	Alternation of the residues holds only for $d = 1$ or $N = 2$	12
6	Explicit two-sided bounds on $\sigma_0, \sigma_1, r_0, r_1$ (general chain)	15
6.1	The Green's function functionals	15
6.2	The principal eigenvalue	15
6.3	The second pole and the gap	16
6.4	The residues r_0 and r_1	17
7	Corner to corner in the reflecting box: general dimension	19
8	Exact reductions of the lattice Green's function at the corner	24
9	Two dimensions: logarithmic asymptotics with an explicit remainder	26
9.1	Statement	26
9.2	Three auxiliary lemmas	27
9.3	Proof of Theorem 9.1	28
9.4	The constant terms: limits of the remainders	30
10	Three dimensions: the Watson-type constant with explicit error	31
10.1	Statement	31
10.2	The functions K and J	33
10.3	Row sums and the exponentially small parts	34
10.4	Proof of Theorem 10.1(a),(b)	35
10.5	Second order: closed form of K , the $O(1)$ remainder and the constant term	36
10.6	Proof of Theorem 10.1(c),(d)	41
11	Explicit consequences in two and three dimensions	42
12	The neglected poles: a rigorous tail bound	45
13	Computer-assisted certification	46
14	Numerical observations and what is not proved	48

Part I. Structure (any finite symmetric chain)

1 Setting, conventions, standing assumptions

1.1 An abstract symmetric chain

Let Ω be a finite set with $n = |\Omega| \geq 2$ elements and let $L = (L_{xy})_{x,y \in \Omega}$ be a real matrix with

$$\begin{aligned} \text{(A1)} \quad L &= L^\top; & \text{(A2)} \quad L_{xy} &\leq 0 \ (x \neq y), \quad L\mathbf{1} = 0; \\ \text{(A3)} \quad & \text{the graph } \{x \sim y \iff L_{xy} < 0\} \text{ is connected.} \end{aligned} \tag{1.1}$$

Then $\langle f, Lf \rangle = \frac{1}{2} \sum_{x \neq y} |L_{xy}| (f(x) - f(y))^2 \geq 0$, so L is positive semidefinite, and by (A3) its kernel is spanned by $\mathbf{1}$. We write e_x for the coordinate vectors, $\langle \cdot, \cdot \rangle$ for the Euclidean inner product on $\mathbb{R}^\Omega$, and

$$0 = \lambda^{(0)} < \lambda^{(1)} < \dots < \lambda^{(M)}$$

for the *distinct* eigenvalues of L , with orthogonal spectral projectors $E_0, \dots, E_M$ and multiplicities $m_i = \text{rank } E_i$; thus $E_0 = \frac{1}{n} \mathbf{1}\mathbf{1}^\top$, $m_0 = 1$.

Fix $q > 0$. The *continuous-time walk* is the Markov chain with generator $-qL$; the *discrete-time walk* has transition matrix $P = I - qL$, which is a symmetric stochastic matrix as soon as $q \max_x L_{xx} \leq 1$. For a target $a \in \Omega$ let $T = T_a$ be the first hitting time of a , and write $\Omega' = \Omega \setminus \{a\}$ and

$$L_a = (L_{xy})_{x,y \in \Omega'}$$

for the principal submatrix (the *killed generator*). The substochastic matrix of the discrete walk on Ω' is $Q = I - qL_a$. For $x \in \Omega'$,

$$\begin{aligned} \mathbb{P}_x(T > t) &= (e^{-qtL_a} \mathbf{1})(x) \quad (t \geq 0, \text{ continuous time}), \\ \mathbb{P}_x(T > t) &= (Q^t \mathbf{1})(x) \quad (t = 0, 1, 2, \dots, \text{ discrete time}), \end{aligned} \tag{1.2}$$

which is the definition $S(t) = e_{x_0}^\top Q^t \mathbf{1}$ of the article. All statements below are written for L (“unit rate”); the activity q only rescales: continuous-time rates are $q\sigma$ and discrete-time multipliers are $\nu = 1 - q\sigma$.

1.2 The reflecting box

For integers $d \geq 1$, $N \geq 2$ let $\Omega = \{1, \dots, N\}^d$, $n = N^d$, and

$$(Lf)(x) = \frac{1}{2d} \sum_{y \sim x} (f(x) - f(y)), \tag{1.3}$$

the sum being over lattice neighbours $y \in \Omega$ of x . Then $P = I - qL$ is exactly the lazy walk of the article: each of the $2d$ directions is chosen with probability $q/(2d)$ and a move leaving Ω is cancelled. L satisfies (1.1), and $L_{xx} \leq 1$, so P is stochastic for every $q \in (0, 1]$.

Lemma 1.1 (spectrum of the reflecting box). *For $k \in \{0, \dots, N-1\}^d$ put $\theta_j = \pi j/N$,*

$$\phi_k(x) = \prod_{i=1}^d \gamma_{k_i} \cos(\theta_{k_i}(x_i - \tfrac{1}{2})), \quad \gamma_0 = N^{-1/2}, \quad \gamma_j = (2/N)^{1/2} \ (j \geq 1), \quad \lambda_k = \frac{1}{d} \sum_{i=1}^d (1 - \cos \theta_{k_i}). \tag{1.4}$$

Then $\{\phi_k\}$ is an orthonormal basis of $\mathbb{R}^\Omega$ and $L\phi_k = \lambda_k \phi_k$.

Proof. For $d = 1$: $g(x) = \cos(\theta_j(x - \frac{1}{2}))$ satisfies $g(x-1) + g(x+1) = 2\cos\theta_j g(x)$ for all real x , and $g(0) = g(1)$, $g(N+1) = g(N)$ because $\cos(\theta_j(N + \frac{1}{2})) = \cos(\pi j + \theta_j/2) = \cos(\pi j - \theta_j/2)$. Hence $(Lg)(x) = \frac{1}{2}(2g(x) - g(x-1) - g(x+1)) = (1 - \cos\theta_j)g(x)$ for $1 \leq x \leq N$, the boundary rows being covered by the two reflection identities. The N eigenvalues $1 - \cos\theta_j$ are distinct, so the eigenvectors are orthogonal, and $\sum_{x=1}^N \cos^2(\theta_j(x - \frac{1}{2})) = N/2$ for $j \geq 1$ (and N for $j = 0$) gives the normalisation. For general d , $L = \frac{1}{d} \sum_i L^{(i)}$ with $L^{(i)}$ the one-dimensional operator acting on coordinate i , and the tensor products of the one-dimensional eigenvectors form an orthonormal eigenbasis. $\square$

The main geometry (“corner to corner”, CC) is $a = (N, \dots, N)$, $x_0 = (1, \dots, 1)$. For the corner target

$$n\phi_k(a)^2 = \prod_{i=1}^d \varepsilon_{k_i} \cos^2 \frac{\theta_{k_i}}{2}, \quad n\phi_k(a)\phi_k(x_0) = (-1)^{|k|} n\phi_k(a)^2, \quad \varepsilon_0 = 1, \quad \varepsilon_j = 2 \quad (j \geq 1), \quad (1.5)$$

with $|k| = k_1 + \dots + k_d$, because $\cos(\theta_j(N - \frac{1}{2})) = (-1)^j \cos(\theta_j/2)$. In particular $\phi_k(a) \neq 0$ for every k .

2 Structure of the killed generator

Fix a target a . Put $w_i = \langle e_a, E_i e_a \rangle = \|E_i e_a\|^2 \geq 0$, so that $\sum_{i=0}^M w_i = 1$ and $w_0 = 1/n$.

Definition 2.1 (visible spectrum). An eigenvalue $\lambda^{(i)}$ of L is *visible from a* if $w_i > 0$, i.e. if some eigenfunction of L with that eigenvalue does not vanish at a . We list the visible eigenvalues as $0 = \Lambda_0 < \Lambda_1 < \dots < \Lambda_p$ with weights $w_0 = \frac{1}{n}, w_1, \dots, w_p > 0$ (re-indexed), and put

$$u_i = \frac{E_i e_a}{\sqrt{w_i}} \quad (0 \leq i \leq p), \quad K = \text{span}\{u_0, \dots, u_p\}, \quad K_0 = K \cap e_a^\perp.$$

The *secular function* and the *cross function* of a site x are

$$G(s) = \langle e_a, (L - s)^{-1} e_a \rangle = \sum_{i=0}^p \frac{w_i}{\Lambda_i - s}, \quad G_x(s) = \langle e_x, (L - s)^{-1} e_a \rangle = \sum_{i=0}^p \frac{\sqrt{w_i} u_i(x)}{\Lambda_i - s}. \quad (2.1)$$

K is the cyclic subspace generated by e_a under L : the u_i are orthonormal eigenvectors of L and $e_a = \sum_i \sqrt{w_i} u_i$. Note $u_0 = n^{-1/2} \mathbf{1}$, so $\mathbf{1} \in K$. We identify $\mathbb{R}^{\Omega'}$ with $e_a^\perp = \{v \in \mathbb{R}^\Omega : v(a) = 0\}$; under this identification $L_a = \Pi L \Pi|_{e_a^\perp}$ with $\Pi = I - e_a e_a^\top$.

Theorem 2.2 (structure of L_a). *Assume (1.1). Then:*

(i) G has exactly p zeros in $\mathbb{C}$. They are real and simple, and they strictly interlace the visible eigenvalues:

$$0 = \Lambda_0 < \sigma_0 < \Lambda_1 < \sigma_1 < \Lambda_2 < \dots < \sigma_{p-1} < \Lambda_p. \quad (2.2)$$

(ii) $e_a^\perp = K_0 \oplus K^\perp$ orthogonally, $\dim K_0 = p$, and both summands are invariant under L_a . On $K^\perp$ (the eigenfunctions of L that are orthogonal to K ; all of them vanish at a) L_a coincides with L . On K_0 , L_a has the simple eigenvalues $\sigma_0, \dots, \sigma_{p-1}$ with eigenvectors

$$\psi_j = \sum_{i=0}^p \frac{\sqrt{w_i}}{\Lambda_i - \sigma_j} u_i, \quad \psi_j(x) = G_x(\sigma_j), \quad \|\psi_j\|^2 = G'(\sigma_j), \quad (L - \sigma_j)\psi_j = e_a. \quad (2.3)$$

(iii) As a multiset, $\text{spec}(L_a) = \{\sigma_0, \dots, \sigma_{p-1}\} \uplus \{\lambda^{(i)} \text{ with multiplicity } m_i - 1 \text{ if } \lambda^{(i)} \text{ is visible, } m_i \text{ if it is not}\}$. Equivalently

$$\det(L_a - s) = G(s) \det(L - s) \quad (s \in \mathbb{C} \setminus \text{spec } L). \quad (2.4)$$

Proof. (i) $\mathcal{N}(s) := G(s) \prod_{l=0}^p (\Lambda_l - s) = \sum_i w_i \prod_{l \neq i} (\Lambda_l - s)$ is a polynomial of degree p (its leading coefficient is $(-1)^p \sum_i w_i = (-1)^p$), so G has at most p zeros in $\mathbb{C}$. On each interval $(\Lambda_j, \Lambda_{j+1})$, $0 \leq j \leq p-1$, G is continuous with $G'(s) = \sum_i w_i (\Lambda_i - s)^{-2} > 0$, $G(s) \rightarrow -\infty$ as $s \downarrow \Lambda_j$ and $G(s) \rightarrow +\infty$ as $s \uparrow \Lambda_{j+1}$. Hence each of these p intervals contains exactly one zero σ_j , it is simple, and there are no others.

(ii) K is spanned by eigenvectors of L , hence L -invariant, and so is $K^\perp$ because L is symmetric. As $e_a \in K$, every $v \in K^\perp$ satisfies $v(a) = \langle v, e_a \rangle = 0$, i.e. $K^\perp \subset e_a^\perp$. If $v \in e_a^\perp$ and $v = v_K + v_\perp$ with $v_K \in K$, $v_\perp \in K^\perp$, then $v_K(a) = v(a) - v_\perp(a) = 0$; this proves the orthogonal decomposition, and $\dim K_0 = \dim K - 1 = p$ since $e_a \in K$ is not orthogonal to K . For $v \in K^\perp$: $L_a v = \Pi L v = L v$ because $L v \in K^\perp \subset e_a^\perp$. For $v \in K_0$: $L v \in K$ and $\Pi L v = L v - (L v)(a) e_a \in K \cap e_a^\perp = K_0$. So both spaces are L_a -invariant.

Now let ψ_j be defined by the first formula in (2.3). Then $\psi_j \in K$, $\psi_j \neq 0$, $(L - \sigma_j)\psi_j = \sum_i \sqrt{w_i} u_i = e_a$, and $\psi_j(a) = \langle e_a, \psi_j \rangle = \sum_i w_i / (\Lambda_i - \sigma_j) = G(\sigma_j) = 0$, so $\psi_j \in K_0$ and $L_a \psi_j = \Pi(\sigma_j \psi_j + e_a) = \sigma_j \psi_j$. The p numbers σ_j are distinct and $\dim K_0 = p$, so these are all the eigenvalues of $L_a|_{K_0}$ and they are simple. Finally $\psi_j(x) = \langle e_x, \psi_j \rangle = G_x(\sigma_j)$ and $\|\psi_j\|^2 = \sum_i w_i (\Lambda_i - \sigma_j)^{-2} = G'(\sigma_j)$ by orthonormality of the u_i .

(iii) $L|_{K^\perp}$ has the eigenvalue $\lambda^{(i)}$ with multiplicity $\dim(E_i \mathbb{R}^\Omega \cap K^\perp) = m_i - \dim(E_i \mathbb{R}^\Omega \cap K)$, and $E_i \mathbb{R}^\Omega \cap K$ is spanned by u_i if $\lambda^{(i)}$ is visible and is $\{0\}$ otherwise. Together with (ii) this gives the multiset statement. Both sides of (2.4) are then the same polynomial divided by $\prod_i (\lambda^{(i)} - s)^{m_i}$: indeed $\det(L_a - s) = \prod_j (\sigma_j - s) \prod_{i \text{ vis}} (\lambda^{(i)} - s)^{m_i - 1} \prod_{i \text{ invis}} (\lambda^{(i)} - s)^{m_i}$ and $\mathcal{N}(s) = \prod_j (\sigma_j - s)$ (same zeros, same leading coefficient $(-1)^p$). ((2.4) is also Cramer's rule for the (a, a) entry of $(L - s)^{-1}$.) $\square$

Remark 2.3 (degeneracies). Nothing in Theorem 2.2 requires simple eigenvalues. Two kinds of coincidence can occur and are harmless. (a) A visible eigenvalue of multiplicity $m_i \geq 2$ remains an eigenvalue of L_a with multiplicity $m_i - 1$; its eigenfunctions vanish at a . (b) A zero σ_j of G may coincide with an *invisible* eigenvalue $\lambda^{(i)}$ of L ; then σ_j is an eigenvalue of L_a of multiplicity $m_i + 1$, but the eigenvector $\psi_j \in K_0$ is still canonically defined and orthogonal to the other m_i . In both cases the extra eigenfunctions lie in $K^\perp$ and are invisible to the first-passage law (Theorem 3.1).

3 Spectral representation of the first-passage law

Theorem 3.1 (first-passage representation). *Assume (1.1), let a be the target and $x \in \Omega'$ the start. Define the residues*

$$r_j(x) = \frac{\psi_j(x) \langle \psi_j, \mathbf{1} \rangle}{\|\psi_j\|^2} = -\frac{G_x(\sigma_j)}{\sigma_j G'(\sigma_j)}, \quad j = 0, \dots, p-1. \quad (3.1)$$

Then, for the walk with activity q ,

$$\text{continuous time: } \mathbb{P}_x(T > t) = \sum_{j=0}^{p-1} r_j(x) e^{-q\sigma_j t}, \quad f(t) = \sum_{j=0}^{p-1} r_j(x) q\sigma_j e^{-q\sigma_j t} \quad (t \geq 0); \quad (3.2)$$

$$\text{discrete time: } \mathbb{P}_x(T > t) = \sum_{j=0}^{p-1} r_j(x) \nu_j^t, \quad f(t) = \sum_{j=0}^{p-1} r_j(x) (1 - \nu_j) \nu_j^{t-1} \quad (t \geq 1), \quad (3.3)$$

with $\nu_j = 1 - q\sigma_j$ (and the convention $0^0 = 1$). Moreover, for every $s \in \mathbb{C}$ that is not a zero of G ,

$$\sum_{j=0}^{p-1} \frac{r_j(x) \sigma_j}{\sigma_j - s} = \frac{G_x(s)}{G(s)}, \quad (3.4)$$

so that $\mathbb{E}_x e^{-uT} = G_x(-u/q)/G(-u/q)$ in continuous time and $\mathbb{E}_x z^T = G_x(\zeta)/G(\zeta)$ with $\zeta = -(1-z)/(qz)$ in discrete time. Only the zeros of G occur: eigenvalues of L_a belonging to $K^\perp$ carry no weight, and σ_j is absent from the law started at x if and only if $G_x(\sigma_j) = 0$.

Proof. The vector $\mathbf{1}_{\Omega'} = \mathbf{1} - e_a$ lies in K (both terms do) and vanishes at a , so $\mathbf{1}_{\Omega'} \in K_0$. By Theorem 2.2(ii) the ψ_j form an orthogonal basis of K_0 , hence $\mathbf{1}_{\Omega'} = \sum_j \frac{\langle \psi_j, \mathbf{1}_{\Omega'} \rangle}{\|\psi_j\|^2} \psi_j$. Since $\psi_j(a) = 0$ and $\mathbf{1} = \sqrt{n} u_0$,

$$\langle \psi_j, \mathbf{1}_{\Omega'} \rangle = \langle \psi_j, \mathbf{1} \rangle = \sqrt{n} \cdot \frac{\sqrt{w_0}}{\Lambda_0 - \sigma_j} = -\frac{1}{\sigma_j},$$

which with (2.3) gives the second expression in (3.1). Applying e^{-qtL_a} , respectively $(I - qL_a)^t$, to this expansion and using (1.2) gives the survival formulae; $f = -\frac{d}{dt} \mathbb{P}_x(T > t)$, respectively $f(t) = \mathbb{P}_x(T > t - 1) - \mathbb{P}_x(T > t)$, gives the densities.

For (3.4): for $x \neq a$, $G_x(s) = -\frac{1}{s} \langle e_x, (1 - L/s)^{-1} e_a \rangle = O(s^{-2})$ as $s \rightarrow \infty$, while $G(s) = -1/s + O(s^{-2})$, so the rational function G_x/G vanishes at infinity; its poles are the simple zeros σ_j of G (the poles Λ_i of G_x are poles of G of the same order and cancel), with residues $G_x(\sigma_j)/G'(\sigma_j) = -r_j \sigma_j$. Hence $G_x(s)/G(s) = \sum_j (-r_j \sigma_j)/(s - \sigma_j)$. Finally $\mathbb{E}_x e^{-uT} = \int_0^\infty e^{-ut} f(t) dt = \sum_j r_j q \sigma_j / (u + q \sigma_j)$, and $\mathbb{E}_x z^T = \sum_j r_j (1 - \nu_j) z / (1 - \nu_j z) = \sum_j r_j \sigma_j / (\sigma_j - \zeta)$. $\square$

Remark 3.2 (relation to the renewal ratio of the article). With

$$\hat{P}(y, u|x) = \int_0^\infty e^{-ut} p_t(x, y) dt = \langle e_x, (u + qL)^{-1} e_y \rangle$$

for the walk without target, $G_x(-u/q) = q \hat{P}(a, u|x)$, so (3.4) is the renewal identity $\hat{F}(u) = \hat{P}(a, u|x_0)/\hat{P}(a, u|a)$, and Theorem 3.1 proves that its poles are exactly $u = -q\sigma_j$ with σ_j the eigenvalues of the killed generator in the sector K_0 . The a_j, ν_j of the computations of Sections 5–6 of the article are $a_j = r_j \sigma_j$, $\nu_j = \sigma_j$.

4 Interlacing

Theorem 4.1 (interlacing). *Assume (1.1). Let $\lambda_1 \leq \dots \leq \lambda_n$ and $\mu_1 \leq \dots \leq \mu_{n-1}$ be the eigenvalues of L and of L_a , counted with multiplicity.*

- (a) (Cauchy) $\lambda_k \leq \mu_k \leq \lambda_{k+1}$ for $1 \leq k \leq n-1$.
- (b) (strict interlacing in the visible sector) *The eigenvalues of L_a that carry first-passage weight are the zeros of G , and they satisfy (2.2): between two consecutive visible eigenvalues of L there is exactly one, and it is different from both.*
- (c) (equality cases) *An eigenvalue μ of L_a is also an eigenvalue of L if and only if L has an eigenfunction with eigenvalue μ vanishing at a ; the multiplicity of μ in $\text{spec}(L_a)$ is then given by Theorem 2.2(iii) and Remark 2.3.*
- (d) (symmetry sector) *Let Γ_a be the group of permutations g of Ω with $gLg^{-1} = L$ and $ga = a$, and $\mathcal{F} = \{f : f \circ g = f \ \forall g \in \Gamma_a\}$. Then $K \subset \mathcal{F}$. Hence the σ_j are eigenvalues of the killed walk restricted to the Γ_a -symmetric sector, and they strictly interlace the visible distinct eigenvalues of the reflecting walk restricted to that sector.*

Proof. (a) is the Cauchy interlacing theorem for a principal submatrix of a real symmetric matrix (R. A. Horn and C. R. Johnson, *Matrix Analysis*, Cambridge Univ. Press: Theorem 4.3.17 in the 2nd edition, 2013; Theorem 4.3.8 in the 1st edition, 1985). It also follows from the Courant–Fischer min–max principle, since L_a is the compression of L to the hyperplane $e_a^\perp$. (b) is Theorem 2.2(i) together with Theorem 3.1. (c): by Theorem 2.2(ii)–(iii), $\text{spec}(L_a) = \{\sigma_j\} \uplus \text{spec}(L|_{K^\perp})$ and no σ_j is a visible eigenvalue of L ; so $\mu \in \text{spec}(L_a) \cap \text{spec}(L)$ iff $\mu \in \text{spec}(L|_{K^\perp})$ (using Remark 2.3(b) when $\mu = \sigma_j$ is an invisible eigenvalue), iff L has an eigenfunction for μ in $K^\perp$; and an eigenfunction v of L with eigenvalue $\lambda^{(i)}$ lies in $K^\perp$ iff $0 = \langle v, u_i \rangle \sqrt{w_i} = \langle v, E_i e_a \rangle = \langle v, e_a \rangle = v(a)$. (d) $ge_a = e_a$ and g commutes with every E_i , so $gE_i e_a = E_i e_a$. $\square$

Corollary 4.2 (the reflecting box). *Let L be (1.3) and $a \in \Omega$ arbitrary.*

- (a) *An eigenvalue λ of L is visible from a iff there is k with $\lambda_k = \lambda$ and $\prod_i \cos(\theta_{k_i}(a_i - \frac{1}{2})) \neq 0$, i.e. iff for no i the number $k_i(2a_i - 1)/N$ is an odd integer.*
- (b) *For a corner target every eigenvalue of L is visible. Hence the number of poles of the first-passage law is $p = \#\{\lambda_k : k \in \{0, \dots, N-1\}^d\} - 1$ (number of distinct values), the poles satisfy $\Lambda_j < \sigma_j < \Lambda_{j+1}$ for the increasing list $\Lambda_0 < \dots < \Lambda_p$ of all distinct eigenvalues of L , and $\Gamma_a \supseteq S_d$ (coordinate permutations), so all ψ_j are symmetric functions of the coordinates. A distinct eigenvalue of multiplicity m (permutation images and accidental degeneracies alike) stays in $\text{spec}(L_a)$ with multiplicity $m - 1$ and without first-passage weight.*
- (c) *For $d \geq 2$, $N \geq 2$ and the corner target: $\Lambda_1 = \frac{1}{d}(1 - \cos \frac{\pi}{N})$ with multiplicity d (the modes $k = e_i$), $\Lambda_2 = 2\Lambda_1$ (the modes $k = e_i + e_j$, $i \neq j$), and*

$$W := n w_1 = 2d \cos^2 \frac{\pi}{2N}, \quad 0 < \sigma_0 < \Lambda_1 < \sigma_1 < 2\Lambda_1 < \sigma_j \quad (j \geq 2). \quad (4.1)$$

Proof. (a),(b): $w(\lambda) = \sum_{k: \lambda_k = \lambda} \phi_k(a)^2$ and Lemma 1.1; for the corner use (1.5). (c) The smallest non-zero value of λ_k is attained exactly at $k = e_i$, $1 \leq i \leq d$. Any other $k \neq 0$ has either at least two non-zero components, giving $\lambda_k \geq \frac{2}{d}(1 - \cos \frac{\pi}{N}) = 2\Lambda_1$ with equality iff $k = e_i + e_j$ ($i \neq j$), or exactly one non-zero component $k_i \geq 2$ (which requires $N \geq 3$), giving $\lambda_k \geq \frac{1}{d}(1 - \cos \frac{2\pi}{N}) = \frac{8}{d} \sin^2 \frac{\pi}{2N} \cos^2 \frac{\pi}{2N} = 2\Lambda_1 \cdot 2 \cos^2 \frac{\pi}{2N} > 2\Lambda_1$ because $2 \cos^2 \frac{\pi}{2N} > 1$ for $N \geq 3$. So $\Lambda_2 = 2\Lambda_1$, and its eigenspace is spanned by the $\phi_{e_i+e_j}$. The weight is $n \sum_i \phi_{e_i}(a)^2 = d \cdot 2 \cos^2 \frac{\pi}{2N}$ by (1.5). $\square$

5 Residues: normalisation, moments, signs

Throughout this section $x \in \Omega'$ is the start and $r_j = r_j(x)$.

Theorem 5.1 (residues). *Assume (1.1).*

- (a) $\sum_{j=0}^{p-1} r_j = 1$.
- (b) *In continuous time $\mathbb{E}_x[T^\ell] = \ell! q^{-\ell} \sum_j r_j \sigma_j^{-\ell}$ for $\ell \geq 1$; in both time conventions $\mathbb{E}_x T = \frac{1}{q} \sum_j r_j / \sigma_j$.*
- (c) *If the graph of L_a on Ω' is connected, then σ_0 is the smallest eigenvalue of L_a , it is a simple eigenvalue of L_a , ψ_0 has a strict sign on Ω' , and*

$$r_0(y) = \frac{\psi_0(y) \sum_z \psi_0(z)}{\sum_z \psi_0(z)^2} > 0 \quad \text{for every } y \in \Omega'.$$

- (d) *Let $D = \text{dist}(x, a) \geq 1$ be the graph distance in the graph of L . Then*

$$\sum_j r_j \sigma_j^i = 0 \quad (1 \leq i \leq D-1), \quad (-1)^{D-1} \sum_j r_j \sigma_j^D > 0. \quad (5.1)$$

Consequently the sequence $(r_0, r_1, \dots, r_{p-1})$ (zeros deleted) has at least $D - 1$ sign changes; in particular at least D residues are non-zero, $p \geq D$, and under the hypothesis of (c) at least $\lceil (D - 1)/2 \rceil$ residues are negative.

Proof. (a) is (3.2) at $t = 0$. (b) follows by integrating $t^{\ell-1} \mathbb{P}_x(T > t)$ over $[0, \infty)$, respectively summing $\mathbb{P}_x(T > t) = \sum_j r_j \nu_j^t$ over $t \geq 0$, which gives $\sum_j r_j / (1 - \nu_j)$ (here $|\nu_j| < 1$ since $0 < q\sigma_j < 2$: $\text{spec } L \subset [0, 2 \max_x L_{xx}]$ by Gershgorin and $q \max_x L_{xx} \leq 1$, and $\sigma_j < \Lambda_p$).

(c) If $|\Omega'| = 1$ (i.e. $n = 2$, $\Omega' = \{y\}$), then L_a is the 1×1 matrix (L_{yy}) with $L_{yy} = -L_{ya} > 0$, $p = \dim K_0 = 1$, so $\sigma_0 = L_{yy}$ is the only eigenvalue of L_a , $\psi_0 \neq 0$ is a non-zero multiple of e_y , and $r_0(y) = 1$ by (a); all claims hold trivially. Let now $|\Omega'| \geq 2$ and $c = \max_x L_{xx}$. The matrix $A = cI - L_a$ is real symmetric, entrywise non-negative and of order ≥ 2 , and it is irreducible because its off-diagonal zero pattern is that of L_a , whose graph is connected. By the Perron–Frobenius theorem (Horn–Johnson, Theorem 8.4.4: for an irreducible non-negative matrix of order at least 2 the spectral radius $\rho(A)$ is positive, is an algebraically simple eigenvalue, and has an entrywise positive eigenvector ψ) and because the eigenvalues of the symmetric matrix A are real and bounded in modulus by $\rho(A)$, the largest eigenvalue of A is $\rho(A)$ and it is simple; i.e. the smallest eigenvalue $\mu_1 = c - \rho(A)$ of L_a is simple and $\psi > 0$ on Ω' . As μ_1 is simple and $K_0, K^\perp$ are L_a -invariant, ψ belongs to one of them; but $\langle \psi, \mathbf{1}_{\Omega'} \rangle > 0$ and $\mathbf{1}_{\Omega'} \in K_0$, so $\psi \notin K^\perp$. Hence $\psi \in K_0$ is a multiple of some ψ_j and $\mu_1 = \sigma_j$; being the minimum, $\mu_1 = \sigma_0$. The formula for r_0 is (3.1).

(d) Let $\beta = L_a \mathbf{1}_{\Omega'}$. Since the rows of L sum to zero, $\beta(y) = -L_{ya} \geq 0$ for $y \in \Omega'$, and β is supported on the neighbours of a . By the expansion in the proof of Theorem 3.1, $\sum_j r_j \sigma_j^i = (L_a^i \mathbf{1}_{\Omega'})(x) = (L_a^{i-1} \beta)(x) = \sum_y (L_a^{i-1})_{xy} \beta(y)$ for $i \geq 1$. The entry $(L_a^{i-1})_{xy}$ is a sum over sequences $x = z_0, z_1, \dots, z_{i-1} = y$ in Ω' , with consecutive points equal or adjacent, of $\prod_l L_{z_l z_{l+1}}$; it vanishes if $\text{dist}(x, y) > i - 1$. Every neighbour y of a has $\text{dist}(x, y) \geq D - 1$, so the sum is zero for $i \leq D - 1$. For $i = D$ only geodesics of length $D - 1$ from x to neighbours y of a contribute; each contributes a product of $D - 1$ negative numbers times $\beta(y) > 0$, and at least one exists (cut the last step off a geodesic from x to a ; its vertices other than a lie in Ω'). This proves (5.1).

Suppose the non-zero residues had only $S \leq D - 2$ sign changes. Choose $c_1 < \dots < c_S$, none equal to a σ_j , separating consecutive maximal blocks of equal sign, and put $P(\sigma) = \sigma \prod_{l=1}^S (c_l - \sigma)$. Then $r_j P(\sigma_j)$ has the same sign for all j with $r_j \neq 0$ (recall $\sigma_j > 0$), and it is not identically zero because $\sum_j r_j = 1$. Thus $\sum_j r_j P(\sigma_j) \neq 0$. But P is a linear combination of $\sigma, \dots, \sigma^{S+1}$ with $S + 1 \leq D - 1$, so $\sum_j r_j P(\sigma_j) = 0$ by (5.1); contradiction. The remaining assertions are immediate (S sign changes need $S + 1$ non-zero terms; with $r_0 > 0$ and at least $D - 1$ sign changes, at least $\lfloor (D - 1)/2 \rfloor$ terms are negative). $\square$

Corollary 5.2 (one dimension: complete alternation). *For $d = 1$, target $a = N$ and start $x = 1$: $p = N - 1$, all residues are non-zero and $\text{sign } r_j = (-1)^j$.*

Proof. The N eigenvalues $1 - \cos \theta_j$ are distinct and visible from $a = N$ by (1.5), so $p = N - 1$; $D = N - 1$, so the p residues have at least $p - 1$ sign changes, which forces all to be non-zero and alternating; $r_0 > 0$ by Theorem 5.1(c). $\square$

Corollary 5.3 (corner to corner, $d \geq 2$). *For the box with $a = (N, \dots, N)$, $x_0 = (1, \dots, 1)$: $D = d(N - 1)$, the continuous-time density satisfies $f(t) = ct^{D-1}(1 + o(1))$ as $t \downarrow 0$ with $c > 0$, and the residue sequence has at least $d(N - 1) - 1$ sign changes.*

Proposition 5.4 (stationary start; sizes of the residues). *Assume (1.1) and put*

$$\varrho_j = \frac{1}{n} \sum_{x \in \Omega'} r_j(x) = \frac{1}{n \sigma_j^2 G'(\sigma_j)} \quad (0 \leq j \leq p - 1). \quad (5.2)$$

(a) $\varrho_j > 0$ for every j and $\sum_j \varrho_j = 1 - \frac{1}{n}$. *If the start is uniformly distributed on Ω (the stationary law), then $\mathbb{P}(T > t) = \sum_j \varrho_j e^{-q \sigma_j^2 t}$ in continuous time: the first-passage law is a mixture of exponentials (plus an atom $\frac{1}{n}$ at 0), its density is completely monotone, and it has no interior mode. In discrete time $\mathbb{P}(T > t) = \sum_j \varrho_j \nu_j^t$.*

(b) *For every start $x \in \Omega'$,*

$$\sum_{j=0}^{p-1} \frac{r_j(x)^2}{n \varrho_j} \leq 1, \quad \text{hence} \quad |r_j(x)| \leq \sqrt{n \varrho_j} \quad \text{and} \quad \sum_j |r_j(x)| \leq \sqrt{n - 1}.$$

(c) (factorisation) For s not a zero of G and $s \neq 0$: $\frac{G_x(s)}{G(s)} = [-nsG_x(s)] \cdot \left[\frac{-1}{nsG(s)}\right]$, where $-1/(nsG(s)) = \frac{1}{n} + \sum_j \varrho_j \sigma_j / (\sigma_j - s)$ is the transform of the stationary-start law and $-nsG_x(-u/q) = u \int_0^\infty e^{-ut} n p_t(x, a) dt$ is the transform of the derivative of the normalised arrival profile $n p_t(x, a)$ of the walk without target.

Proof. (a),(b): By (3.1) and (2.3), $\sum_x r_j(x) = \langle \psi_j, \mathbf{1} \rangle^2 / \|\psi_j\|^2 = 1/(\sigma_j^2 G'(\sigma_j)) > 0$, and $\sum_j \langle \psi_j, \mathbf{1} \rangle^2 / \|\psi_j\|^2 = \|\mathbf{1}_{\Omega'}\|^2 = n - 1$ because the ψ_j are an orthogonal basis of $K_0 \ni \mathbf{1}_{\Omega'}$. Averaging (3.2), (3.3) over $x \in \Omega$ (with $T = 0$ for $x = a$) gives the mixture representation. Next, $r_j(x)^2 = \pi_j(x) \langle \psi_j, \mathbf{1} \rangle^2 / \|\psi_j\|^2 = \pi_j(x) n \varrho_j$ with $\pi_j(x) := \psi_j(x)^2 / \|\psi_j\|^2$, and $\sum_j \pi_j(x)$ is the squared norm of the orthogonal projection of e_x onto K_0 , hence ≤ 1 . The last inequality is Cauchy–Schwarz: $\sum_j |r_j| \leq (\sum_j \pi_j(x))^{1/2} (\sum_j n \varrho_j)^{1/2}$. (c) $\frac{1}{n} \sum_{x \in \Omega} G_x(s) = \frac{1}{n} \langle \mathbf{1}, (L - s)^{-1} e_a \rangle = -\frac{1}{ns}$, so the average over $x \in \Omega$ of G_x/G (which is 1 for $x = a$) equals $-1/(nsG(s))$; by (3.4) and (5.2) it also equals $\frac{1}{n} + \sum_j \varrho_j \sigma_j / (\sigma_j - s)$. Finally $G_x(-u/q) = q \int_0^\infty e^{-ut} p_t(x, a) dt$ for the walk with generator $-qL$. $\square$

Proposition 5.5 (quasi-stationary averages). *Assume (1.1) and that the graph of L_a on Ω' is connected, and let $\nu(x) = \psi_0(x) / \sum_z \psi_0(z)$, $x \in \Omega'$, be the normalised principal eigenvector of L_a (the quasi-stationary distribution). Then*

$$\sum_{x \in \Omega'} \nu(x) r_j(x) = \delta_{j0} \quad (0 \leq j \leq p-1), \quad \sum_{x \in \Omega'} \nu(x) m(x) = \frac{1}{\sigma_0}, \quad (5.3)$$

where $m(x) = \mathbb{E}_x T$ for the unit-rate walk. Equivalently, started from ν the hitting time is exactly exponential: $\mathbb{P}_\nu(T > t) = e^{-q\sigma_0 t}$ (continuous time), $\mathbb{P}_\nu(T > t) = \nu_0^t$ (discrete time). Consequently

$$\min_{x \in \Omega'} r_0(x) \leq 1 \leq \max_{x \in \Omega'} r_0(x), \quad \min_{x \in \Omega'} m(x) \leq \frac{1}{\sigma_0} \leq \max_{x \in \Omega'} m(x),$$

and each of these four inequalities is an equality if and only if $\mathbf{1}_{\Omega'}$ is an eigenvector of L_a , i.e. iff L_{ya} takes the same value for all $y \in \Omega'$.

Proof. By (3.1) and the orthogonality of the ψ_j , $\sum_x \psi_0(x) r_j(x) = \langle \psi_0, \psi_j \rangle \langle \psi_j, \mathbf{1} \rangle / \|\psi_j\|^2 = \delta_{j0} \langle \psi_0, \mathbf{1} \rangle$; divide by $\langle \psi_0, \mathbf{1} \rangle > 0$ (Theorem 5.1(c)). Inserting this into (3.2), (3.3) gives the exponential (geometric) law, and with Theorem 5.1(b) the mean $\sum_x \nu(x) m(x) = \sum_j r_j^\nu / \sigma_j = 1/\sigma_0$. The two-sided inequalities hold because ν is a probability vector with strictly positive entries, and equality in any of them forces $r_0(\cdot)$, respectively $m(\cdot)$, to be constant on Ω' . By (3.1), $r_0(\cdot)$ is constant iff ψ_0 is, i.e. iff $\mathbf{1}_{\Omega'}$ is an eigenvector of L_a ; and $m = L_a^{-1} \mathbf{1}_{\Omega'}$ (Lemma 6.1) is constant iff $L_a \mathbf{1}_{\Omega'}$ is a multiple of $\mathbf{1}_{\Omega'}$. Finally $(L_a \mathbf{1}_{\Omega'})(y) = -L_{ya}$ because the rows of L sum to zero. $\square$

5.1 Reflection-symmetric pairs: the sign of r_1

Theorem 5.6 (sign rule for a symmetric start–target pair). *Assume (1.1) and let R be a permutation of Ω with $RLR^{-1} = L$, $R^2 = \text{id}$ and $Ra = x_0 \neq a$. Put $v_\pm = \frac{1}{2}(e_a \pm e_{x_0})$ and*

$$G^\pm(s) = \langle v_\pm, (L - s)^{-1} v_\pm \rangle = \sum_i \frac{w_i^\pm}{\lambda^{(i)} - s}, \quad w_i^\pm = \|E_i v_\pm\|^2.$$

(a) $G = G^+ + G^-$, $G_{x_0} = G^+ - G^-$, and for every j

$$r_j(x_0) = \frac{2G^-(\sigma_j)}{\sigma_j G'(\sigma_j)} = -\frac{2G^+(\sigma_j)}{\sigma_j G'(\sigma_j)}. \quad (5.4)$$

Thus $r_j(x_0) > 0$ iff $G^-(\sigma_j) > 0$ iff $G^+(\sigma_j) < 0$.

(b) Suppose in addition that (H1) the first visible eigenvalue Λ_1 is purely odd, $E_{\Lambda_1} v_+ = 0$, and (H2) L has at least two eigenvalues, counted with multiplicity, in $(0, \Lambda_1]$. Then $r_1(x_0) < 0$.

Proof. (a) R commutes with $(L - s)^{-1}$ and maps $v_{\pm}$ to $\pm v_{\pm}$, so $\langle v_+, (L - s)^{-1} v_- \rangle = 0$. With $e_a = v_+ + v_-$, $e_{x_0} = v_+ - v_-$ this gives the two decompositions, and (5.4) follows from (3.1) and $G^+(\sigma_j) = -G^-(\sigma_j)$.

(b) Let L_{a,x_0} be the principal submatrix of L on $\Omega'' = \Omega \setminus \{a, x_0\}$ and $\eta_1 > 0$ its smallest eigenvalue (L_a is positive definite by Theorem 2.2(iii), hence so is its principal submatrix L_{a,x_0}). Applying Theorem 4.1(a) twice ($L \rightarrow L_a \rightarrow L_{a,x_0}$) gives $\eta_1 \leq \mu_2 \leq \lambda_3$, the third smallest eigenvalue of L counted with multiplicity; by (H2), $\lambda_3 \leq \Lambda_1$. So

$$0 < \eta_1 \leq \Lambda_1. \quad (5.5)$$

By (H2), $n \geq 3$, so $\Omega'' \neq \emptyset$. With $c = \max_x L_{xx}$ the matrix $cI - L_{a,x_0}$ is non-negative and symmetric, hence (Horn–Johnson, Theorem 8.3.1, valid for every order ≥ 1 : the spectral radius of a non-negative matrix is an eigenvalue with a non-negative eigenvector) has a non-negative eigenvector $\chi \neq 0$ for its spectral radius, which for a symmetric matrix is its largest eigenvalue $c - \eta_1$. R preserves Ω'' and commutes with L_{a,x_0} , so $\psi = \chi + \chi \circ R$ is a non-negative, non-zero, R -invariant eigenvector of L_{a,x_0} for η_1 . Extend ψ by zero to Ω . Then $L\psi = \eta_1\psi + \alpha(e_a + e_{x_0})$ with $\alpha = (L\psi)(a) = \sum_y L_{ay}\psi(y) \leq 0$ (the coefficients at a and x_0 agree by R -invariance). If $\alpha = 0$, ψ would be an eigenvector of L with eigenvalue $\eta_1 > 0$, hence orthogonal to $\mathbf{1}$, contradicting $\psi \geq 0$, $\psi \neq 0$. So $\alpha < 0$ and, applying E_i ,

$$(\lambda^{(i)} - \eta_1) E_i \psi = 2\alpha E_i v_+ \quad \text{for all } i.$$

If $\lambda^{(i)} = \eta_1$ this gives $E_i v_+ = 0$; otherwise $w_i^+ / (\lambda^{(i)} - \eta_1) = \langle E_i v_+, E_i \psi \rangle / (2\alpha)$. Summing,

$$G^+(\eta_1) = \frac{1}{2\alpha} \sum_{i: \lambda^{(i)} \neq \eta_1} \langle E_i v_+, \psi \rangle = \frac{1}{2\alpha} \langle v_+, \psi \rangle = \frac{\psi(a) + \psi(x_0)}{4\alpha} = 0.$$

By (H1) the poles of G^+ are 0 (with $w_0^+ = 1/n$) and visible eigenvalues $\geq \Lambda_2$; let $\Lambda^+ \geq \Lambda_2$ be the smallest positive one (there is one, otherwise $G^+(s) = -1/(ns)$ would have no zero). On $(0, \Lambda^+)$, G^+ increases strictly from $-\infty$ to $+\infty$, and by (5.5) its zero there is $\eta_1 \leq \Lambda_1$. Since $\Lambda_1 < \sigma_1 < \Lambda_2 \leq \Lambda^+$ we get $G^+(\sigma_1) > G^+(\eta_1) = 0$, and (5.4) gives $r_1(x_0) < 0$. $\square$

Corollary 5.7 (corner to corner). *For the reflecting box with $d \geq 2$, every $N \geq 2$, $a = (N, \dots, N)$ and $x_0 = (1, \dots, 1)$:*

$$r_0 > 0, \quad r_1 < 0.$$

Proof. Ω' is connected, so $r_0 > 0$ by Theorem 5.1(c). Take $R : x_i \mapsto N + 1 - x_i$ for all i . Then $\phi_k \circ R = (-1)^{|k|} \phi_k$; the eigenspace of Λ_1 is spanned by the ϕ_{e_i} , which are odd, so (H1) holds; it has dimension $d \geq 2$ and Λ_1 is the smallest non-zero eigenvalue, so (H2) holds. $\square$

Remark 5.8 (which residues are positive). In the situation of Theorem 5.6(a), let $0 = \Lambda_0^+ < \Lambda_1^+ < \dots$ be the poles of G^+ and $\rho_i^+ \in (\Lambda_i^+, \Lambda_{i+1}^+)$ its zeros. The ρ_i^+ are eigenvalues of the walk with *both* a and x_0 absorbing in the R -even sector (same argument as Theorem 2.2, applied to v_+), and $G^+ > 0$ exactly on $\bigcup_i (\rho_i^+, \Lambda_{i+1}^+)$. Hence

$$r_j(x_0) < 0 \iff \sigma_j \in \bigcup_i (\rho_i^+, \Lambda_{i+1}^+), \quad r_j(x_0) = 0 \iff \sigma_j \in \{\rho_i^+\}.$$

For the box, G^+ (G^-) is the part of (2.1) with $|k|$ even (odd), and $\mathbb{E}_{x_0} T = \frac{2n}{q} G^-(0)$ (Lemma 6.1 below). In one dimension the rule is consistent with the alternation of Corollary 5.2. In $d \geq 2$ with $N \geq 3$ the residues do *not* alternate (Theorem 5.10, Proposition 14.1); formula (5.4) and the rule were checked against brute-force residues for all poles (script 12).

5.2 Alternation of the residues holds only for $d = 1$ or $N = 2$

In one dimension the residues alternate in sign (Corollary 5.2). The natural conjecture that they also alternate for the corner-to-corner geometry in $d \geq 2$ is *false* (already for $d = 2$, $N = 3$ the signs are $++-$, Proposition 14.1). The following two statements are the true replacement: a monotonicity rule across every eigenvalue of pure parity, and, as a consequence, a complete answer for the box.

Proposition 5.9 (monotonicity of the signs across an eigenvalue of pure parity). *In the situation of Theorem 5.6(a) call a visible eigenvalue Λ_i purely even if $E_{\Lambda_i}v_- = 0$ and purely odd if $E_{\Lambda_i}v_+ = 0$ (E_{Λ_i} the spectral projector of L for Λ_i). Let $1 \leq i \leq p-1$ and $r_j = r_j(x_0)$.*

- (a) *If Λ_i is purely even, then $G^-(\sigma_{i-1}) < G^-(\sigma_i)$; hence $r_{i-1} \geq 0$ implies $r_i > 0$, and $r_i \leq 0$ implies $r_{i-1} < 0$.*
- (b) *If Λ_i is purely odd, then $G^+(\sigma_{i-1}) < G^+(\sigma_i)$; hence $r_{i-1} \leq 0$ implies $r_i < 0$, and $r_i \geq 0$ implies $r_{i-1} > 0$.*

In particular the pair (r_{i-1}, r_i) cannot have the signs $(+, -)$ in case (a) nor $(-, +)$ in case (b); and if $\Lambda_i, \Lambda_{i+1}, \dots, \Lambda_{i+l-1}$ ($i+l-1 \leq p-1$) are all purely even (all purely odd), then the sequence $\text{sign } r_{i-1}, \dots, \text{sign } r_{i+l-1} \in \{-1, 0, 1\}$ is non-decreasing (non-increasing), with at most one zero.

Proof. As in the proof of Theorem 5.6(a), R commutes with every spectral projector and $Rv_{\pm} = \pm v_{\pm}$, so $\langle E_i v_+, E_i v_- \rangle = 0$ and $w(\lambda^{(i)}) = \|E_i e_a\|^2 = w_i^+ + w_i^-$ for every eigenvalue. Hence every pole of G^+ or of G^- is a visible eigenvalue. (a) If $w^-(\Lambda_i) = 0$, the only visible eigenvalue in the open interval $I = (\Lambda_{i-1}, \Lambda_{i+1})$ is not a pole of G^- , so G^- is real-analytic on I with $(G^-)'(s) = \sum_l w_l^-(\lambda^{(l)} - s)^{-2} > 0$ (some w_l^- is positive because $\sum_l w_l^- = \|v_-\|^2 = \frac{1}{2}$). By (2.2), $\sigma_{i-1} < \Lambda_i < \sigma_i$ both lie in I , so $G^-(\sigma_{i-1}) < G^-(\sigma_i)$, and by (5.4) $\text{sign } r_j = \text{sign } G^-(\sigma_j)$ because $\sigma_j G'(\sigma_j) > 0$. (b) is the same argument for G^+ , whose poles in I could only be Λ_i (note $0 = \Lambda_0 \notin I$), with $\text{sign } r_j = -\text{sign } G^+(\sigma_j)$. The last statement follows by applying (a) (resp. (b)) to consecutive indices. $\square$

Theorem 5.10 (the residues alternate if and only if $d = 1$ or $N = 2$). *Consider the reflecting box $\{1, \dots, N\}^d$, $d \geq 1$, $N \geq 2$, with $a = (N, \dots, N)$, $x_0 = (1, \dots, 1)$, $D = d(N-1)$, and say that the residues alternate if $\text{sign } r_j = (-1)^j$ for all $0 \leq j \leq p-1$.*

- (a) *If $d = 1$ or $N = 2$, then $p = D$ and the residues alternate.*
- (b) *Let $d \geq 2$ and $N \geq 3$, and put $i = p-3$ if $N \neq 4$, $i = p-4$ if $N = 4$. Then $i \geq 1$, and the two consecutive eigenvalues Λ_i, Λ_{i+1} are both purely even or both purely odd in the sense of Proposition 5.9: purely even if D is even and $N \neq 4$, or if D is odd and $N = 4$; purely odd otherwise. Consequently the sequence $(\text{sign } r_{i-1}, \text{sign } r_i, \text{sign } r_{i+1})$ is non-decreasing in the first case and non-increasing in the second; in particular the residues do not alternate.*
- (c) *(failures at small indices) Moreover, writing “ $(r_l, r_{l+1}) \neq (+, -)$ ” for “ $r_l > 0 > r_{l+1}$ does not hold”: for $d = 2$, $N \geq 3$, $(r_2, r_3) \neq (+, -)$; for $d \geq 4$, $N \geq 4$, $(r_4, r_5) \neq (+, -)$; for $d = 3$, $N \geq 7$, $(r_6, r_7) \neq (+, -)$. In each of these cases alternation would require exactly the excluded signs.*

The proof uses no computer assistance.

Proof. Let $u = \frac{\pi}{2N}$, $x = \sin^2 u$ and $\varsigma_j = \frac{1 - \cos \theta_j}{1 - \cos \theta_1} = \frac{\sin^2(ju)}{\sin^2 u}$ for $0 \leq j \leq N-1$. Then $\varsigma_0 = 0$, $\varsigma_1 = 1$, $\varsigma_2 = 4 \cos^2 u = 4(1-x) =: t$, $\varsigma_3 = (3-4x)^2$ (from $\sin 3u = \sin u(3-4 \sin^2 u)$), and ς_j is strictly increasing in j because $ju \in [0, \frac{\pi}{2}]$. By (1.4) and Corollary 4.2(c), $\lambda_k = \Lambda_1 V(k)$

with $V(k) = \sum_i \varsigma_{k_i}$; every eigenvalue is visible from the corner, so $\Lambda_0 < \Lambda_1 < \dots < \Lambda_p$ is the increasing list of the distinct values of V (times Λ_1). With $R : x_i \mapsto N + 1 - x_i$ we have $\phi_k \circ R = (-1)^{|k|} \phi_k$; if all k with $\lambda_k = \Lambda_i$ have $|k|$ even (odd), then R acts as $+1$ (-1) on the range of E_{Λ_i} and $E_{\Lambda_i} v_{\mp} = \frac{1}{2}(E_{\Lambda_i} e_a \mp R E_{\Lambda_i} e_a) = 0$, i.e. Λ_i is purely even (purely odd) in the sense of Proposition 5.9.

(a) For $d = 1$ this is Corollary 5.2. For $N = 2$: $k \in \{0, 1\}^d$ and $V(k) = |k| \in \{0, 1, \dots, d\}$, so $p = d = D$. By Theorem 5.1(d) the p residues have at least $D - 1 = p - 1$ sign changes, which forces all of them to be non-zero and of alternating sign, and $r_0 > 0$ by Theorem 5.1(c) (Ω' is connected).

(b) We count from the top of the spectrum. For $0 \leq m \leq N - 1$ let $\Delta_m = \varsigma_{N-1} - \varsigma_{N-1-m}$, so $\Delta_0 = 0$. Since $\sin^2 A - \sin^2 B = \sin(A + B) \sin(A - B)$ and $(2N - 2m - 1)u = \pi - (2m + 1)u$,

$$\Delta_m - \Delta_{m-1} = \frac{\sin^2((N - m)u) - \sin^2((N - m - 1)u)}{\sin^2 u} = \frac{\sin((2m + 1)u)}{\sin u} > 0 \quad (1 \leq m \leq N - 1), \quad (5.6)$$

because $0 < (2m + 1)u < 2Nu = \pi$; so Δ_m is strictly increasing in m . For $k \in \{0, \dots, N - 1\}^d$ put $m_l = N - 1 - k_l$ and $\Delta(m) = \sum_l \Delta_{m_l}$; then $V(k) = d\varsigma_{N-1} - \Delta(m)$ and $|k| = D - |m|$. Hence, if $0 = \Delta^{(0)} < \Delta^{(1)} < \dots$ are the distinct values of Δ on $\{0, \dots, N - 1\}^d$, then $\Lambda_{p-j} = \Lambda_1(d\varsigma_{N-1} - \Delta^{(j)})$, and if all m with $\Delta(m) = \Delta^{(j)}$ have $|m|$ of one parity, then Λ_{p-j} is purely even if $D - |m|$ is even and purely odd if it is odd. Below a vector m is described by the multiset of its non-zero entries, and n_l denotes the number of entries equal to l . In this proof let

$$\alpha = \Delta_1 = \frac{\sin 3u}{\sin u} = 3 - 4x, \quad \beta = \Delta_2 - \Delta_1 = \frac{\sin 5u}{\sin u} = 5 - 20x + 16x^2$$

(multiple-angle formulas). For $N \geq 3$ we have $0 < x \leq \frac{1}{4}$, hence $2\alpha - \beta = 1 + 12x - 16x^2 > 0$; and $\beta - \alpha = 2(1 - 8x + 8x^2)$ is positive for $x < \sin^2 \frac{\pi}{8} = \frac{2 - \sqrt{2}}{4}$, i.e. for $N \geq 5$, and zero for $N = 4$.

Case $N \geq 5$. Here $0 < \alpha < \beta < 2\alpha$. Let $\Delta(m) \leq \alpha + \beta = \Delta_2$. As Δ_l increases with l , all entries of m are ≤ 2 ; if m has an entry 2, then $\Delta(m) \geq \Delta_2$ with equality only for $m = \{2\}$; otherwise $\Delta(m) = n_1\alpha$, and $n_1\alpha \leq \alpha + \beta < 3\alpha$ forces $n_1 \leq 2$. So the four smallest values of Δ are $0 < \alpha < 2\alpha < \alpha + \beta$, attained only by $\emptyset$, $\{1\}$, $\{1, 1\}$ (which exists because $d \geq 2$) and $\{2\}$. In particular Λ_{p-2} ($\Delta = 2\alpha$) and Λ_{p-3} ($\Delta = \alpha + \beta$) are pure, both with $|m| = 2$.

Case $N = 3$. Here $x = \frac{1}{4}$, $\Delta_1 = 2$, $\Delta_2 = 3$, so $\Delta(m) = 2n_1 + 3n_2$; the four smallest values are $0 < 2 < 3 < 4$, attained only by $\emptyset$, $\{1\}$, $\{2\}$, $\{1, 1\}$. So Λ_{p-2} ($\Delta = 3$) and Λ_{p-3} ($\Delta = 4$) are pure, both with $|m| = 2$.

Case $N = 4$. Here $x = \sin^2 \frac{\pi}{8}$, $\alpha = \beta = 1 + \sqrt{2}$ and $\Delta_3 - \Delta_2 = \sin \frac{7\pi}{8} / \sin \frac{\pi}{8} = 1$, so $\Delta_1 = \alpha$, $\Delta_2 = 2\alpha$, $\Delta_3 = 2\alpha + 1$. Thus $\Delta(m) = c\alpha + n_3$ with $c = n_1 + 2n_2 + 2n_3$, while $|m| = n_1 + 2n_2 + 3n_3 = c + n_3$. Since α is irrational, $\Delta(m)$ determines (c, n_3) and hence $|m|$; so every eigenvalue is pure when $N = 4$. As $2n_3 \leq c$ and $\alpha > 1$, the values of Δ below 3α are those with $c \leq 2$, namely $0 < \alpha < 2\alpha < 2\alpha + 1$, and the value 3α is attained (by $\{1, 2\}$, as $d \geq 2$). So Λ_{p-3} ($\Delta = 2\alpha + 1$, $(c, n_3) = (2, 1)$) and Λ_{p-4} ($\Delta = 3\alpha$, $(c, n_3) = (3, 0)$) are pure, both with $|m| = 3$.

Conclusion. The eigenvalue $\Lambda_0 = 0$ belongs to $m = (N - 1, \dots, N - 1)$, with $\Delta(m) = d\Delta_{N-1} \geq 2\Delta_2$, which exceeds all the values found above ($2\alpha + 2\beta > \alpha + \beta$ for $N \geq 5$, $6 > 4$ for $N = 3$, $4\alpha > 3\alpha$ for $N = 4$). Hence $p \geq 4$ ($N \neq 4$), resp. $p \geq 5$ ($N = 4$), so $1 \leq i$ and $i + 1 \leq p - 2$. The eigenvalues Λ_i, Λ_{i+1} are pure with $|m| = 2$ ($N \neq 4$), resp. $|m| = 3$ ($N = 4$), so both are purely even iff $D - 2$, resp. $D - 3$, is even, and both are purely odd otherwise. By the last statement of Proposition 5.9 (with $l = 2$), $(\text{sign } r_{i-1}, \text{sign } r_i, \text{sign } r_{i+1})$ is non-decreasing, resp. non-increasing, whereas alternation would require $(+, -, +)$ or $(-, +, -)$.

(c) $d = 2$, $N \geq 3$. Here $x \leq \sin^2 \frac{\pi}{6} = \frac{1}{4}$, so $t \geq 3$. If $V(k) \leq t$, then no component of k is ≥ 3 (otherwise $V(k) \geq \varsigma_3 > \varsigma_2 = t$), so $V(k) \in \{0, 1, 2, t, t + 1, 2t\}$. Hence the four smallest distinct values are $0 < 1 < 2 < t$, i.e. $\Lambda_3 = t\Lambda_1$ with eigenspace spanned by $\phi_{(2,0)}, \phi_{(0,2)}$: Λ_3 is purely even,

and $3 \leq p-1$ because $t+1$ and $2t$ are two further values. By Proposition 5.9(a), $r_2 \geq 0$ implies $r_3 > 0$. Alternation would require $r_2 > 0 > r_3$.

$d \geq 4$, $N \geq 4$. Here $0 < x \leq \sin^2 \frac{\pi}{8} < 0.1465$, so $3 < t < 4$ and $\varsigma_3 = (3-4x)^2 > 5.8$. Let $V(k) \leq 4$. Then k has no component ≥ 3 (as $\varsigma_3 > 4$) and at most one component equal to 2 (as $2t > 6$); if it has one, the other components vanish (as $t+1 > 4$) and $V(k) = t$; otherwise $V(k) = |k| \in \{0, 1, 2, 3, 4\}$. So the six smallest distinct values are $0 < 1 < 2 < 3 < t < 4$, $\Lambda_5 = 4\Lambda_1$, and its eigenspace is spanned by the ϕ_k with k a permutation of $(1, 1, 1, 1, 0, \dots, 0)$, $|k| = 4$: Λ_5 is purely even. Further values exist ($t+1, 2t$), so $5 \leq p-1$. By Proposition 5.9(a), $r_4 \geq 0$ implies $r_5 > 0$; alternation would require $r_4 > 0 > r_5$.

$d = 3$, $N \geq 7$. Here $0 < x \leq \sin^2 \frac{\pi}{14} < 0.0496$, so $3.8 < t < 4$ and $\varsigma_3 - 2t = 1 - 16x + 16x^2 > 0$. If $V(k) \leq 2t$, then k has no component ≥ 3 , so with l components equal to 2 and j equal to 1 ($j+l \leq 3$), $V(k) = lt + j \in \{0, 1, 2, 3, t, t+1, t+2, 2t, 2t+1, 3t\}$. As $3 < t$ and $t+2 < 2t$, the eight smallest distinct values are $0 < 1 < 2 < 3 < t < t+1 < t+2 < 2t$, so $\Lambda_7 = 2t\Lambda_1$ with eigenspace spanned by the ϕ_k , k a permutation of $(2, 2, 0)$, $|k| = 4$: purely even; and $7 \leq p-1$ ($2t+1$ and $3t$ are further values). By Proposition 5.9(a), $r_6 \geq 0$ implies $r_7 > 0$; alternation would require $r_6 > 0 > r_7$. $\square$

Remark 5.11 (the sign pattern as the sign of one polynomial; general chain). For an arbitrary chain satisfying (1.1) and a start $x \neq a$ with $D = \text{dist}(x, a)$, write $G_x(s) = \sum_{i=0}^p \omega_i / (\Lambda_i - s)$, $\omega_i = \langle e_x, E_{\Lambda_i} e_a \rangle$. Then

$$\mathcal{N}_x(s) := G_x(s) \prod_{l=0}^p (\Lambda_l - s) = \sum_{i=0}^p \omega_i \prod_{l \neq i} (\Lambda_l - s)$$

is a real polynomial of degree *exactly* $p - D$, with leading coefficient $(-1)^p (L^D)_{xa}$ of sign $(-1)^{p+D}$, and

$$\text{sign } r_j(x) = (-1)^j \text{sign } \mathcal{N}_x(\sigma_j) \quad (0 \leq j \leq p-1), \quad \text{sign } \mathcal{N}_x(\Lambda_i) = (-1)^i \text{sign } \omega_i \quad (0 \leq i \leq p). \quad (5.7)$$

Indeed, for $|s| > \|L\|$ the Neumann series gives $G_x(s) = -\sum_{m \geq 0} (L^m)_{xa} s^{-m-1}$, where $(L^m)_{xa} = 0$ for $m < D$ and $(L^D)_{xa} \neq 0$ is a sum over geodesics of products of D negative entries (as in the proof of Theorem 5.1(d)); multiplying by $\prod_l (\Lambda_l - s) = (-1)^{p+1} s^{p+1} (1 + O(s^{-1}))$ gives the degree and the leading coefficient. The first relation in (5.7) follows from (3.1), $\sigma_j G'(\sigma_j) > 0$ and the fact that exactly $j+1$ factors of $\prod_l (\Lambda_l - \sigma_j)$ are negative; the second from the definition. Consequences: (1) $\text{sign } r_j = (-1)^j$ for all j iff $\mathcal{N}_x > 0$ at every σ_j ; in particular this holds whenever $p = D$ (then $\mathcal{N}_x$ is the constant $(-1)^p (L^D)_{xa} > 0$), which is case (a) of Theorem 5.10. (2) If $r_j r_{j+1} > 0$ then $\mathcal{N}_x$ has a zero in (σ_j, σ_{j+1}) , so at most $p - D$ of the $p - 1$ consecutive pairs fail to alternate; this is again the bound “at least $D - 1$ sign changes” of Theorem 5.1(d) when no residue vanishes. (3) The sequence $((-1)^i \text{sign } \omega_i)_{1 \leq i \leq p}$ (zeros deleted) has at most $p - D$ sign changes. (4) If the residues alternate, then $\mathcal{N}_x > 0$ at σ_{i-1} and σ_i , so every $i \in \{1, \dots, p-1\}$ with $(-1)^i \omega_i \leq 0$ forces two zeros of $\mathcal{N}_x$ (with multiplicity) in (σ_{i-1}, σ_i) ; hence alternation implies $2 \# \{1 \leq i \leq p-1 : (-1)^i \omega_i \leq 0\} \leq p - D$. For the corner-to-corner box $\omega_i = \frac{1}{n} \sum_{k: \lambda_k = \Lambda_i} (-1)^{|k|} \prod_l \varepsilon_{kl} \cos^2 \frac{\theta_{kl}}{2}$, and this counting criterion (which needs only the ordered spectrum, no pole) excludes alternation in 95 of the 104 pairs (d, N) with $d \geq 2$, $N \geq 3$ examined in script 18, but not in all of them (not for $(d, N) = (3, 4), (3, 20), (3, 21), (3, 23)$ and $(d, 4)$, $4 \leq d \leq 8$); it is therefore not a substitute for Theorem 5.10, whose proof uses the finer information of Proposition 5.9.

Part II. Explicit bounds (any finite symmetric chain)

6 Explicit two-sided bounds on $\sigma_0, \sigma_1, r_0, r_1$ (general chain)

This section is still for a general L satisfying (1.1); all hypotheses are stated in terms of a few explicit functionals of the Green's function at the target. Their values for the box are computed in Sections 7–10; the resulting explicit statements are collected in Section 11.

6.1 The Green's function functionals

With the visible spectrum $0 = \Lambda_0 < \Lambda_1 < \dots < \Lambda_p$ and weights w_i of Definition 2.1, write $G(s) = -\frac{1}{ns} + g(s)$ and, for a start $x \in \Omega'$, $G_x(s) = -\frac{1}{ns} + g_x(s)$, where

$$g(s) = \sum_{i=1}^p \frac{w_i}{\Lambda_i - s}, \quad g_x(s) = \sum_{i=1}^p \frac{\sqrt{w_i} u_i(x)}{\Lambda_i - s} \quad (6.1)$$

are the regular parts ($g(0) = \langle e_a, L^+ e_a \rangle$ and $g_x(0) = \langle e_x, L^+ e_a \rangle$ with L^+ the Moore–Penrose inverse). Define

$$\tau = n g(0) = n \sum_{i \geq 1} \frac{w_i}{\Lambda_i}, \quad K_2 = n g'(0) = n \sum_{i \geq 1} \frac{w_i}{\Lambda_i^2}, \quad m(x) = n(g(0) - g_x(0)), \quad (6.2)$$

$$\bar{X} = \Lambda_1 \tau, \quad \kappa = \frac{K_2}{\tau^2}, \quad W = n w_1, \quad \mu_x = \frac{m(x) - \tau}{\tau}. \quad (6.3)$$

Lemma 6.1 (meaning of τ and m). *For the unit-rate walk ($q = 1$), $m(x) = \mathbb{E}_x T_a$ for every x (with $m(a) = 0$), and $\tau = \frac{1}{n} \sum_{x \in \Omega} \mathbb{E}_x T_a$ is the mean hitting time from the uniform (stationary) distribution. For activity q both are divided by q , in continuous and in discrete time. Moreover $\kappa \leq 1/\bar{X}$.*

Proof. Let $h(x) = n(g(0) - g_x(0)) = n\langle e_a - e_x, L^+ e_a \rangle$. Since $LL^+ = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, $(Lh)(x) = -n(LL^+ e_a)(x) = -n(\delta_{xa} - \frac{1}{n}) = 1$ for $x \neq a$, and $h(a) = 0$. Hence $L_a(h|_{\Omega'}) = \mathbf{1}_{\Omega'}$, i.e. $h|_{\Omega'} = L_a^{-1}\mathbf{1}_{\Omega'}$, which is $\int_0^\infty e^{-tL_a}\mathbf{1} dt = \mathbb{E}.T_a$ by (1.2) (and $\sum_{t \geq 0} Q^t \mathbf{1} = (qL_a)^{-1}\mathbf{1}$ in discrete time). Averaging over x : $\frac{1}{n} \sum_x h(x) = ng(0) - \langle \mathbf{1}, L^+ e_a \rangle = ng(0)$. Finally $K_2 = n \sum_{i \geq 1} w_i \Lambda_i^{-2} \leq \Lambda_1^{-1} \tau$, so $\kappa \leq 1/(\Lambda_1 \tau)$. $\square$

We shall use repeatedly the elementary inequalities, valid for $0 \leq s < \Lambda \leq \Lambda'$:

$$\frac{1}{\Lambda'} + \frac{s}{\Lambda'^2} \leq \frac{1}{\Lambda' - s} = \frac{1}{\Lambda'} + \frac{s}{\Lambda'(\Lambda' - s)} \leq \frac{1}{\Lambda'} + \frac{s}{\Lambda'^2} \cdot \frac{1}{1 - s/\Lambda}, \quad \frac{1}{\Lambda'^2} \leq \frac{1}{(\Lambda' - s)^2} \leq \frac{1}{\Lambda'^2(1 - s/\Lambda)^2}. \quad (6.4)$$

6.2 The principal eigenvalue

Theorem 6.2 (principal killed eigenvalue). *Assume (1.1). Then*

$$\tau \leq \frac{1}{\sigma_0} \leq \tau + \frac{1}{\Lambda_1}, \quad \text{i.e.} \quad \frac{\bar{X}}{1 + \bar{X}} \leq \sigma_0 \tau \leq 1. \quad (6.5)$$

If $\bar{X} > 1$, put $\beta = \bar{X}/(\bar{X} - 1)$. Then

$$1 - \beta\kappa \leq \frac{2}{1 + \sqrt{1 + 4\beta\kappa}} \leq \sigma_0 \tau \leq \frac{2}{1 + \sqrt{1 + 4\kappa}} \leq 1 - \kappa + 2\kappa^2. \quad (6.6)$$

Proof. σ_0 is the zero of G in $(0, \Lambda_1)$, i.e. $1 = \sigma_0 n g(\sigma_0)$. By (6.4) with $\Lambda = \Lambda_1$, for $0 \leq s < \Lambda_1$,

$$\tau + sK_2 \leq n g(s) \leq \tau + \frac{sK_2}{1 - s/\Lambda_1}, \quad n g(s) \leq \frac{\tau}{1 - s/\Lambda_1}.$$

The first inequality at $s = \sigma_0$ gives $1 \geq \sigma_0 \tau$ (hence $\sigma_0 \leq 1/\tau$) and, with $t = \sigma_0 \tau$, $1 \geq t + \kappa t^2$, i.e. $t \leq 2/(1 + \sqrt{1 + 4\kappa})$. The third gives $1 - \sigma_0/\Lambda_1 \leq \sigma_0 \tau$, i.e. $1/\sigma_0 \leq \tau + 1/\Lambda_1$. If $\bar{X} > 1$ then $\sigma_0/\Lambda_1 \leq 1/\bar{X} < 1$, and the second inequality gives $1 \leq t + \beta \kappa t^2$, i.e. $t \geq 2/(1 + \sqrt{1 + 4\beta\kappa})$. The outer inequalities in (6.6) are $\frac{2}{1 + \sqrt{1 + 4y}} \geq \frac{1}{1 + y} \geq 1 - y$ and $\frac{2}{1 + \sqrt{1 + 4y}} = \frac{\sqrt{1 + 4y} - 1}{2y} \leq 1 - y + 2y^2$ (from $\sqrt{1 + z} \leq 1 + \frac{z}{2} - \frac{z^2}{8} + \frac{z^3}{16}$, $z \geq 0$). $\square$

Remark 6.3. (6.5) says that the quasi-stationary mean exit time $1/\sigma_0$ lies between the stationary mean hitting time and that time plus the relaxation time $1/\Lambda_1$ of the visible sector. For reversible chains inequalities of this type go back to D. Aldous and M. Brown, *Inequalities for rare events in time-reversible Markov chains I*, in: Stochastic Inequalities, IMS Lecture Notes 22 (1992), 1–16; the proof above is self-contained and gives the refinement (6.6), whose two sides agree to first order in κ .

Corollary 6.4 ($\sigma_0 \cdot \text{MFPT}$). *For every start $x \in \Omega'$: $\sigma_0 m(x) = \sigma_0 \tau (1 + \mu_x)$, hence, if $\bar{X} > 1$,*

$$(1 - \beta\kappa)(1 + \mu_x) \leq \sigma_0 m(x) \leq (1 - \kappa + 2\kappa^2)(1 + \mu_x) \quad (\text{for } 1 + \mu_x \geq 0),$$

and always $\frac{\bar{X}}{1 + \bar{X}}(1 + \mu_x) \leq \sigma_0 m(x) \leq 1 + \mu_x$.

6.3 The second pole and the gap

Assume $p \geq 2$ and put

$$h(s) = \sum_{i=2}^p \frac{w_i}{\Lambda_i - s}, \quad Y_0 = n\Lambda_1 h(0) = \bar{X} - W, \quad \vartheta = n\Lambda_1^2 h'(0) = \Lambda_1^2 K_2 - W, \quad \varrho = \frac{\Lambda_1}{\Lambda_2} < 1. \quad (6.7)$$

Theorem 6.5 (second pole). *Assume (1.1), $p \geq 2$, and write $\sigma_1 = \Lambda_1(1 + \delta)$, $\delta > 0$. Then*

$$\delta = \frac{W}{Y(\sigma_1) - \frac{1}{1 + \delta}}, \quad Y(s) := n\Lambda_1 h(s). \quad (6.8)$$

(i) *If $Y_0 + \vartheta > 1$, then $\delta \leq \delta_+ := \frac{W}{Y_0 + \vartheta - 1}$.*

(ii) *If moreover $(1 + \delta_+)\varrho < 1$, put $\omega = \frac{1 + \delta_+}{1 - (1 + \delta_+)\varrho}$. Then $\delta \geq \delta_- := \frac{W}{Y_0 + \omega\vartheta - \frac{1}{1 + \delta_+}}$, and*

$$Y_0 + \vartheta \leq Y(\sigma_1) \leq Y_0 + \omega\vartheta, \quad \vartheta \leq n\Lambda_1^2 h'(\sigma_1) \leq \frac{\omega^2 \vartheta}{(1 + \delta_+)^2}.$$

Proof. $G(\sigma_1) = 0$ reads $-\frac{1}{n\sigma_1} - \frac{w_1}{\sigma_1 - \Lambda_1} + h(\sigma_1) = 0$; multiplying by $n\Lambda_1$ gives $W/\delta = Y(\sigma_1) - \Lambda_1/\sigma_1$, which is (6.8). (i) By (6.4) (first inequality, with $\Lambda' = \Lambda_i$, $i \geq 2$, and $s = \sigma_1 < \Lambda_2$), $h(\sigma_1) \geq h(0) + \sigma_1 h'(0) \geq h(0) + \Lambda_1 h'(0)$, so $Y(\sigma_1) \geq Y_0 + \vartheta$; and $\Lambda_1/\sigma_1 < 1$. Hence $W/\delta > Y_0 + \vartheta - 1 > 0$. (ii) By (i), $\sigma_1 \leq (1 + \delta_+)\Lambda_1 < \Lambda_2$, and (6.4) (second inequality, with $\Lambda = \Lambda_2$) gives $h(\sigma_1) \leq h(0) + \sigma_1 h'(0)/(1 - \sigma_1/\Lambda_2)$, i.e. $Y(\sigma_1) \leq Y_0 + \frac{(1 + \delta)\vartheta}{1 - (1 + \delta)\varrho} \leq Y_0 + \omega\vartheta$, the last step because $y \mapsto (1 + y)/(1 - (1 + y)\varrho)$ is increasing. Also $\Lambda_1/\sigma_1 \geq 1/(1 + \delta_+)$. Hence $W/\delta \leq Y_0 + \omega\vartheta - \frac{1}{1 + \delta_+}$ (the right-hand side is positive, being $\geq W/\delta$). The bounds on $h'(\sigma_1)$ follow from the last inequality in (6.4). $\square$

Corollary 6.6 (spectral gap). *Always $\sigma_1 - \sigma_0 > \Lambda_1 - \frac{1}{\tau} = \Lambda_1(1 - \bar{X}^{-1})$ and $\sigma_1 - \sigma_0 < \Lambda_2$. Under the hypotheses of Theorem 6.5(ii) and $\bar{X} > 1$,*

$$\Lambda_1 \left(1 + \delta_- - \frac{1}{\bar{X}}\right) \leq \sigma_1 - \sigma_0 \leq \Lambda_1 \left(1 + \delta_+ - \frac{1}{1 + \bar{X}}\right), \quad \bar{X}(1 + \delta_-) \leq \frac{\sigma_1}{\sigma_0} \leq (1 + \bar{X})(1 + \delta_+).$$

Proof. Interlacing (2.2), (6.5) and Theorem 6.5. $\square$

6.4 The residues r_0 and r_1

For the start x let $K_2^{(x)} = n \sum_{i \geq 1} \|E_i e_x\|^2 / (\lambda^{(i)})^2$ (the functional K_2 with x as target) and

$$\kappa_x = \frac{\sqrt{K_2 K_2^{(x)}}}{\tau^2}, \quad A_x = -n\sqrt{w_1} u_1(x) = -n \langle e_x, E_{\Lambda_1} e_a \rangle, \quad \vartheta_x = \Lambda_1^2 \sqrt{(K_2 - \frac{W}{\Lambda_1^2}) K_2^{(x)}}.$$

(In a symmetric situation such as corner to corner, $K_2^{(x_0)} = K_2$, so $\kappa_{x_0} = \kappa$, and $\vartheta_{x_0} \leq \sqrt{\vartheta(\vartheta + W)}$; for CC one may take $\vartheta_{x_0} = \vartheta$, see the proof.)

Theorem 6.7 (the residue r_0). *Assume (1.1) and $\bar{X} > 1$, $\beta = \bar{X}/(\bar{X} - 1)$, $t = \sigma_0 \tau$. Then*

$$r_0(x) = \frac{1 + t\mu_x + \varepsilon_1}{1 + \varepsilon_2}, \quad |\varepsilon_1| \leq \beta \kappa_x t^2, \quad \kappa t^2 \leq \varepsilon_2 \leq \beta^2 \kappa t^2. \quad (6.9)$$

In particular, with $t_- \leq t \leq t_+$ the bounds of (6.6), if $\mu_x \geq 0$ and $\beta \kappa_x t_+^2 \leq 1$,

$$\frac{1 + t_- \mu_x - \beta \kappa_x t_+^2}{1 + \beta^2 \kappa t_+^2} \leq r_0(x) \leq \frac{1 + t_+ \mu_x + \beta \kappa_x t_+^2}{1 + \kappa t_-^2}.$$

Moreover the stationary-start weight (5.2) of the principal pole is $\varrho_0 = 1/(1 + \varepsilon_2)$, so that

$$1 - \beta^2 \kappa \leq \varrho_0 \leq 1, \quad \sum_{j \geq 1} \varrho_j \leq \beta^2 \kappa. \quad (6.10)$$

Proof. By (3.1), $r_0 = -G_x(\sigma_0)/(\sigma_0 G'(\sigma_0))$ with $G_x(s) = -\frac{1}{ns} + g_x(s)$, $G'(s) = \frac{1}{ns^2} + g'(s)$; multiplying numerator and denominator by $n\sigma_0$,

$$r_0 = \frac{1 - n\sigma_0 g_x(\sigma_0)}{1 + n\sigma_0^2 g'(\sigma_0)}.$$

Now $n g_x(\sigma_0) = n g_x(0) + \sigma_0 n \sum_{i \geq 1} \frac{\sqrt{w_i} u_i(x)}{\Lambda_i(\Lambda_i - \sigma_0)}$ and $-n\sigma_0 g_x(0) = \sigma_0(m(x) - \tau) = t\mu_x$. By Cauchy-Schwarz and $\sqrt{w_i} |u_i(x)| = |\langle E_i e_x, E_i e_a \rangle| \leq \sqrt{w_i} \|E_i e_x\|$,

$$n \sum_{i \geq 1} \frac{\sqrt{w_i} |u_i(x)|}{\Lambda_i(\Lambda_i - \sigma_0)} \leq \frac{1}{1 - \sigma_0/\Lambda_1} n \sum_{i \geq 1} \frac{\sqrt{w_i} \|E_i e_x\|}{\Lambda_i^2} \leq \beta \sqrt{K_2 K_2^{(x)}},$$

which gives $|\varepsilon_1| \leq \sigma_0^2 \beta \sqrt{K_2 K_2^{(x)}} = \beta \kappa_x t^2$. Finally $\varepsilon_2 = n\sigma_0^2 g'(\sigma_0) \in [\sigma_0^2 K_2, \beta^2 \sigma_0^2 K_2]$ by (6.4). The two-sided bound follows because the numerator lies in $[1 + t_- \mu_x - \beta \kappa_x t_+^2, 1 + t_+ \mu_x + \beta \kappa_x t_+^2]$, the lower end being ≥ 0 by assumption, and the denominator in $[1 + \kappa t_-^2, 1 + \beta^2 \kappa t_+^2]$. For (6.10): $\varrho_0 = 1/(n\sigma_0^2 G'(\sigma_0)) = 1/(1 + n\sigma_0^2 g'(\sigma_0)) = 1/(1 + \varepsilon_2) \geq 1 - \beta^2 \kappa t^2 \geq 1 - \beta^2 \kappa$, and $\sum_{j \geq 1} \varrho_j = 1 - \frac{1}{n} - \varrho_0$ by Proposition 5.4(a). $\square$

Theorem 6.8 (the residue r_1). *Assume (1.1), $p \geq 2$, and let $\delta = \sigma_1/\Lambda_1 - 1$, $\varsigma = 1/(1 + \delta)$. Then*

$$r_1(x) = -\frac{\delta [A_x + \delta (Y_x(\sigma_1) - \varsigma)]}{(1 + \delta)W + \delta^2 [\varsigma + (1 + \delta) n \Lambda_1^2 h'(\sigma_1)]}, \quad Y_x(s) := n \Lambda_1 \sum_{i \geq 2} \frac{\sqrt{w_i} u_i(x)}{\Lambda_i - s}. \quad (6.11)$$

Moreover $Y_x(0) = A_x - \Lambda_1(m(x) - \tau)$, and under the hypotheses of Theorem 6.5(ii), $|Y_x(\sigma_1) - Y_x(0)| \leq \omega\vartheta_x$. Consequently, with $y_- = Y_x(0) - \omega\vartheta_x - 1$, $y_+ = Y_x(0) + \omega\vartheta_x - \frac{1}{1+\delta_+}$,

$$\mathcal{N}_- = A_x + \min(\delta_- y_-, \delta_+ y_-), \quad \mathcal{N}_+ = A_x + \max(\delta_- y_+, \delta_+ y_+),$$

$$\mathcal{D}_- = (1 + \delta_-)W + \delta_-^2 \left[\frac{1}{1 + \delta_+} + (1 + \delta_-)\vartheta \right], \quad \mathcal{D}_+ = (1 + \delta_+)W + \delta_+^2 \left[1 + \frac{\omega^2 \vartheta}{1 + \delta_+} \right],$$

one has, provided $\mathcal{N}_- > 0$,

$$\frac{\delta_- \mathcal{N}_-}{\mathcal{D}_+} \leq -r_1(x) \leq \frac{\delta_+ \mathcal{N}_+}{\mathcal{D}_-}. \quad (6.12)$$

Proof. Since $\sqrt{w_1}u_1(x) = -A_x/n$,

$$n\Lambda_1 G_x(\sigma_1) = -\frac{\Lambda_1}{\sigma_1} + \frac{-A_x}{\Lambda_1 - \sigma_1} \Lambda_1 + Y_x(\sigma_1) = -\varsigma + \frac{A_x}{\delta} + Y_x(\sigma_1),$$

$$n\Lambda_1 \sigma_1 G'(\sigma_1) = \frac{\Lambda_1}{\sigma_1} + \frac{\sigma_1 \Lambda_1 n w_1}{(\sigma_1 - \Lambda_1)^2} + \sigma_1 n \Lambda_1 h'(\sigma_1) = \varsigma + \frac{(1 + \delta)W}{\delta^2} + (1 + \delta) n \Lambda_1^2 h'(\sigma_1).$$

Dividing and multiplying through by δ^2 gives (6.11). From $ng_x(0) = -(m(x) - \tau)$ and $g_x(0) = \sqrt{w_1}u_1(x)/\Lambda_1 + \sum_{i \geq 2} \sqrt{w_i}u_i(x)/\Lambda_i$ we get $Y_x(0) = A_x - \Lambda_1(m(x) - \tau)$. As in the proof of Theorem 6.7, using (6.4) with $\Lambda = \Lambda_2$,

$$|Y_x(\sigma_1) - Y_x(0)| \leq n\Lambda_1 \sigma_1 \sum_{i \geq 2} \frac{\sqrt{w_i}|u_i(x)|}{\Lambda_i(\Lambda_i - \sigma_1)} \leq \frac{(1 + \delta)}{1 - (1 + \delta)\varrho} \Lambda_1^2 n \sum_{i \geq 2} \frac{\sqrt{w_i} \|E_i e_x\|}{\Lambda_i^2} \leq \omega\vartheta_x.$$

(The last step is Cauchy–Schwarz. In a symmetric situation $x_0 = Ra$, $\|E_i e_{x_0}\| = \|E_i e_a\| = \sqrt{w_i}$, the middle expression equals $\frac{1+\delta}{1-(1+\delta)\varrho} \Lambda_1^2 n \sum_{i \geq 2} w_i \Lambda_i^{-2} \leq \omega\vartheta$, so one may take $\vartheta_{x_0} = \vartheta$.) The bracket in the numerator of (6.11) therefore lies in $[\mathcal{N}_-, \mathcal{N}_+]$ because $\delta \in [\delta_-, \delta_+]$, $\varsigma \in [\frac{1}{1+\delta_+}, 1]$ and $Y_x(\sigma_1) - \varsigma \in [y_-, y_+]$; the denominator lies in $[\mathcal{D}_-, \mathcal{D}_+]$ by Theorem 6.5(ii). This gives (6.12). $\square$

Remark 6.9 (using bounds on the functionals instead of their exact values). In applications K_2 (hence κ and ϑ) is only known between two bounds. The proofs above use the functionals monotonically, so the statements remain true in the following form. Let $\kappa_- \leq \kappa \leq \kappa_+$ and $\vartheta_- \leq \vartheta \leq \vartheta_+$ (and $\vartheta_x \leq \vartheta_+$). (a) In (6.6): $2/(1 + \sqrt{1 + 4\beta\kappa_+}) \leq \sigma_0\tau \leq 2/(1 + \sqrt{1 + 4\kappa_-})$; in (6.9) and (6.10) κ may be replaced by κ_+ in the upper bounds for $|\varepsilon_1|$, ε_2 , $\sum_{j \geq 1} \varrho_j$. (b) In Theorem 6.5: if $Y_0 + \vartheta_- > 1$ then $\delta \leq \hat{\delta}_+ := W/(Y_0 + \vartheta_- - 1)$ [because $Y(\sigma_1) \geq Y_0 + \vartheta \geq Y_0 + \vartheta_-$]; if moreover $(1 + \hat{\delta}_+)\varrho < 1$ and $\hat{\omega} := (1 + \hat{\delta}_+)/(1 - (1 + \hat{\delta}_+)\varrho)$, then $\delta \geq \hat{\delta}_- := W/(Y_0 + \hat{\omega}\vartheta_+ - \frac{1}{1+\hat{\delta}_+})$, $Y(\sigma_1) \leq Y_0 + \hat{\omega}\vartheta_+$ and $\vartheta_- \leq n\Lambda_1^2 h'(\sigma_1) \leq \hat{\omega}^2 \vartheta_+/(1 + \hat{\delta}_+)^2$ [because $y \mapsto (1 + y)/(1 - (1 + y)\varrho)$ is increasing and $\delta \leq \hat{\delta}_+$]. (c) In Theorem 6.8: (6.12) holds with $\delta_{\pm}, \omega$ replaced by $\hat{\delta}_{\pm}, \hat{\omega}$, with $\omega\vartheta_x$ replaced by $\hat{\omega}\vartheta_+$ in $y_{\pm}$, with ϑ replaced by ϑ_- in $\mathcal{D}_-$ and by ϑ_+ in $\mathcal{D}_+$, and with any lower (upper) bound for $Y_x(0)$ in y_- (y_+) [the denominator of (6.11) is increasing in δ and in $h'(\sigma_1)$, and $\delta^2\varsigma = \delta^2/(1 + \delta)$ is increasing in δ].

Remark 6.10 (first-order content). When $\bar{X} \rightarrow \infty$ with W , ϑ , A_x , $\Lambda_1(m(x) - \tau)$ bounded and ϱ bounded away from 1 (this is the situation of the box, Section 11), the theorems of this section give, with explicit error terms,

$$\sigma_0\tau = 1 - \kappa + O(\bar{X}^{-3}), \quad \frac{\sigma_1}{\Lambda_1} = 1 + \frac{W}{\bar{X}}(1 + O(\bar{X}^{-1})), \quad r_0 = 1 + \mu_x + O(\bar{X}^{-2}), \quad r_1 = -\frac{A_x}{\bar{X}}(1 + O(\bar{X}^{-1})),$$

which are the relations “ $\nu_0 = \text{MFPT}^{-1}(1 + O(1/X))$, $\nu_1 = \mu(1 + W/X + \dots)$, $a_0 \approx \nu_0$, $a_1 \approx -A\nu_0$ ” obtained by formal perturbation theory in Section 6 of the article ($a_j = r_j\sigma_j$), here with proofs and constants.

Part III. The reflecting box: corner to corner

7 Corner to corner in the reflecting box: general dimension

From now on L is the generator (1.3) of the reflecting box, $d \geq 1$, $N \geq 2$, $n = N^d$, the target is the corner $a = (N, \dots, N)$ and the start is the opposite corner $x_0 = (1, \dots, 1)$ (“CC”). We write

$$u = \frac{\pi}{2N}, \quad c_k = \cos \theta_k = \cos \frac{\pi k}{N}, \quad w_k = n \phi_k(a)^2 = \prod_{i=1}^d \varepsilon_{k_i} \cos^2(k_i u) \quad (k \in \{0, \dots, N-1\}^d),$$

so that, by Lemma 1.1, (1.5), (6.2) and Corollary 4.2,

$$\begin{aligned} \tau &= \sum_{k \neq 0} \frac{w_k}{\lambda_k}, \quad m := m(x_0) = \sum_{k \neq 0} \frac{(1 - (-1)^{|k|}) w_k}{\lambda_k}, \quad K_2 = \sum_{k \neq 0} \frac{w_k}{\lambda_k^2}, \\ \Lambda_1 &= \frac{2}{d} \sin^2 u, \quad W = 2d \cos^2 u. \end{aligned} \tag{7.1}$$

By Lemma 6.1, τ/q is the mean hitting time of the corner from a uniformly distributed start and m/q is the corner-to-corner MFPT of the article. The point reflection $R : x_i \mapsto N + 1 - x_i$ exchanges a and x_0 and commutes with L ; consequently $\|E_i e_{x_0}\| = \|E_i e_a\|$ for all i , and in the notation of Section 6

$$K_2^{(x_0)} = K_2, \quad \kappa_{x_0} = \kappa, \quad A_{x_0} = W, \quad \vartheta_{x_0} \text{ may be replaced by } \vartheta, \quad \varrho = \frac{1}{2} \quad (d \geq 2), \tag{7.2}$$

where $A_{x_0} = -n \sum_i \phi_{e_i}(a) \phi_{e_i}(x_0) = d \cdot 2 \cos^2 u$ by (1.5).

Lemma 7.1 (elementary inequalities). (a) $\sin v \geq \frac{2}{\pi} v$ for $0 \leq v \leq \frac{\pi}{2}$, and $v^2(1 - \frac{v^2}{3}) \leq \sin^2 v \leq v^2$ for $0 \leq v \leq 1$.

(b) $(\frac{\sin v}{v})^2 \geq \cos v$ for $0 < v \leq \frac{\pi}{2}$.

(c) arsinh is concave on $[0, \infty)$; hence $\operatorname{arsinh} \rho \geq \rho \operatorname{arsinh}(\rho_0)/\rho_0$ for $0 \leq \rho \leq \rho_0$.

(d) $0 \leq x - \operatorname{arsinh}(\sin x) \leq \frac{x^3}{3}$ and $\operatorname{arsinh}(\sin x) \geq \frac{2 \operatorname{arsinh}(1)}{\pi} x$ for $0 \leq x \leq \frac{\pi}{2}$.

(e) $(1 - y)^4(1 + 2y) \leq 1$ for $0 \leq y \leq 1$.

(f) $1 - \frac{y}{2} - \frac{y^2}{2} \leq \sqrt{1 - y} \leq 1 - \frac{y}{2}$ for $0 \leq y \leq 1$.

Proof. (a) Jordan’s inequality (concavity of $\sin$ on $[0, \frac{\pi}{2}]$). $\sin v \leq v$, and $\sin v \geq v - \frac{v^3}{6} \geq 0$ gives $\sin^2 v \geq v^2(1 - \frac{v^2}{6})^2 \geq v^2(1 - \frac{v^2}{3})$. (b) Let $h(v) = \sin^2 v - v^2 \cos v$. Then $h(0) = 0$ and $h'(v) = 2 \cos v(\sin v - v) + v^2 \sin v \geq -\frac{v^3}{3} \cos v + v^2 \sin v = v^2(\sin v - \frac{v}{3} \cos v) \geq 0$, because $\sin v - v \geq -\frac{v^3}{6}$ and $\tan v \geq v$. (c) $\operatorname{arsinh}''(\rho) = -\rho(1 + \rho^2)^{-3/2} \leq 0$. (d) Let $A(x) = \operatorname{arsinh}(\sin x)$; $A'(x) = \cos x / \sqrt{1 + \sin^2 x} \in [0, 1]$ is decreasing on $[0, \frac{\pi}{2}]$, so A is concave with $A(0) = 0$, $A \leq x$, and $A(x) \geq xA(\frac{\pi}{2})/\frac{\pi}{2}$. With $s = \sin x$: $A'(x) = \sqrt{(1 - s^2)/(1 + s^2)} \geq 1 - s^2$ (square both sides: $1 \geq (1 - s^2)(1 + s^2)$), so $\frac{d}{dx}(\frac{x^3}{3} - x + A(x)) \geq x^2 - \sin^2 x \geq 0$. (e) $(1 - y)^2(1 + 2y) = 1 - 3y^2 + 2y^3 \leq 1$ and $(1 - y)^2 \leq 1$. (f) The upper bound is concavity. For the lower bound, $F(y) = \sqrt{1 - y} - 1 + \frac{y}{2} + \frac{y^2}{2}$ has $F(0) = F(1) = 0$ and $F'(y) = \frac{1}{2} + y - \frac{1}{2\sqrt{1 - y}}$, which vanishes on $(0, 1)$ only where $(1 + 2y)^2(1 - y) = 1$, i.e. $y(3 - 4y^2) = 0$, i.e. at $y = \sqrt{3}/2$; since $F''(0) = \frac{3}{4} > 0$, F increases on $[0, \frac{\sqrt{3}}{2}]$ and decreases afterwards, hence $F \geq 0$. $\square$

Lemma 7.2 (arrival profile; $m - \tau$). Let $u_1(s) = 1 + 2 \sum_{k=1}^{N-1} (-1)^k \cos^2(ku) e^{-(1-c_k)s}$. For the unit-rate walk, $n p_t(x_0, a) = u_1(t/d)^d$,

$$0 \leq 1 - n p_t(x_0, a) \leq \min \{1, W e^{-\Lambda_1 t}\} \quad (t \geq 0), \quad (7.3)$$

and therefore

$$\frac{d(N^2 - 1)}{3} \leq m - \tau = \int_0^\infty (1 - n p_t(x_0, a)) dt \leq \frac{1 + \ln W}{\Lambda_1}, \quad (7.4)$$

with equality on the left if and only if $d = 1$. Equivalently, with $\mu := \mu_{x_0} = (m - \tau)/\tau$,

$$\frac{\mu_-(N)}{\bar{X}} \leq \mu \leq \frac{1 + \ln W}{\bar{X}}, \quad \mu_-(N) := \frac{2}{3} (N^2 - 1) \sin^2 \frac{\pi}{2N} \geq \max \left\{ 1, \frac{\pi^2}{6} \left(1 - \frac{1}{N^2} \right) \left(1 - \frac{\pi^2}{12N^2} \right) \right\}, \quad (7.5)$$

and $\mu_-(N) \rightarrow \pi^2/6$. In particular $m - \tau \geq 1/\Lambda_1$: the corner-to-corner mean first-passage time exceeds the stationary-start mean hitting time by at least one relaxation time.

Proof. For $d = 1$ the spectral theorem and Lemma 1.1 give

$$N p_s^{(1)}(1, N) = \sum_k N \phi_k(1) \phi_k(N) e^{-(1-c_k)s} = u_1(s),$$

using $N \phi_k(1) \phi_k(N) = \varepsilon_k (-1)^k \cos^2(ku)$. Hence $u_1 \geq 0$. Moreover $1 - u_1(s) = 2 \sum_{k=1}^{N-1} (-1)^{k+1} b_k$ with $b_k = \cos^2(ku) e^{-(1-c_k)s}$, and $b_1 > b_2 > \dots > b_{N-1} > 0$ since both factors decrease in k . An alternating sum of decreasing positive terms lies between 0 and its first term: $0 \leq 1 - u_1(s) \leq 2 \cos^2(u) e^{-(1-c_1)s}$. In dimension d , $L = \frac{1}{d} \sum_i L^{(i)}$ with commuting one-dimensional operators, so $e^{-tL} = \bigotimes_i e^{-(t/d)L^{(i)}}$ and $n p_t(x_0, a) = u_1(t/d)^d$. With $v = u_1(t/d) \in [0, 1]$, $0 \leq 1 - v^d \leq \min\{1, d(1 - v)\} \leq \min\{1, 2d \cos^2(u) e^{-(1-c_1)t/d}\}$, which is (7.3). Next, $L^+ = \int_0^\infty (e^{-tL} - E_0) dt$, so by (6.2) $m - \tau = -n g_{x_0}(0) = -n \langle e_{x_0}, L^+ e_a \rangle = \int_0^\infty (1 - n p_t(x_0, a)) dt$. The integrand is non-negative, continuous and equal to 1 at $t = 0$, so $m > \tau$. Since $W = 2d \cos^2 u \geq 2d \cos^2 \frac{\pi}{4} = d \geq 1$, with $t_* = \Lambda_1^{-1} \ln W \geq 0$ we get $m - \tau \leq t_* + \int_{t_*}^\infty W e^{-\Lambda_1 t} dt = (1 + \ln W)/\Lambda_1$.

For the lower bound, $v = u_1(t/d) \in [0, 1]$ gives $1 - v^d \geq 1 - v$, with strict inequality when $d \geq 2$ and $0 < v < 1$ (which holds for all $t > 0$: $u_1(s) < 1$ by the first part since $1 - u_1(s) \geq 2(b_1 - b_2) > 0$ for $N \geq 3$ and $= 2b_1 > 0$ for $N = 2$, and $u_1(s) > 0$ for $s > 0$ because the chain is irreducible). Hence

$$\begin{aligned} m - \tau &\geq \int_0^\infty (1 - u_1(t/d)) dt = d \int_0^\infty (1 - u_1(s)) ds = d \sum_{k=1}^{N-1} (-1)^{k+1} \frac{2 \cos^2(ku)}{1 - c_k} \\ &= d \sum_{k=1}^{N-1} (-1)^{k+1} \cot^2(ku) = \frac{d(N^2 - 1)}{3}, \end{aligned}$$

the last step by Lemma 8.2 below (whose proof does not use the present lemma); the integration of the finite sum is termwise. Multiplying by $\Lambda_1 = \frac{2}{d} \sin^2 u$ gives $\Lambda_1(m - \tau) \geq \mu_-(N)$. Finally $\mu_-(2) = \frac{2}{3} \cdot 3 \cdot \frac{1}{2} = 1$, and for $N \geq 3$, by Lemma 7.1(a), $\mu_-(N) \geq \frac{2}{3} (N^2 - 1) u^2 (1 - \frac{u^2}{3}) = \frac{\pi^2}{6} (1 - N^{-2}) (1 - \frac{\pi^2}{12N^2})$, which is increasing in N and equals $1.3285\dots > 1$ at $N = 3$. $\square$

Remark 7.3. For $d = 1$ the lower bound in (7.4) is the exact value: $m_1 = N(N - 1)$, $\tau_1 = \frac{1}{3} (N - 1)(2N - 1)$, $m_1 - \tau_1 = \frac{1}{3} (N^2 - 1)$. For $d = 2$ and $d = 3$ the lower bound is the leading term of the exact reductions (8.6), (8.9); the true limits of $\Lambda_1(m - \tau)$ are $\pi \ln 2 = 2.1776$ and 2.5194 (Theorem 11.3), between $\pi^2/6 = 1.6449$ and $1 + \ln 2d$.

Lemma 7.4 (second moment functional). For $d \geq 1$, $N \geq 2$,

$$W + \binom{d}{2} \cos^4 u \leq \Lambda_1^2 K_2 \leq Z_d(N) := \sum_{\substack{k \in \mathbb{Z}^d \setminus \{0\} \\ |k|_\infty \leq N-1}} \frac{1}{|k|^4}, \quad (7.6)$$

where $|k|$ is the Euclidean norm. $Z_d(N)$ increases to $Z_d := \sum_{k \in \mathbb{Z}^d \setminus 0} |k|^{-4} < \infty$ for $d \leq 3$; $Z_1 = 2\zeta(4) = \pi^4/45$, and (certified enclosures, Lemma 7.6 and Section 13)

$$6.02681 < Z_2 < 6.02682, \quad 16.53231 < Z_3 < 16.53232. \quad (7.7)$$

For every d , $Z_d(N) \leq 2d3^{d-1} \sum_{j=1}^{N-1} j^{d-5}$.

Proof. By (7.1), $\Lambda_1^2 K_2 = \sum_{k \neq 0} \prod_i \varepsilon_{k_i} \cdot \prod_i \cos^2(k_i u) (\Lambda_1/\lambda_k)^2$ and $\lambda_k/\Lambda_1 = \sum_i \sin^2(k_i u)/\sin^2 u$. For an integer $1 \leq j \leq N-1$ we have $ju \in (0, \frac{\pi}{2})$ and, by Lemma 7.1(a),(b), $\sin^2(ju)/\sin^2 u \geq \sin^2(ju)/u^2 \geq j^2 \cos(ju)$. Hence

$$\lambda_k/\Lambda_1 \geq \sum_i k_i^2 \cos(k_i u) \geq |k|^2 \prod_i \cos(k_i u), \quad \text{i.e.} \quad \prod_i \cos^2(k_i u) (\Lambda_1/\lambda_k)^2 \leq |k|^{-4}.$$

Since $\prod_i \varepsilon_{k_i}$ is the number of points of $\mathbb{Z}^d$ obtained from k by sign changes, the upper bound follows. For the lower bound keep the terms $k = e_i$ ($\lambda_k = \Lambda_1$, total weight W) and $k = e_i + e_j$, $i < j$ ($\lambda_k = 2\Lambda_1$, weight $4\cos^4 u$ each). The number of $k \in \mathbb{Z}^d$ with $|k|_\infty = j$ is $(2j+1)^d - (2j-1)^d \leq 2d(2j+1)^{d-1} \leq 2d3^{d-1}j^{d-1}$, and $|k| \geq j$ for them; this gives the last bound and the convergence for $d \leq 3$. An elementary enclosure, $6.02679 < Z_2 < 6.02682$ and $16.4629 < Z_3 < 16.5484$ (script 06), is obtained from the partial sums over $|k|_\infty \leq R$ and the tail bound $\sum_{|k|_\infty > R} |k|^{-4} \leq (1 + \frac{\sqrt{d}}{2(R+1)})^4 \int_{|x| \geq R+\frac{1}{2}} |x|^{-4} dx$ (compare each term with the integral over the unit cube centred at k : for x in that cube $|x| \leq |k| + \frac{\sqrt{d}}{2} \leq |k|(1 + \frac{\sqrt{d}}{2(R+1)})$, and the cubes are disjoint and lie outside the ball of radius $R + \frac{1}{2}$); the integral equals $\pi/(R + \frac{1}{2})^2$ for $d = 2$ and $4\pi/(R + \frac{1}{2})$ for $d = 3$. The sharp enclosures (7.7) are proved in Lemma 7.6. $\square$

Remark 7.5. $Z_2 = 4\zeta(2)G = 6.0268120\dots$ (G Catalan's constant) by the Lorenz–Hardy formula $\sum_{(m,n) \neq 0} (m^2 + n^2)^{-s} = 4\zeta(s)\beta(s)$ (see e.g. I. J. Zucker and M. M. Robertson, J. Phys. A 8 (1975) 874, or J. M. Borwein et al., *Lattice Sums Then and Now*, Cambridge Univ. Press 2013, Section 1.3); the closed form is not used, only (7.7) (the certified value of Lemma 7.6 agrees with it to 39 digits). Numerically $\Lambda_1^2 K_2 \rightarrow Z_d$ for $d = 2, 3$ (the ratio is 0.998 at $d = 2$, $N = 60$), so (7.6) is asymptotically sharp.

Lemma 7.6 (Epstein zeta values by theta splitting). *For $d \geq 1$ and real $s > d/2$ let $Z_d(s) = \sum_{k \in \mathbb{Z}^d \setminus 0} |k|^{-2s}$ (so $Z_d = Z_d(2)$), and for $c > 0$, $\alpha \in \mathbb{R}$ let $E_\alpha(c) = \int_1^\infty t^{\alpha-1} e^{-ct} dt = c^{-\alpha} \Gamma(\alpha, c)$ (incomplete gamma function). Then*

$$\pi^{-s} \Gamma(s) Z_d(s) = \frac{1}{s - \frac{d}{2}} - \frac{1}{s} + \sum_{k \in \mathbb{Z}^d \setminus 0} \left[E_s(\pi|k|^2) + E_{\frac{d}{2}-s}(\pi|k|^2) \right], \quad (7.8)$$

where all terms of the series are positive and, for $s \geq 1$ and every integer $R \geq 1$ with $\pi R^2 > s-1$,

$$\sum_{|k|_\infty > R} \left[E_s(\pi|k|^2) + E_{\frac{d}{2}-s}(\pi|k|^2) \right] \leq \frac{4d(1.1)^{d-1} \cdot 1.001 e^{-\pi(R+1)^2}}{\pi R^2 - s + 1}. \quad (7.9)$$

Moreover, for $Z'_d = \sum_{|k|^2 \geq 2} (|k|^2(|k|^2 - 1))^{-1}$ ($d \leq 3$) and every integer $J \geq 2$,

$$Z'_d = \sum_{j=2}^J (Z_d(j) - 2d) + \sum_{|k|^2 \geq 2} \frac{|k|^{-2J}}{|k|^2 - 1}, \quad \sum_{|k|_\infty > R} \frac{|k|^{-2J}}{|k|^2 - 1} \leq \frac{4d3^{d-1}}{2J + 2 - d} R^{d-2-2J}. \quad (7.10)$$

Evaluating (7.8)–(7.10) in ball arithmetic ($R = 7$; $J = 8$, $R = 40$; script `14_certify_epstein.py`) gives

$$\begin{aligned} Z_2 &= 6.02681203969194012\dots, & Z_3 &= 16.5323159597616696\dots, \\ Z'_2 &= 3.22591076832683889\dots, & Z'_3 &= 14.7022972380072310\dots \end{aligned} \quad (7.11)$$

Here the decimal expansions are truncated, not rounded, and every digit shown is certified: each value lies strictly between the printed number and the printed number plus one unit in its last place (the script asserts exactly these inequalities).

Proof. For $c > 0$, $\Gamma(s)c^{-s} = \int_0^\infty t^{s-1}e^{-ct}dt$. With $c = \pi|k|^2$, summing over $k \neq 0$ (Tonelli), $\pi^{-s}\Gamma(s)Z_d(s) = \int_0^\infty t^{s-1}(\Theta(t) - 1)dt$, where $\Theta(t) = \theta(t)^d$ and $\theta(t) = \sum_{j \in \mathbb{Z}} e^{-\pi j^2 t}$. Jacobi's transformation $\theta(1/t) = \sqrt{t}\theta(t)$ ($t > 0$; DLMF 20.7.32 with $z = 0$, $\tau = it$, or Poisson summation) gives $\Theta(1/t) = t^{d/2}\Theta(t)$. Substituting $t \mapsto 1/t$ in the part $\int_0^1$,

$$\int_0^1 t^{s-1}(\Theta(t)-1)dt = \int_1^\infty t^{-s-1}(t^{d/2}\Theta(t)-1)dt = \int_1^\infty t^{\frac{d}{2}-s-1}(\Theta(t)-1)dt + \int_1^\infty (t^{\frac{d}{2}-s-1}-t^{-s-1})dt,$$

and the last integral equals $\frac{1}{s-d/2} - \frac{1}{s}$ because $s > \frac{d}{2}$. Since $\Theta(t) - 1 = \sum_{k \neq 0} e^{-\pi|k|^2 t}$, Tonelli's theorem gives (7.8). For the tail: $\frac{d}{2} - s - 1 \leq s - 1$, so each bracket is at most $2E_s(c)$, and for $c > s - 1 \geq 0$, $E_s(c) = e^{-c} \int_0^\infty (1+x)^{s-1} e^{-cx} dx \leq e^{-c} \int_0^\infty e^{-(c-s+1)x} dx = e^{-c}/(c-s+1)$. For $|k|_\infty > R$ we have $c = \pi|k|^2 > \pi R^2$, hence the left side of (7.9) is at most $\frac{2}{\pi R^{2-s+1}} \sum_{|k|_\infty > R} e^{-\pi|k|^2}$, and $\sum_{|k|_\infty > R} e^{-\pi|k|^2} \leq d\theta(1)^{d-1} \cdot 2 \sum_{j \geq R+1} e^{-\pi j^2} \leq d(1.1)^{d-1} \cdot \frac{2e^{-\pi(R+1)^2}}{1-e^{-2\pi(R+1)}}$, using $\theta(1) = 1 + 2 \sum_{j \geq 1} e^{-\pi j^2} < 1.0865$, $j^2 - (R+1)^2 \geq 2(R+1)(j-R-1)$ and $(1-e^{-4\pi})^{-1} < 1.001$. For (7.10): $\frac{1}{n(n-1)} = \sum_{j=2}^J n^{-j} + \frac{n^{-J}}{n-1}$ for $n \geq 2$, and exactly $2d$ points of $\mathbb{Z}^d$ have $|k|^2 = 1$. For $|k|_\infty = j > R$: $|k|^2 - 1 \geq \frac{1}{2}|k|^2$ and $|k| \geq j$, and there are at most $2d3^{d-1}j^{d-1}$ such points (proof of Lemma 7.4); so the tail is at most $4d3^{d-1} \sum_{j > R} j^{d-3-2J} \leq 4d3^{d-1} \int_R^\infty x^{d-3-2J} dx$. The script evaluates the finitely many remaining terms with 300-bit balls ($\Gamma(\alpha, c)$ by Arb), adds the tail bounds, and checks the result against (i) the closed form $Z_2 = 4\zeta(2)G$, (ii) direct summation of $Z_d(4)$, $Z_d(6)$, (iii) a second choice of R and J . $\square$

Lemma 7.7 (size of τ in general dimension). *For every $d \geq 1$, $N \geq 2$,*

$$\tau \geq 2n\left(1 - \frac{1}{n}\right)^2, \quad \text{hence} \quad \bar{X} = \Lambda_1 \tau \geq \frac{4}{d} N^{d-2} (1 - N^{-d})^2. \quad (7.12)$$

For $d \geq 3$ also $\tau \leq \frac{d^2 3^{d-1}}{d-2} N^d$, so that $\bar{X}$ is of exact order N^{d-2} .

Proof. Let $v = e_a - \frac{1}{n}\mathbf{1} \perp \mathbf{1}$. Then $\langle v, L^+ v \rangle = \langle e_a, L^+ e_a \rangle = \tau/n$, $\langle v, v \rangle = 1 - \frac{1}{n}$ and $\langle v, Lv \rangle = L_{aa} = \frac{d}{2d} = \frac{1}{2}$ (a corner has d neighbours). On $\mathbf{1}^\perp$, L is positive definite with inverse L^+ , so by Cauchy-Schwarz $\langle v, v \rangle^2 = \langle (L^+)^{1/2} v, L^{1/2} v \rangle^2 \leq \langle v, L^+ v \rangle \langle v, Lv \rangle$, which is the first claim; the second uses $\Lambda_1 = \frac{2}{d} \sin^2 u \geq \frac{2}{dN^2}$ (Lemma 7.1(a)). For the upper bound, $\lambda_k = \frac{2}{d} \sum_i \sin^2(k_i u) \geq \frac{2|k|^2}{dN^2}$ and $w_k \leq \prod_i \varepsilon_{k_i}$ give $\tau \leq \frac{dN^2}{2} \sum_{0 < |k|_\infty < N} |k|^{-2} \leq \frac{dN^2}{2} 2d3^{d-1} \sum_{j=1}^{N-1} j^{d-3} \leq \frac{d^2 3^{d-1}}{d-2} N^d$. $\square$

Lemma 7.8 (monotone coupling). *Let $d \geq 1$, $N \geq 2$, target $a = (N, \dots, N)$, and order Ω coordinatewise: $x \preceq y$ iff $x_i \leq y_i$ for all i . If $x \preceq y$ then, in continuous and in discrete time and for every activity $q \in (0, 1]$,*

$$\mathbb{P}_x(T > t) \geq \mathbb{P}_y(T > t) \quad \text{for all } t \geq 0.$$

Consequently $m(x) \geq m(y)$ and $r_0(x) \geq r_0(y)$ (with $m(a) = r_0(a) = 0$): the mean first-passage time and the principal residue are coordinatewise non-increasing, and both are maximal at the opposite corner $x_0 = (1, \dots, 1)$.

Proof. Run two copies $(X_s), (Y_s)$ of the reflecting walk *without* target from x and y , driven by the same jump times and the same proposed directions $\pm e_i$. A proposed move $+e_i$ maps the i -th coordinates to $\min(X^i + 1, N)$, $\min(Y^i + 1, N)$, a proposed move $-e_i$ to $\max(X^i - 1, 1)$, $\max(Y^i - 1, 1)$ (cancelled moves are exactly the cases where min/max is active), and the other coordinates are unchanged. Both maps are non-decreasing, so $X_s \preceq Y_s$ for all s . At the hitting time T^X of a by X we have $a = X_{T^X} \preceq Y_{T^X}$, hence $Y_{T^X} = a$ because a is the maximum of Ω ; thus $T^Y \leq T^X$ pathwise, which proves the inequality. Integrating over t gives $m(x) \geq m(y)$ (Lemma 6.1). By (3.2) and $\sigma_0 < \sigma_j$ ($j \geq 1$), $r_0(x) = \lim_{t \rightarrow \infty} e^{\sigma_0 t} \mathbb{P}_x(T > t)$ for the unit-rate walk, so $r_0(x) \geq r_0(y)$. $\square$

Theorem 7.9 ($r_0 > 1$ and $\sigma_0 \cdot \text{MFPT} > 1$, all d and N). *For the reflecting box with corner target and start at the opposite corner, for every $d \geq 1$ and $N \geq 2$ with $(d, N) \neq (1, 2)$:*

$$r_0 = \max_{x \in \Omega'} r_0(x) > 1, \quad \sigma_0 m = \max_{x \in \Omega'} \sigma_0 m(x) > 1,$$

and quantitatively $\sigma_0 m \geq \frac{\bar{X} + \mu_-}{\bar{X} + 1} = 1 + \frac{\mu_- - 1}{\bar{X} + 1}$ with $\mu_- = \mu_-(N) \geq 1$ from (7.5). (For $(d, N) = (1, 2)$: $r_0 = \sigma_0 m = 1$.)

Proof. By Proposition 5.5 the ψ_0 -weighted averages over Ω' of $r_0(\cdot)$ and of $\sigma_0 m(\cdot)$ are equal to 1, with $\psi_0 > 0$ (Theorem 5.1(c); Ω' is connected). By Lemma 7.8 both functions attain their maximum at x_0 , so $r_0(x_0) \geq 1$ and $\sigma_0 m(x_0) \geq 1$, with equality only if the function is constant on Ω' . By Proposition 5.5 this happens iff $\mathbf{1}_{\Omega'}$ is an eigenvector of L_a , i.e. iff $-L_{ya}$ is the same for all $y \in \Omega'$, i.e. iff every site of Ω' is a neighbour of a . The corner a has d neighbours and $|\Omega'| = N^d - 1$, and $N^d - 1 = d$ only for $(d, N) = (1, 2)$. The quantitative bound follows from $\sigma_0 m = (1 + \mu) \sigma_0 \tau$, $\sigma_0 \tau \geq \bar{X}/(1 + \bar{X})$ (6.5) and $\mu \geq \mu_-/\bar{X}$ (7.5), all valid for every $d \geq 1$; it gives a second proof of $\sigma_0 m \geq 1$ that does not use the coupling. $\square$

Theorem 7.10 (corner to corner, explicit in $\bar{X}$). *Let $d \geq 2$, $N \geq 2$, and for the corner-to-corner geometry let $\bar{X} = \Lambda_1 \tau$, $\mu = (m - \tau)/\tau$,*

$$W = 2d \cos^2 \frac{\pi}{2N}, \quad \ell_W = 1 + \ln W, \quad \vartheta_0 = \binom{d}{2} \cos^4 \frac{\pi}{2N}, \quad \mu_- = \mu_-(N) = \frac{2}{3}(N^2 - 1) \sin^2 \frac{\pi}{2N} (\geq 1),$$

and let Z be any number with $\Lambda_1^2 K_2 \leq Z$ (for instance $Z = Z_d(N)$, or $Z = Z_d$ if $d \leq 3$). Assume $\bar{X} > 1$ and put $\beta = \bar{X}/(\bar{X} - 1)$. Then:

(i) (principal eigenvalue)

$$1 - \frac{\beta Z}{\bar{X}^2} \leq \frac{2}{1 + \sqrt{1 + 4\beta Z/\bar{X}^2}} \leq \sigma_0 \tau \leq \frac{2}{1 + \sqrt{1 + 4W/\bar{X}^2}} \leq 1 - \frac{W}{\bar{X}^2} + \frac{2W^2}{\bar{X}^4}.$$

(ii) (principal eigenvalue times MFPT) $\mu_-/\bar{X} \leq \mu \leq \ell_W/\bar{X}$ and $\sigma_0 m = (1 + \mu) \sigma_0 \tau$; in particular

$$\left(1 + \frac{\mu_-}{\bar{X}}\right) \left(1 - \frac{\beta Z}{\bar{X}^2}\right) \leq \sigma_0 m < 1 + \frac{\ell_W}{\bar{X}};$$

$$\text{without any assumption on } \bar{X}: \quad \frac{\bar{X} + \mu_-}{1 + \bar{X}} \leq \sigma_0 m < 1 + \frac{\ell_W}{\bar{X}}.$$

Since $\mu_- \geq 1$, the last lower bound is ≥ 1 : always $\tau \leq \frac{1}{\sigma_0} \leq \tau + \frac{1}{\Lambda_1} \leq m$ (see also Theorem 7.9).

(iii) (second pole) $\sigma_1 = \Lambda_1(1 + \delta)$ with $0 < \delta < 1$. If $\bar{X} > W + 1 - \vartheta_0$ then $\delta \leq \delta_+ := \frac{W}{\bar{X} - W - 1 + \vartheta_0}$. If moreover $\delta_+ < 1$, put $\omega = \frac{2(1 + \delta_+)}{1 - \delta_+}$; then $\delta \geq \delta_- := \frac{W}{\bar{X} - W + \omega(Z - W) - \frac{1}{1 + \delta_+}}$.

(iv) (gap) $\Lambda_1(1 - \frac{1}{\bar{X}}) < \sigma_1 - \sigma_0 < 2\Lambda_1$, and under the hypotheses of (iii)

$$\Lambda_1 \left(1 + \delta_- - \frac{1}{\bar{X}}\right) \leq \sigma_1 - \sigma_0 \leq \Lambda_1 \left(1 + \delta_+ - \frac{1}{1 + \bar{X}}\right).$$

(v) (residue of the principal pole) $r_0 > 0$ and

$$\begin{aligned} \left(1 + \frac{\mu_-}{\bar{X}}\right) \left(1 - \frac{\beta(1+\beta)Z}{\bar{X}^2}\right) &\leq (1+\mu) \left(1 - \frac{\beta(1+\beta)Z}{\bar{X}^2}\right) \leq r_0, \\ r_0 &\leq 1 + \mu + \frac{\beta Z}{\bar{X}^2} \leq 1 + \frac{\ell_W}{\bar{X}} + \frac{\beta Z}{\bar{X}^2} \end{aligned}$$

(the first inequality provided $\beta(1+\beta)Z \leq \bar{X}^2$). Moreover $r_0 > 1$ without any hypothesis on $\bar{X}$ (Theorem 7.9).

(vi) (residue of the second pole) $r_1 < 0$. Under the hypotheses of (iii) let

$$y_- = W - \ell_W - \omega(Z - W) - 1, \quad y_+ = W - \mu_- + \omega(Z - W) - \frac{1}{1+\delta_+},$$

$$\mathcal{N}_- = W + \min(\delta_- y_-, \delta_+ y_+), \quad \mathcal{N}_+ = W + \max(\delta_- y_+, \delta_+ y_+),$$

$$\mathcal{D}_- = (1 + \delta_-)W + \delta_-^2 \left[\frac{1}{1+\delta_+} + (1 + \delta_-)\vartheta_0 \right], \quad \mathcal{D}_+ = (1 + \delta_+)W + \delta_+^2 \left[1 + \frac{\omega^2(Z-W)}{1+\delta_+} \right].$$

Then $-r_1 \leq \delta_+ \mathcal{N}_+ / \mathcal{D}_-$, and if $\mathcal{N}_- > 0$ also $-r_1 \geq \delta_- \mathcal{N}_- / \mathcal{D}_+$.

Proof. By Lemma 7.4, $W/\bar{X}^2 \leq \kappa = \Lambda_1^2 K_2 / \bar{X}^2 \leq Z/\bar{X}^2$ and $\vartheta_0 \leq \vartheta = \Lambda_1^2 K_2 - W \leq Z - W$; by Lemma 7.2, $\mu_-/\bar{X} \leq \mu \leq \ell_W/\bar{X}$ with $\mu_- \geq 1$; and (7.2) holds. (i) is Theorem 6.2 with Remark 6.9(a). (ii) $\sigma_0 m = \sigma_0 \tau(1 + \mu)$ (Corollary 6.4); combine (i), $\sigma_0 \tau \leq 1$ and (6.5): $\sigma_0 m \geq (1 + \frac{\mu_-}{\bar{X}}) \sigma_0 \tau$ with $\sigma_0 \tau \geq \max\{0, 1 - \beta Z/\bar{X}^2\}$, respectively $\sigma_0 \tau \geq \bar{X}/(1 + \bar{X})$; the upper inequality is strict because $\sigma_0 \tau < 1$ (the first inequality in the proof of Theorem 6.2 is strict since $K_2 > 0$). Finally $(\bar{X} + \mu_-)/(1 + \bar{X}) \geq 1$ because $\mu_- \geq 1$, i.e. $\frac{1}{\sigma_0} \leq \tau + \frac{1}{\Lambda_1} \leq \tau + \frac{\mu_-}{\Lambda_1} \leq m$ by (6.5) and (7.5). (iii) $\delta < 1$ is (4.1). The rest is Theorem 6.5 in the form of Remark 6.9(b) with $Y_0 = \bar{X} - W$, $\vartheta_- = \vartheta_0$, $\vartheta_+ = Z - W$, $\varrho = \frac{1}{2}$, for which $\hat{\omega} = (1 + \delta_+)/ (1 - \frac{1+\delta_+}{2}) = \omega$. (iv) Corollary 6.6. (v) $r_0 > 0$ is Corollary 5.7. By (6.9) with $\kappa_{x_0} = \kappa$, $t = \sigma_0 \tau \leq 1$: $r_0 \leq 1 + t\mu + \varepsilon_1 \leq 1 + \mu + \beta\kappa \leq 1 + \mu + \beta Z/\bar{X}^2$. For the lower bound let $\kappa' = Z/\bar{X}^2$ with $\beta\kappa' \leq 1$. By (i), $1 - t \leq \beta\kappa'$, so the numerator of (6.9) is at least $1 + \mu - \mu\beta\kappa' - \beta\kappa' = (1 + \mu)(1 - \beta\kappa') \geq 0$, and $1/(1 + \varepsilon_2) \geq 1 - \beta^2\kappa'$; the product of $(1 + \mu)(1 - \beta\kappa')$ with a number in $[1 - \beta^2\kappa', 1] \cap (0, 1]$ is at least $(1 + \mu)(1 - \beta\kappa' - \beta^2\kappa')$. (If $\beta\kappa' > 1$ this lower bound is negative and holds trivially because $r_0 > 0$; so the middle inequality needs no hypothesis.) If $1 - \beta(1 + \beta)\kappa' \geq 0$ we may further replace $1 + \mu$ by its lower bound $1 + \mu_-/\bar{X}$. The last upper bound uses $\mu \leq \ell_W/\bar{X}$. (vi) $r_1 < 0$ is Corollary 5.7. The bounds are Theorem 6.8 in the form of Remark 6.9(c), with $A_{x_0} = W$ and $Y_{x_0}(0) = W - \Lambda_1(m - \tau) \in [W - \ell_W, W - \mu_-]$ by (7.4), (7.5). $\square$

Corollary 7.11 (all dimensions $d \geq 3$). For $d \geq 3$ and $N \geq 2$ with $\bar{X} > 1$: $0 < 1 - \sigma_0 \tau \leq \beta Z_d(N)/\bar{X}^2$ and $0 < \sigma_0 m - 1 < (1 + \ln 2d)/\bar{X}$, where $\bar{X} \geq \frac{4}{d} N^{d-2} (1 - N^{-d})^2$ and $Z_d(N) \leq Z_3$ ($d = 3$), $Z_4(N) \leq 216(1 + \ln N)$, $Z_d(N) \leq \frac{2d3^{d-1}}{d-4} N^{d-4}$ ($d \geq 5$). In particular $\sigma_0 \tau = 1 - O(N^{-2})$ for $d = 3$, $1 - O(N^{-4} \ln N)$ for $d = 4$ and $1 - O(N^{-d})$ for $d \geq 5$, with explicit constants.

Proof. Theorem 7.10(i),(ii), Theorem 7.9, Lemma 7.7 and Lemma 7.4 ($\sum_{j=1}^{N-1} j^{-1} \leq 1 + \ln N$, $\sum_{j=1}^{N-1} j^{d-5} \leq N^{d-4}/(d-4)$ for $d \geq 5$). $\square$

8 Exact reductions of the lattice Green's function at the corner

The functionals τ and m are d -fold sums (7.1). One of the d summations can always be carried out in closed form. Throughout, $\varepsilon'_0 = \frac{1}{2}$, $\varepsilon'_k = 1$ ($k \geq 1$), so that $w_k = \prod_i \varepsilon'_{k_i} (1 + c_{k_i})$.

Lemma 8.1 (one summation in closed form). *Let $N \geq 1$ and $\varphi > 0$. Then*

$$\sum_{k=0}^{N-1} \varepsilon'_k \frac{1+c_k}{\cosh \varphi - c_k} = N \left[\coth \frac{\varphi}{2} \coth(N\varphi) - 1 \right], \quad (8.1)$$

$$\sum_{k=0}^{N-1} \varepsilon'_k (-1)^k \frac{1+c_k}{\cosh \varphi - c_k} = \frac{N \coth \frac{\varphi}{2}}{\sinh(N\varphi)}, \quad (8.2)$$

$$\frac{1}{\pi} \int_0^\pi \frac{1+\cos \theta}{\cosh \varphi - \cos \theta} d\theta = \coth \frac{\varphi}{2} - 1. \quad (8.3)$$

Proof. For $r = e^{-\varphi} \in (0, 1)$ the Poisson kernel identity $\sum_{m \in \mathbb{Z}} r^{|m|} e^{im\theta} = \frac{1-r^2}{1-2r \cos \theta + r^2}$ reads

$$\frac{1}{\cosh \varphi - \cos \theta} = \frac{1}{\sinh \varphi} \sum_{m \in \mathbb{Z}} e^{-|m|\varphi} e^{im\theta}, \quad (8.4)$$

the series converging absolutely and uniformly in θ . Let $\theta_l = \pi l/N$, $l = 0, \dots, 2N-1$, and let j be an integer with $0 \leq j \leq 2N$. Since $\sum_{l=0}^{2N-1} e^{im\theta_l}$ equals $2N$ if $2N \mid m$ and 0 otherwise,

$$\begin{aligned} \frac{1}{2N} \sum_{l=0}^{2N-1} \frac{\cos(j\theta_l)}{\cosh \varphi - \cos \theta_l} &= \frac{1}{\sinh \varphi} \sum_{i \in \mathbb{Z}} e^{-|2Ni+j|\varphi} = \frac{1}{\sinh \varphi} \left[e^{-j\varphi} + \frac{(e^{-j\varphi} + e^{j\varphi})e^{-2N\varphi}}{1 - e^{-2N\varphi}} \right] \\ &= \frac{\cosh((N-j)\varphi)}{\sinh \varphi \sinh(N\varphi)} =: \gamma_j \end{aligned}$$

(we used $|2Ni+j| = 2Ni+j$ for $i \geq 0$ and $= 2N|i| - j$ for $i \leq -1$, as $0 \leq j \leq 2N$). The summand is invariant under $l \mapsto 2N-l$ and $1+\cos \theta_N = 0$, so for any function F , $\sum_{k=0}^{N-1} \varepsilon'_k (1+c_k) F(c_k) = \frac{1}{2} \sum_{l=0}^{2N-1} (1+\cos \theta_l) F(\cos \theta_l)$. With $F(c) = 1/(\cosh \varphi - c)$ this gives for the left side of (8.1) the value $N(\gamma_0 + \gamma_1) = N \frac{\cosh(N\varphi) + \cosh((N-1)\varphi)}{\sinh \varphi \sinh N\varphi} = N \frac{\cosh((N-\frac{1}{2})\varphi)}{\sinh \frac{\varphi}{2} \sinh N\varphi}$, which is the right side because $\cosh((N-\frac{1}{2})\varphi) = \cosh N\varphi \cosh \frac{\varphi}{2} - \sinh N\varphi \sinh \frac{\varphi}{2}$. For (8.2) use $(-1)^l = \cos(N\theta_l)$ and $\cos(N\theta) \cos \theta = \frac{1}{2} [\cos((N+1)\theta) + \cos((N-1)\theta)]$: the left side equals $N[\gamma_N + \frac{1}{2}(\gamma_{N-1} + \gamma_{N+1})] = N \frac{1+\cosh \varphi}{\sinh \varphi \sinh N\varphi} = N \coth \frac{\varphi}{2} / \sinh(N\varphi)$. Finally, the integrand of (8.3) is even, so the left side is $\frac{1}{2\pi} \int_{-\pi}^\pi$; inserting (8.4) and integrating term by term, only $m=0$ and $m=\pm 1$ contribute and give $(1+e^{-\varphi})/\sinh \varphi = 2/(e^\varphi - 1) = \coth \frac{\varphi}{2} - 1$. $\square$

Lemma 8.2 (cotangent sums). *For $N \geq 2$:* $\sum_{k=1}^{N-1} \cot^2 \frac{\pi k}{2N} = \frac{(N-1)(2N-1)}{3}$, $\sum_{k=1}^{N-1} (-1)^k \cot^2 \frac{\pi k}{2N} = -\frac{N^2-1}{3}$, $\sum_{k=1}^{N-1} \cos^2 \frac{\pi k}{2N} = \frac{N-1}{2}$, $\prod_{k=1}^{N-1} \sin \frac{\pi k}{2N} = \frac{\sqrt{N}}{2^{N-1}}$.

Proof. $(1+c_k)/(1-c_k) = \cot^2(\pi k/2N)$. Subtract the $k=0$ term $1/(\cosh \varphi - 1)$ from (8.1) and from (8.2) and let $\varphi \downarrow 0$. With $\coth x = \frac{1}{x} + \frac{x}{3} + O(x^3)$, $1/\sinh x = \frac{1}{x} - \frac{x}{6} + O(x^3)$ and $1/(\cosh \varphi - 1) = \frac{2}{\varphi^2} - \frac{1}{6} + O(\varphi^2)$: $N[\coth \frac{\varphi}{2} \coth N\varphi - 1] = \frac{2}{\varphi^2} + \frac{2N^2}{3} + \frac{1}{6} - N + O(\varphi^2)$ and $N \coth \frac{\varphi}{2} / \sinh N\varphi = \frac{2}{\varphi^2} - \frac{N^2}{3} + \frac{1}{6} + O(\varphi^2)$, which gives the two cotangent sums. The third identity follows from $\cos^2 \frac{\pi k}{2N} + \cos^2 \frac{\pi(N-k)}{2N} = 1$. The last one follows from the classical $\prod_{k=1}^{M-1} \sin \frac{\pi k}{M} = M/2^{M-1}$ with $M = 2N$ and the symmetry $k \mapsto 2N-k$. $\square$

Proposition 8.3 ($d=2$: single sums). *Let $d=2$, $x_k = \frac{\pi k}{2N}$, $y_k = 2N \operatorname{arsinh}(\sin x_k)$ and*

$$\Phi(x) = \frac{\cos^2 x \sqrt{1+\sin^2 x}}{\sin x} \quad (0 < x \leq \frac{\pi}{2}). \quad (8.5)$$

Then for every $N \geq 2$

$$\tau_2 = 4N \sum_{k=1}^{N-1} \Phi(x_k) \coth y_k - \frac{2}{3}(N^2 - 1), \quad m_2 = 4N \sum_{k=1}^{N-1} \Phi(x_k) \left[\coth y_k - \frac{(-1)^k}{\sinh y_k} \right]. \quad (8.6)$$

Proof. Here $\lambda_k = \frac{1}{2}(2 - c_{k_1} - c_{k_2})$ and $\tau_2 = 2 \sum_{k \neq 0} \varepsilon'_{k_1} \varepsilon'_{k_2} \frac{(1+c_{k_1})(1+c_{k_2})}{2-c_{k_1}-c_{k_2}}$. For $k_1 \geq 1$ define $\varphi > 0$ by $\cosh \varphi = 2 - c_{k_1}$; then $\sinh^2 \frac{\varphi}{2} = \frac{1-c_{k_1}}{2} = \sin^2 x_{k_1}$, so $\varphi = 2 \operatorname{arsinh}(\sin x_{k_1})$, $N\varphi = y_{k_1}$, $\coth \frac{\varphi}{2} = \sqrt{1 + \sin^2 x_{k_1}} / \sin x_{k_1}$, and $1 + c_{k_1} = 2 \cos^2 x_{k_1}$. By (8.1) the sum over k_2 is $N[\coth \frac{\varphi}{2} \coth y_{k_1} - 1]$. For $k_1 = 0$ the sum over $k_2 \geq 1$ is $\frac{1}{2} \cdot 2 \sum_{k_2 \geq 1} \cot^2 x_{k_2} = \frac{(N-1)(2N-1)}{3}$. Hence $\tau_2 = \frac{2}{3}(N-1)(2N-1) + 4N \sum_{k \geq 1} \Phi(x_k) \coth y_k - 4N \sum_{k \geq 1} \cos^2 x_k$, and Lemma 8.2 gives the first formula. In the same way, by (8.2), $\tau_2 - m_2 = \sum_{k \neq 0} (-1)^{|k|} w_k / \lambda_k = -\frac{2}{3}(N^2 - 1) + 4N \sum_{k \geq 1} (-1)^k \Phi(x_k) / \sinh y_k$. $\square$

Remark 8.4. The formula for m_2 is the single-sum MFPT identity proved in Theorem 4.2 of the article; it is equivalent, for even N , to the corner-to-corner resistance formula of J. W. Essam and F. Y. Wu, J. Phys. A 42 (2009) 025205. The formula for the stationary-start time τ_2 is its analogue for the diagonal Green's function.

Proposition 8.5 ($d = 3$: double sums). *Let $d = 3$. For $(\theta_1, \theta_2) \in [0, \pi]^2 \setminus \{0\}$ define $\phi = \phi(\theta_1, \theta_2) > 0$ by $\sinh^2 \frac{\phi}{2} = \sin^2 \frac{\theta_1}{2} + \sin^2 \frac{\theta_2}{2}$ (equivalently $\cosh \phi = 3 - \cos \theta_1 - \cos \theta_2$) and*

$$H = (1 + \cos \theta_1)(1 + \cos \theta_2) \left(\coth \frac{\phi}{2} - 1 \right), \quad P_N = (1 + \cos \theta_1)(1 + \cos \theta_2) \coth \frac{\phi}{2} (\coth(N\phi) - 1). \quad (8.7)$$

Then, with $\sum'$ denoting the sum over $(k_1, k_2) \in \{0, \dots, N-1\}^2 \setminus \{0\}$ and all functions evaluated at $(\theta_{k_1}, \theta_{k_2})$,

$$\tau_3 = (N-1)(2N-1) + 3N \sum' \varepsilon'_{k_1} \varepsilon'_{k_2} H + 3N \sum' \varepsilon'_{k_1} \varepsilon'_{k_2} P_N, \quad (8.8)$$

$$m_3 - \tau_3 = (N^2 - 1) - 3N \sum' \varepsilon'_{k_1} \varepsilon'_{k_2} (-1)^{k_1+k_2} \frac{(1+c_{k_1})(1+c_{k_2}) \coth \frac{\phi}{2}}{\sinh(N\phi)}. \quad (8.9)$$

Proof. $\tau_3 = 3 \sum_{k \neq 0} \prod_i \varepsilon'_{k_i} (1+c_{k_i}) / (3-c_{k_1}-c_{k_2}-c_{k_3})$. For $(k_1, k_2) \neq 0$ apply (8.1) to the sum over k_3 with $\cosh \phi = 3 - c_{k_1} - c_{k_2} > 1$: it equals $N[\coth \frac{\phi}{2} \coth N\phi - 1] = N[(\coth \frac{\phi}{2} - 1) + \coth \frac{\phi}{2} (\coth N\phi - 1)]$. For $k_1 = k_2 = 0$ the sum over $k_3 \geq 1$ is $\sum \cot^2 x_{k_3} = \frac{(N-1)(2N-1)}{3}$. This is (8.8); (8.9) follows in the same way from (8.2) and Lemma 8.2, since $\tau_3 - m_3 = \sum_{k \neq 0} (-1)^{|k|} w_k / \lambda_k$. $\square$

9 Two dimensions: logarithmic asymptotics with an explicit remainder

9.1 Statement

Let γ be Euler's constant, $\kappa_2 = 2 \operatorname{arsinh} 1 = \ln(3 + 2\sqrt{2})$, and

$$I_\Psi = -\frac{1}{2} + \frac{3}{2} \ln 2 - \ln \pi, \quad \Lambda_* = \sum_{j \geq 1} \frac{1}{j(e^{2\pi j} - 1)}, \quad \Sigma_* = \sum_{j \geq 1} \frac{(-1)^{j+1}}{j \sinh(\pi j)}, \quad (9.1)$$

$$c_\tau = \frac{8}{\pi}(\gamma + I_\Psi + 2\Lambda_*) - \frac{2}{3} = -0.727904612791689\dots, \quad c_{m\tau} = \frac{2}{3} + \frac{8}{\pi}\Sigma_* = 0.882542400610606\dots \quad (9.2)$$

Theorem 9.1 ($d = 2$, corner target). *For every $N \geq 2$ the unit-rate stationary-start mean hitting time τ_2 of the corner and the corner-to-corner mean first-passage time m_2 of the $N \times N$ reflecting box satisfy*

$$\tau_2 = \frac{8}{\pi} N^2 \ln N + c_\tau N^2 + R_\tau(N), \quad -2.14 \leq R_\tau(N) \leq 3.53, \quad (9.3)$$

$$m_2 - \tau_2 = c_{m\tau} N^2 + R_{m\tau}(N), \quad -9.06 \leq R_{m\tau}(N) \leq 7.72. \quad (9.4)$$

Consequently $m_2 = \frac{8}{\pi} N^2 \ln N + C_2 N^2 + R_m(N)$ with $C_2 = c_\tau + c_{m\tau} = 0.154637787818917\dots$ and $|R_m(N)| \leq 11.3$. (For activity q divide τ_2 and m_2 by q .) The proof is analytic except for the numerical constants: the bound $\sup_{[0, \pi/2]} |\Psi''| \leq 2.35$ of Lemma 9.5(d) and the values of four elementary convergent series are certified by interval arithmetic (Section 13).

Lemma 9.2 (closed forms). $\Lambda_* = -\frac{\pi}{12} - \ln \Gamma(\frac{1}{4}) + \ln 2 + \frac{3}{4} \ln \pi$ and $\Sigma_* = \frac{1}{2} \ln 2 - \frac{\pi}{12}$. Hence

$$c_\tau = \frac{8}{\pi} \left(\gamma - \frac{1}{2} + \frac{7}{2} \ln 2 + \frac{1}{2} \ln \pi - 2 \ln \Gamma(\frac{1}{4}) \right) - 2, \quad c_{m\tau} = \frac{4 \ln 2}{\pi},$$

and C_2 is the constant of Theorem 4.5 of the article.

Proof. Expanding $1/(e^{2\pi j} - 1) = \sum_{l \geq 1} e^{-2\pi j l}$ and $1/\sinh(\pi j) = 2 \sum_{l \text{ odd}} e^{-\pi j l}$ and summing over j first (absolute convergence), $\Lambda_* = -\ln \prod_{l \geq 1} (1 - e^{-2\pi l})$ and $\Sigma_* = 2 \ln \prod_{l \text{ odd}} (1 + e^{-\pi l})$. We use the following classical facts, with the notation and equation numbers of the NIST Digital Library of Mathematical Functions (DLMF, dlmf.nist.gov), at $\tau = i$, i.e. nome $q = e^{i\pi\tau} = e^{-\pi}$: (20.5.3) $\theta_3(0, q) = \prod_{l \geq 1} (1 - q^{2l})(1 + q^{2l-1})^2$; (20.9.1)–(20.9.2) $k = \theta_2^2(0, q)/\theta_3^2(0, q)$, $K(k) = \frac{\pi}{2} \theta_3^2(0, q)$, $K'(k) = -i\tau K(k)$; (23.5.5) $K(k) = K'(k) = \Gamma(\frac{1}{4})^2/(4\sqrt{\pi})$ for $k^2 = \frac{1}{2}$; (20.4.6) $\theta_1'(0, q) = \theta_2\theta_3\theta_4(0, q)$; (20.7.5) $\theta_3^4 = \theta_2^4 + \theta_4^4$; (23.15.9), (23.17.8) $\eta(\tau) = (\frac{1}{2}\theta_1'(0, q))^{1/3} = q^{1/12} \prod_{l \geq 1} (1 - q^{2l})$. At $\tau = i$, $K' = K$, so $k^2 = \frac{1}{2}$, $\theta_3(0, e^{-\pi})^2 = \frac{2}{\pi} K = \Gamma(\frac{1}{4})^2/(2\pi^{3/2})$ and $\theta_2^4 = \frac{1}{2}\theta_3^4 = \theta_4^4$; hence $\eta(i)^3 = \frac{1}{2}\theta_2\theta_3\theta_4 = \theta_3^3/(2\sqrt{2})$, i.e. $\eta(i) = \theta_3(0, e^{-\pi})/\sqrt{2} = \Gamma(\frac{1}{4})/(2\pi^{3/4})$. Therefore $\prod_l (1 - e^{-2\pi l}) = e^{\pi/12} \eta(i)$, which gives Λ_* , and $\prod_{l \text{ odd}} (1 + e^{-\pi l})^2 = \theta_3(0, e^{-\pi})/(e^{\pi/12} \eta(i)) = \sqrt{2} e^{-\pi/12}$, which gives Σ_* . $\square$

Remark 9.3. The inequalities (9.3), (9.4) do not depend on Lemma 9.2: the constants are defined by the rapidly convergent series (9.1). In Section 13 the series and the closed forms are both enclosed by ball arithmetic and the enclosures overlap (to 30 digits).

9.2 Three auxiliary lemmas

Lemma 9.4 (harmonic numbers). For $N \geq 2$: $H_{N-1} := \sum_{k=1}^{N-1} \frac{1}{k} = \ln N + \gamma - \frac{1}{2N} - \eta_N$ with $\frac{1}{12(N+1)^2} < \eta_N < \frac{1}{12N^2} + \frac{1}{6N^3} \leq \frac{1}{6N^2}$.

Proof. Let $a_M = H_M - \ln M - \frac{1}{2M}$. Then $a_{M+1} - a_M = \frac{1}{2}(\frac{1}{M} + \frac{1}{M+1}) - \int_M^{M+1} \frac{dx}{x}$ is the error of the trapezoidal rule for the convex function $1/x$ on $[M, M+1]$, which equals $\frac{1}{12} f''(\xi) = \frac{1}{6\xi^3}$ for some $\xi \in (M, M+1)$. So $\frac{1}{6(M+1)^3} < a_{M+1} - a_M < \frac{1}{6M^3}$; a_M increases to γ , and $\gamma - a_N = \sum_{M \geq N} (a_{M+1} - a_M) \in (\frac{1}{6} \sum_{M \geq N+1} M^{-3}, \frac{1}{6} \sum_{M \geq N} M^{-3}) \subset (\frac{1}{12(N+1)^2}, \frac{1}{6N^3} + \frac{1}{12N^2})$ by comparison with $\int x^{-3} dx$. Finally $H_{N-1} = H_N - \frac{1}{N} = \ln N + \gamma - \frac{1}{2N} - (\gamma - a_N)$, and $\frac{1}{12N^2} + \frac{1}{6N^3} \leq \frac{1}{6N^2}$ for $N \geq 2$. $\square$

Lemma 9.5 (the function Ψ). Let $\Psi(x) = \Phi(x) - \frac{1}{x}$ on $(0, \frac{\pi}{2}]$, with Φ from (8.5). Then:

- (a) $\Psi(x) = c(x) + Q(\sin x)$ with $c(x) = \frac{1}{\sin x} - \frac{1}{x}$ and $Q(s) = -\frac{s(1+s^2-s^4)}{1+(1-s^2)\sqrt{1+s^2}}$; Ψ extends to a real-analytic function on a neighbourhood of $[0, \frac{\pi}{2}]$, with $\Psi(0) = 0$ and $\Psi(\frac{\pi}{2}) = -\frac{2}{\pi}$.
- (b) $-1.09x \leq \Psi(x) \leq 0$ on $[0, \frac{\pi}{2}]$.
- (c) $\int_0^{\pi/2} \Psi(x) dx = I_\Psi = -\frac{1}{2} + \frac{3}{2} \ln 2 - \ln \pi$.
- (d) (computer-assisted) $-0.7382 \leq \Psi''(x) \leq 2.3125$ on $[0, \frac{\pi}{2}]$; in particular $|\Psi''| \leq M_2 := 2.35$.

Proof. (a) With $s = \sin x$, $\Phi(x) = \frac{(1-s^2)\sqrt{1+s^2}}{s}$, and $\Phi - \frac{1}{s} = \frac{(1-s^2)\sqrt{1+s^2}-1}{s} = \frac{(1-s^2)^2(1+s^2)-1}{s[(1-s^2)\sqrt{1+s^2}+1]} = Q(s)$ since $(1-s^2)^2(1+s^2)-1 = -s^2(1+s^2-s^4)$. The denominator $D(s) = 1 + (1-s^2)\sqrt{1+s^2}+1$ is analytic and ≥ 1 on $[0, 1]$, and c is analytic for $|x| < \pi$ with $c(0) = 0$. (b) By Abramowitz and Stegun, *Handbook of Mathematical Functions*, formula 4.3.68, $c(x) = \sum_{j \geq 1} \frac{2(2^{2j}-1)|B_{2j}|}{(2j)!} x^{2j-1} = \frac{x}{6} + \frac{7x^3}{360} + \dots$ for $|x| < \pi$, with positive coefficients. Hence c, c', c'' are non-negative and increasing on $[0, \pi)$ and $c(x)/x$ is increasing, so $\frac{x}{6} \leq c(x) \leq x c(\frac{\pi}{2})/\frac{\pi}{2} = \frac{2}{\pi}(1 - \frac{2}{\pi})x$ on $[0, \frac{\pi}{2}]$. On $[0, 1]$, $1 \leq D(s) \leq 2$ and $1 \leq 1+s^2-s^4 \leq \frac{5}{4}$, so $-\frac{5}{4}s \leq Q(s) \leq -\frac{s}{2}$. Therefore $\Psi(x) \leq \frac{2}{\pi}(1 - \frac{2}{\pi})x - \frac{1}{2} \sin x \leq [\frac{2}{\pi}(1 - \frac{2}{\pi}) - \frac{1}{\pi}]x = \frac{1}{\pi}(1 - \frac{4}{\pi})x \leq 0$ (Lemma 7.1(a)) and $\Psi(x) \geq \frac{x}{6} - \frac{5}{4}x = -\frac{13}{12}x \geq -1.09x$. (c) For $0 < \epsilon < \frac{\pi}{2}$ substitute $s = \sin x$ and then $v = s^2$: $\int_{\epsilon}^{\pi/2} \Phi dx = \int_{\sin \epsilon}^1 \frac{\sqrt{1-s^4}}{s} ds = \frac{1}{2} \int_{\sin^2 \epsilon}^1 \frac{\sqrt{1-v^2}}{v} dv$, and $\frac{d}{dv}[\sqrt{1-v^2} + \ln v - \ln(1 + \sqrt{1-v^2})] = \frac{\sqrt{1-v^2}}{v}$. So $\int_{\epsilon}^{\pi/2} \Phi dx = \frac{1}{2}[-\sqrt{1-\sin^4 \epsilon} - 2 \ln \sin \epsilon + \ln(1 + \sqrt{1-\sin^4 \epsilon})] = -\frac{1}{2} - \ln \epsilon + \frac{1}{2} \ln 2 + o(1)$, while $\int_{\epsilon}^{\pi/2} \frac{dx}{x} = \ln \frac{\pi}{2} - \ln \epsilon$. Subtract and let $\epsilon \rightarrow 0$. (d) $\Psi''(x) = c''(x) + Q''(\sin x) \cos^2 x - Q'(\sin x) \sin x$. The script `06_certify_constants.py` splits $[0, \frac{\pi}{2}]$ into 4000 intervals $[\alpha, \beta]$; on each it bounds c'' by $c''(\alpha) \leq c'' \leq c''(\beta)$ (monotonicity from (b)); $c''(x) = \csc x (\csc^2 x + \cot^2 x) - 2x^{-3}$ evaluated at the end points with 200-bit balls, $c''(0) = 0$) and evaluates the explicit rational-algebraic expressions for Q', Q'' in ball arithmetic on the ball containing $\sin([\alpha, \beta])$. The union of the resulting enclosures is contained in $[-0.7382, 2.3125]$. $\square$

Lemma 9.6 (trapezoidal sum). $\sum_{k=1}^{N-1} \Psi(x_k) = \frac{2N}{\pi} I_{\Psi} + \frac{1}{\pi} + e_1(N)$ with $|e_1(N)| \leq \frac{\pi^2 M_2}{48N}$, where $x_k = \frac{\pi k}{2N}$. Moreover

$$\lim_{N \rightarrow \infty} N e_1(N) = \frac{\pi}{24} (\Psi'(\frac{\pi}{2}) - \Psi'(0)) = \frac{\pi}{24} \left(\frac{4}{\pi^2} + \frac{1}{3} \right) = \frac{1}{6\pi} + \frac{\pi}{72}. \quad (9.5)$$

Proof. Let $h = \frac{\pi}{2N}$ and $T_h = h[\frac{\Psi(0)}{2} + \sum_{k=1}^{N-1} \Psi(x_k) + \frac{\Psi(\pi/2)}{2}]$. On each cell $[x_{k-1}, x_k]$ two integrations by parts give $\frac{h}{2}(\Psi(x_{k-1}) + \Psi(x_k)) - \int_{x_{k-1}}^{x_k} \Psi = \int_{x_{k-1}}^{x_k} q_h \Psi''$ with $q_h(x) = \frac{1}{2}(x - x_{k-1})(x_k - x) \geq 0$ and $\int_{x_{k-1}}^{x_k} q_h = \frac{h^3}{12}$. Summing over the N cells, $T_h - \int_0^{\pi/2} \Psi = \int_0^{\pi/2} q_h \Psi''$, which is at most $\frac{\pi}{2} \cdot \frac{h^2}{12} \sup |\Psi''|$ in modulus. Since $\Psi(0) = 0$ and $\Psi(\frac{\pi}{2}) = -\frac{2}{\pi}$, $T_h = h[\sum_k \Psi(x_k) - \frac{1}{\pi}]$, so $e_1(N) = h^{-1}(T_h - I_{\Psi})$ and the bound follows. For the limit, Ψ'' is continuous on $[0, \frac{\pi}{2}]$ (Lemma 9.5(a)) and $q_h \geq 0$, so by the mean value theorem for integrals $\int_{x_{k-1}}^{x_k} q_h \Psi'' = \frac{h^3}{12} \Psi''(\xi_k)$ with some $\xi_k \in [x_{k-1}, x_k]$; hence $h^{-2}(T_h - I_{\Psi}) = \frac{1}{12} \sum_{k=1}^N h \Psi''(\xi_k)$ is a Riemann sum and tends to $\frac{1}{12} \int_0^{\pi/2} \Psi'' = \frac{1}{12} (\Psi'(\frac{\pi}{2}) - \Psi'(0))$. Thus $N e_1(N) = \frac{\pi}{2} h^{-2}(T_h - I_{\Psi}) \rightarrow \frac{\pi}{24} (\Psi'(\frac{\pi}{2}) - \Psi'(0))$. Finally, by Lemma 9.5(a), $\Psi'(x) = c'(x) + Q'(\sin x) \cos x$ with $c'(x) = -\frac{\cos x}{\sin^2 x} + \frac{1}{x^2}$, $c'(0) = \frac{1}{6}$, $c'(\frac{\pi}{2}) = \frac{4}{\pi^2}$ and $Q'(0) = -\frac{1}{2}$ (as $Q(s) = -\frac{s}{2} + O(s^3)$), so $\Psi'(0) = -\frac{1}{3}$ and $\Psi'(\frac{\pi}{2}) = \frac{4}{\pi^2}$. $\square$

9.3 Proof of Theorem 9.1

Write $F(y) = \coth y - 1 = \frac{2}{e^{2y}-1}$ and recall $y_k = 2N \operatorname{arsinh}(\sin x_k)$. By Lemma 7.1(d),

$$\kappa_2 k \leq y_k \leq \pi k, \quad 0 \leq \pi k - y_k \leq \frac{2N}{3} x_k^3 \quad (1 \leq k \leq N-1). \quad (9.6)$$

Introduce the series (certified values in Section 13)

$$S_1 = \sum_{j \geq 1} \frac{j^2}{\sinh^2(\kappa_2 j)} = 0.1398576\dots, \quad S_2 = \sum_{j \geq 1} \frac{2j}{e^{2\kappa_2 j} - 1} = 0.0642886\dots, \\ S_3 = \sum_{j \geq 1} \frac{j^2 \cosh(\kappa_2 j)}{\sinh^2(\kappa_2 j)} = 0.7395811\dots, \quad S_4 = \sum_{j \geq 1} \frac{j}{\sinh(\kappa_2 j)} = 0.5105105\dots$$

Step 1 (the sum of Φ). By Lemmas 9.4 and 9.6, using $\sum_{k=1}^{N-1} \frac{1}{x_k} = \frac{2N}{\pi} H_{N-1}$,

$$\sum_{k=1}^{N-1} \Phi(x_k) = \frac{2N}{\pi} (\ln N + \gamma + I_\Psi) - \frac{2N}{\pi} \eta_N + e_1(N) \quad (9.7)$$

(the terms $-\frac{2N}{\pi} \cdot \frac{1}{2N}$ and $+\frac{1}{\pi}$ cancel).

Step 2 (the exponentially small part). Since $\frac{1}{x_k} = \frac{2N}{\pi k}$, we have $\frac{2N}{\pi} 2\Lambda_* = \sum_{k \geq 1} \frac{1}{x_k} F(\pi k)$, and

$$E_c(N) := \sum_{k=1}^{N-1} \Phi(x_k) F(y_k) - \frac{2N}{\pi} 2\Lambda_* = \underbrace{\sum_{k=1}^{N-1} \Psi(x_k) F(y_k)}_{(1)} + \underbrace{\sum_{k=1}^{N-1} \frac{F(y_k) - F(\pi k)}{x_k}}_{(2)} - \underbrace{\sum_{k \geq N} \frac{2N}{\pi k} F(\pi k)}_{(3)}.$$

F is positive and decreasing with $|F'(y)| = \sinh^{-2} y$ decreasing. By Lemma 9.5(b) and (9.6): $0 \geq (1) \geq -1.09 \sum_k x_k F(\kappa_2 k) \geq -1.09 \frac{\pi}{2N} S_2$. By the mean value theorem and (9.6): $0 \leq (2) \leq \sum_k \frac{1}{x_k} \frac{2N x_k^3}{3} \sinh^{-2}(\kappa_2 k) = \frac{\pi^2}{6N} S_1$. Finally $0 \leq (3) \leq \frac{2N}{\pi} \cdot \frac{1}{N} \sum_{k \geq N} \frac{2e^{-2\pi k}}{1-e^{-2\pi}} = \frac{4e^{-2\pi N}}{\pi(1-e^{-2\pi})^2}$, and $N e^{-2\pi N} \leq 2e^{-4\pi}$ for $N \geq 2$. Hence

$$-c_E^- \leq N E_c(N) \leq c_E^+, \quad c_E^- = 1.09 \frac{\pi}{2} S_2 + \frac{8e^{-4\pi}}{\pi(1-e^{-2\pi})^2} < 0.11009, \quad c_E^+ = \frac{\pi^2}{6} S_1 < 0.23006. \quad (9.8)$$

Step 3 (τ_2). By (8.6), (9.7) and the definition of E_c and c_τ ,

$$\tau_2 = 4N \left[\sum_k \Phi(x_k) + \sum_k \Phi(x_k) F(y_k) \right] - \frac{2}{3} (N^2 - 1) = \frac{8}{\pi} N^2 \ln N + c_\tau N^2 + R_\tau, \\ R_\tau = \frac{2}{3} - \frac{8N^2}{\pi} \eta_N + 4N e_1(N) + 4N E_c(N).$$

By Lemma 9.4, $-\frac{4}{3\pi} < -\frac{8N^2}{\pi} \eta_N < 0$; by Lemma 9.6, $|4N e_1| \leq \frac{\pi^2 M_2}{12}$; with (9.8),

$$-2.1309 < \frac{2}{3} - \frac{4}{3\pi} - \frac{\pi^2 M_2}{12} - 4c_E^- \leq R_\tau \leq \frac{2}{3} + \frac{\pi^2 M_2}{12} + 4c_E^+ < 3.5197.$$

Step 4 ($m_2 - \tau_2$). By (8.6), $m_2 - \tau_2 = \frac{2}{3}(N^2 - 1) + 4N A_N$ with $A_N = \sum_{k=1}^{N-1} (-1)^{k+1} \Phi(x_k) / \sinh y_k$, and $\frac{2N}{\pi} \Sigma_* = \sum_{k \geq 1} (-1)^{k+1} \frac{1}{x_k \sinh(\pi k)}$. As in Step 2, with $y \mapsto 1/\sinh y$ positive, decreasing, and $|\frac{d}{dy} \frac{1}{\sinh y}| = \frac{\cosh y}{\sinh^2 y}$ decreasing,

$$\left| A_N - \frac{2N}{\pi} \Sigma_* \right| \leq \sum_{k=1}^{N-1} \frac{|\Psi(x_k)|}{\sinh y_k} + \sum_{k=1}^{N-1} \frac{1}{x_k} \left(\frac{1}{\sinh y_k} - \frac{1}{\sinh \pi k} \right) + \sum_{k \geq N} \frac{2N}{\pi k \sinh(\pi k)} \\ \leq \frac{1.09\pi}{2N} S_4 + \frac{\pi^2}{6N} S_3 + \frac{4e^{-\pi N}}{\pi(1-e^{-\pi})(1-e^{-2\pi N})}.$$

For $N \geq 2$ the right side is at most c_A/N with $c_A = 1.09 \frac{\pi}{2} S_4 + \frac{\pi^2}{6} S_3 + \frac{8e^{-2\pi}}{\pi(1-e^{-\pi})(1-e^{-4\pi})} < 2.0957$. Therefore $m_2 - \tau_2 = (\frac{2}{3} + \frac{8}{\pi} \Sigma_*) N^2 - \frac{2}{3} + 4N(A_N - \frac{2N}{\pi} \Sigma_*)$, i.e. (9.4) with $|R_{m\tau} + \frac{2}{3}| \leq 4c_A < 8.383$. $\square$

Numerical observation 9.7. In ball arithmetic (script 19; certified comparisons) $R_\tau(N) < R_\tau(N+1)$ and $R_{m\tau}(N) < R_{m\tau}(N+1)$ for every $2 \leq N < 300$, and both sequences keep increasing along $N = 200, 500, 1000, 2000, 5000, 10^4, 3 \cdot 10^4, 10^5, 3 \cdot 10^5, 10^6$: $R_\tau(N)$ increases from $R_\tau(2) = 0.851279 \dots$ towards $0.883035 \dots$ and $R_{m\tau}(N)$ increases from $R_{m\tau}(2) = -0.530169 \dots$ towards $-0.350894 \dots$ (a check of this note obtained the same picture with separately written 40-digit floating-point code up to $N = 10^6$, within the same project). The two limits are proved in Theorem 9.8 below; the monotonicity for *all* N (hence the two-sided bounds $0.85127 < R_\tau(N) <$

0.88304 and $-0.53017 < R_{m\tau}(N) < -0.35089$ for all $N \geq 2$) is only observed. Also observed (certified values of $R_\tau(N)$, $R_{m\tau}(N)$ for the listed $N \leq 10^6$): $(R_\tau(N) - c_\tau^{(0)})N^2$ settles at -0.0670577 and $(R_{m\tau}(N) - c_{m\tau}^{(0)})N^2$ at -1.0025013 (rounded to seven decimals, the same values for every listed N from $3 \cdot 10^4$ to 10^6), and the sum -1.069559 of these two numbers agrees, after rounding to six decimals, with the coefficient C_{-2} of Theorem 4.5 of the article (Eq. (25)); no rate of convergence is proved here. Thus the true remainders are about four times (resp. twenty times) smaller than the bounds proved in Theorem 9.1.

9.4 The constant terms: limits of the remainders

Besides $S_1, \dots, S_4$ we need the four series

$$\begin{aligned} S_5 &= \sum_{k \geq 1} \frac{2k}{e^{2\pi k} - 1} = 0.0037558\dots, & S_6 &= \sum_{k \geq 1} \frac{k^2}{\sinh^2(\pi k)} = 0.0075537\dots, \\ S_7 &= \sum_{k \geq 1} \frac{(-1)^{k+1}k}{\sinh(\pi k)} = 0.0795774\dots, & S_8 &= \sum_{k \geq 1} \frac{(-1)^{k+1}k^2 \cosh(\pi k)}{\sinh^2(\pi k)} = 0.0733219\dots \end{aligned} \quad (9.9)$$

(certified values, script `19_certify_d2_limits.py`).

Theorem 9.8 ($d = 2$: the constant terms). *For the remainders of Theorem 9.1, as $N \rightarrow \infty$,*

$$R_\tau(N) \rightarrow c_\tau^{(0)} := \frac{2}{3} + \frac{\pi}{18} - \frac{2\pi}{3}S_5 + \frac{2\pi^2}{3}S_6 = 0.8830352\dots, \quad (9.10)$$

$$R_{m\tau}(N) \rightarrow c_{m\tau}^{(0)} := -\frac{2}{3} - \frac{2\pi}{3}S_7 + \frac{2\pi^2}{3}S_8 = -0.3508943\dots, \quad (9.11)$$

so that $m_2 = \frac{8}{\pi}N^2 \ln N + C_2N^2 + C_2^{(0)} + o(1)$ with $C_2^{(0)} = c_\tau^{(0)} + c_{m\tau}^{(0)} = 0.5321408\dots$. The limits are proved analytically (no rate is claimed); only the decimal values are computer-assisted.

Proof. R_τ . By Step 3 of the proof of Theorem 9.1, $R_\tau = \frac{2}{3} - \frac{8N^2}{\pi}\eta_N + 4Ne_1(N) + 4NE_c(N)$. By Lemma 9.4, $\frac{N^2}{12(N+1)^2} < N^2\eta_N < \frac{1}{12} + \frac{1}{6N}$, so $\frac{8N^2}{\pi}\eta_N \rightarrow \frac{2\pi}{3}$; by (9.5), $4Ne_1(N) \rightarrow \frac{2}{3\pi} + \frac{\pi}{18}$. It remains to show $NE_c(N) \rightarrow -\frac{\pi}{6}S_5 + \frac{\pi^2}{6}S_6$. In the notation of Step 2, $NE_c(N) = \sum_{k=1}^{N-1} a_k(N) + \sum_{k=1}^{N-1} b_k(N) - N \cdot (3)$ with

$$a_k(N) = N\Psi(x_k)F(y_k), \quad b_k(N) = \frac{N}{x_k}(F(y_k) - F(\pi k)),$$

and the estimates of Step 2 are the N -independent, summable majorants

$$|a_k(N)| \leq 1.09 \frac{\pi k}{2} F(\kappa_2 k), \quad 0 \leq b_k(N) \leq \frac{N}{x_k} \cdot \frac{2Nx_k^3}{3} \cdot \frac{1}{\sinh^2(\kappa_2 k)} = \frac{\pi^2 k^2}{6 \sinh^2(\kappa_2 k)}$$

(their sums are $1.09\frac{\pi}{2}S_2$ and $\frac{\pi^2}{6}S_1$), while $0 \leq N \cdot (3) \leq \frac{4Ne^{-2\pi N}}{\pi(1-e^{-2\pi})^2} \rightarrow 0$. Fix k and let $N \rightarrow \infty$. Then $x_k = \frac{\pi k}{2N} \rightarrow 0$; since $\operatorname{arsinh}(\sin x) = x - \frac{x^3}{3} + O(x^5)$ as $x \rightarrow 0$ (from $\sin x = x - \frac{x^3}{6} + O(x^5)$) and $\operatorname{arsinh} s = s - \frac{s^3}{6} + O(s^5)$,

$$y_k = 2N \operatorname{arsinh}(\sin x_k) \rightarrow \pi k, \quad \frac{N}{x_k}(\pi k - y_k) = 2N^2 x_k^2 \cdot \frac{x_k - \operatorname{arsinh}(\sin x_k)}{x_k^3} \rightarrow \frac{\pi^2 k^2}{2} \cdot \frac{1}{3} = \frac{\pi^2 k^2}{6}.$$

As Ψ is differentiable at 0 with $\Psi(0) = 0$, $\Psi'(0) = -\frac{1}{3}$ (proof of Lemma 9.6), $N\Psi(x_k) = \frac{\pi k}{2} \cdot \frac{\Psi(x_k)}{x_k} \rightarrow -\frac{\pi k}{6}$, and F is continuous, so $a_k(N) \rightarrow -\frac{\pi k}{6}F(\pi k)$. By the mean value theorem $F(y_k) - F(\pi k) =$

$(\pi k - y_k) \sinh^{-2}(\xi)$ with $\xi \in [y_k, \pi k]$, so $b_k(N) \rightarrow \frac{\pi^2 k^2}{6} \sinh^{-2}(\pi k)$. By dominated convergence for series (Tannery's theorem; the terms with $k \geq N$ are set to zero),

$$\sum_{k=1}^{N-1} a_k(N) \rightarrow -\frac{\pi}{6} \sum_{k \geq 1} k F(\pi k) = -\frac{\pi}{6} S_5, \quad \sum_{k=1}^{N-1} b_k(N) \rightarrow \frac{\pi^2}{6} S_6.$$

Hence $R_\tau \rightarrow \frac{2}{3} - \frac{2}{3\pi} + \frac{2}{3\pi} + \frac{\pi}{18} + 4(-\frac{\pi}{6} S_5 + \frac{\pi^2}{6} S_6)$, which is (9.10).

$R_{m\tau}$. By Step 4, $R_{m\tau} = -\frac{2}{3} + 4N(A_N - \frac{2N}{\pi} \Sigma_*)$ and, since $\Phi(x) = \Psi(x) + \frac{1}{x}$ and $\frac{1}{x_k} = \frac{2N}{\pi k}$,

$$N\left(A_N - \frac{2N}{\pi} \Sigma_*\right) = \sum_{k=1}^{N-1} (-1)^{k+1} (a'_k(N) + b'_k(N)) - N \sum_{k \geq N} \frac{(-1)^{k+1} 2N}{\pi k \sinh(\pi k)},$$

$$a'_k(N) = \frac{N\Psi(x_k)}{\sinh y_k}, \quad b'_k(N) = \frac{N}{x_k} \left(\frac{1}{\sinh y_k} - \frac{1}{\sinh \pi k} \right).$$

By the estimates of Step 4, $|a'_k(N)| \leq 1.09 \frac{\pi k}{2} / \sinh(\kappa_2 k)$ and $0 \leq b'_k(N) \leq \frac{\pi^2 k^2 \cosh(\kappa_2 k)}{6 \sinh^2(\kappa_2 k)}$ (summable, independent of N), and the last sum is at most $\frac{4Ne^{-\pi N}}{\pi(1-e^{-\pi})(1-e^{-2\pi N})} \rightarrow 0$ in modulus. For fixed k , exactly as above, $a'_k(N) \rightarrow -\frac{\pi k}{6 \sinh(\pi k)}$ and, with $\frac{d}{dy} \frac{1}{\sinh y} = -\frac{\cosh y}{\sinh^2 y}$, $b'_k(N) \rightarrow \frac{\pi^2 k^2 \cosh(\pi k)}{6 \sinh^2(\pi k)}$. By dominated convergence $N(A_N - \frac{2N}{\pi} \Sigma_*) \rightarrow -\frac{\pi}{6} S_7 + \frac{\pi^2}{6} S_8$, which gives (9.11).

Values. The script `19_certify_d2_limits.py` sums the series (9.9) over $k \leq 60$ in ball arithmetic and adds a tail bound (for $k \geq 1$ each summand is at most $8k^2 e^{-\pi k}$ in modulus, the ratio of consecutive bounds is below $\frac{1}{20}$, and so $\sum_{k \geq 60} 8k^2 e^{-\pi k} < 10^{-75}$); it certifies $0.8830352 < c_\tau^{(0)} < 0.8830353$, $-0.3508944 < c_{m\tau}^{(0)} < -0.3508943$ and $0.5321408 < C_2^{(0)} < 0.5321409$. $\square$

Remark 9.9 (closed forms; not used). The classical evaluations of the Eisenstein series at the lemniscatic point, $E_2(i) = \frac{3}{\pi}$ and $E_4(i) = \frac{3\Gamma(\frac{1}{4})^8}{(2\pi)^6}$, together with Ramanujan's identity $q \frac{dE_2}{dq} = \frac{E_2 - E_4}{12}$, give $S_5 = \frac{1}{12} - \frac{1}{4\pi}$, $S_7 = \frac{1}{4\pi}$, $S_6 = \frac{1}{72} (E_4(i) - \frac{9}{\pi^2})$, $S_8 = \frac{1}{8\pi^2} + \frac{\Gamma(\frac{1}{4})^8}{512\pi^6}$, and therefore

$$c_\tau^{(0)} = \frac{3}{4} + \frac{\Gamma(\frac{1}{4})^8}{2304\pi^4}, \quad c_{m\tau}^{(0)} = -\frac{3}{4} + \frac{\Gamma(\frac{1}{4})^8}{768\pi^4}, \quad C_2^{(0)} = \frac{\Gamma(\frac{1}{4})^8}{576\pi^4} = 0.5321408832\dots,$$

the constant C_0 of the asymptotic expansion of Theorem 4.5 of the article, which is thereby proved in the series form (9.10)–(9.11). These closed forms are *not* proved in the present note and are not used anywhere; script `19` verifies each of them against the certified series values to more than 50 digits (status: classical identities, quoted; numerically confirmed).

10 Three dimensions: the Watson-type constant with explicit error

10.1 Statement

With H , P_N , ϕ from (8.7) define, for $\theta \in (0, \pi]$ and $s \in (0, 1]$,

$$J(\theta) = \int_0^\pi H(\theta, \theta') d\theta', \quad K(s) = 4 \int_0^1 \sqrt{\frac{(1-t^2)(1+s^2+t^2)}{s^2+t^2}} dt, \quad C_3 = \frac{3}{\pi^2} \int_0^\pi J(\theta) d\theta, \quad (10.1)$$

and the constants ($F_*(r) = 1/(r(e^{2\pi r} - 1))$, $|k|$ the Euclidean norm on $\mathbb{Z}^2$)

$$\begin{aligned} E_\infty &= \frac{4}{\pi} \sum_{k \in \mathbb{Z}^2 \setminus 0} F_*(|k|) + \frac{16}{\pi} \sum_{j \geq 1} \int_0^\infty F_*(\sqrt{j^2 + v^2}) dv = 0.014720007338 \dots, \\ c_{\tau 3} &= 2 + \frac{12}{\pi} (\gamma - \ln \pi - 2 \ln 2) + 3E_\infty = -5.418839026578 \dots, \\ c_{m\tau 3} &= 1 - \frac{6}{\pi} \sum_{k \in \mathbb{Z}^2 \setminus 0} \frac{(-1)^{k_1+k_2}}{|k| \sinh(\pi|k|)} = 1.531584884077 \dots \end{aligned} \quad (10.2)$$

$$\begin{aligned} \text{For } \kappa = (\kappa_1, \kappa_2) \in \mathbb{R}^2 \setminus \{0\} \text{ put } r = |\kappa|, \quad \chi_+(\kappa) &= \frac{\kappa_1^4 + \kappa_2^4}{6r^2} + \frac{r^2}{6}, \quad \chi_-(\kappa) = \frac{\kappa_1^4 + \kappa_2^4}{6r^2} - \frac{r^2}{2}, \text{ and} \\ p(\kappa) &= \frac{2\pi}{r} \chi_-(\kappa) (\coth(\pi r) - 1) + \frac{2\pi^2 \chi_+(\kappa)}{\sinh^2(\pi r)}, \quad p_a(\kappa) = \frac{2\pi}{r} \frac{\chi_-(\kappa)}{\sinh(\pi r)} + \frac{2\pi^2 \chi_+(\kappa) \cosh(\pi r)}{\sinh^2(\pi r)}, \\ \delta_\infty &= \frac{1}{4} \sum_{k \in \mathbb{Z}^2 \setminus 0} p(k) + \sum_{j \geq 1} \int_0^\infty p(j, v) dv, \quad \alpha_\infty = \frac{1}{4} \sum_{k \in \mathbb{Z}^2 \setminus 0} (-1)^{k_1+k_2} p_a(k) \end{aligned} \quad (10.3)$$

(absolutely convergent; certified values $\delta_\infty = 0.0695430676 \dots$, $\alpha_\infty = -0.2336942394 \dots$).

Theorem 10.1 ($d = 3$, corner target). (a) (the constant) $C_3 = \frac{3}{\pi^3} \int_{[0, \pi]^3} \frac{\prod_{i=1}^3 (1 + \cos \theta_i)}{3 - \sum_i \cos \theta_i} d\theta = 3 \int_0^\infty [e^{-s}(I_0(s) + I_1(s))]^3 ds = \sum_{y \in \{0, 1\}^3} u(y)$, where I_ν are modified Bessel functions and $u(y) = \sum_{t \geq 0} \mathbb{P}_0(S_t = y)$ is the Green's function of the simple random walk on $\mathbb{Z}^3$ (so $u(0)$ is Watson's integral). Moreover (computer-assisted enclosure)

$$4.320460169373 < C_3 < 4.320460169375.$$

(b) (elementary first-order bounds) For every $N \geq 2$,

$$\begin{aligned} C_3 N^3 - 8.51 N^2 - 2.54 &\leq \tau_3 \leq C_3 N^3 - 0.53 N^2 + 3.90, \\ N^2 - 1 &\leq m_3 - \tau_3 \leq \frac{1 + \ln 6}{\Lambda_1} \leq \frac{3}{2} (1 + \ln 6) N^2. \end{aligned} \quad (10.4)$$

(c) (second order with explicit $O(1)$ remainders; computer-assisted constants) For every $N \geq 2$,

$$\begin{aligned} \tau_3 &= C_3 N^3 + c_{\tau 3} N^2 + R_3(N), \quad -17.4 \leq R_3(N) \leq 19.4, \\ m_3 - \tau_3 &= c_{m\tau 3} N^2 + R'_3(N), \quad -82.4 \leq R'_3(N) \leq 80.4. \end{aligned} \quad (10.5)$$

(d) (the constant terms) As $N \rightarrow \infty$,

$$\begin{aligned} R_3(N) &\rightarrow c_{\tau 3}^{(0)} := 1 + \frac{\pi}{12} + \frac{3\zeta(3)}{4\pi} + 3\delta_\infty = 1.7573985 \dots, \\ R'_3(N) &\rightarrow c_{m\tau 3}^{(0)} := -1 - 3\alpha_\infty = -0.2989172 \dots, \end{aligned}$$

so that $m_3 = C_3 N^3 + C'_3 N^2 + C''_3 + o(1)$ with $C'_3 = c_{\tau 3} + c_{m\tau 3} = -3.887254142500 \dots$ and $C''_3 = c_{\tau 3}^{(0)} + c_{m\tau 3}^{(0)} = 1.4584812 \dots$.

Remark 10.2. $u(0) = \frac{\sqrt{6}}{32\pi^3} \Gamma(\frac{1}{24}) \Gamma(\frac{5}{24}) \Gamma(\frac{7}{24}) \Gamma(\frac{11}{24}) = 1.5163860591 \dots$ (G. N. Watson, *Three triple integrals*, Quart. J. Math. Oxford 10 (1939) 266–276; closed form: M. L. Glasser and I. J. Zucker, *Extended Watson integrals for the cubic lattices*, Proc. Natl. Acad. Sci. USA 74 (1977) 1800–1801), and $u(e_1) = u(0) - 1$. The constants $C_3 = 4.32046$ and $c_{\tau 3} + c_{m\tau 3} = -3.887$ are the constants K_3, K'_3 of Theorem 4.8 of the article (numerical estimates in its Supplementary Section S5.3); (c) and (d) prove their values, an explicit $O(1)$ remainder for every N , and the next term C''_3 . The proof of (b) is elementary (Lemmas 10.3–10.6); (c) and (d) need in addition the closed form of K by complete elliptic integrals and a trapezoidal estimate for a function whose second derivative has a logarithmic singularity (Lemmas 10.8–10.11).

10.2 The functions K and J

Lemma 10.3. Let $s = \sin \frac{\theta}{2} \in (0, 1]$.

- (a) $J(\theta) = 2(1 - s^2)(K(s) - \pi)$, and $K(s) > \pi$.
- (b) K is decreasing and $M(s) := K(s) + 4 \ln s$ is increasing on $(0, 1]$, with $M(0+) = 6 \ln 2 - 2$.
- (c) $g_3(s) := 2s^2(K(s) - \pi)$ is increasing on $(0, 1]$, with $g_3(0+) = 0$.
- (d) $J(\theta) = -8 \ln \sin \frac{\theta}{2} + J_b(\theta)$ with $J_b = g_1 - g_3$, $g_1(s) := 2(M(s) - \pi)$. As functions of $\theta \in [0, \pi]$, g_1 and g_3 are increasing and absolutely continuous, with

$$g_1(0) = 12 \ln 2 - 4 - 2\pi = -1.965419\dots, \quad g_3(0) = 0,$$

$$g_1(\pi) = g_3(\pi) = 2(K(1) - \pi) = 2.206121\dots$$

(the value $K(1) = 4.2446532915\dots$ is certified in Section 13).

Proof. (a) In $\int_0^\pi (1 + \cos \theta') \coth \frac{\phi}{2} d\theta'$ substitute $t = \sin \frac{\theta'}{2}$: $1 + \cos \theta' = 2(1 - t^2)$, $d\theta' = 2 dt / \sqrt{1 - t^2}$ and $\coth \frac{\phi}{2} = \sqrt{1 + \sinh^2 \frac{\phi}{2}} / \sinh \frac{\phi}{2} = \sqrt{(1 + s^2 + t^2)/(s^2 + t^2)}$. So this integral is $K(s)$; since $\coth > 1$ and $\int_0^\pi (1 + \cos \theta') d\theta' = \pi$, $K(s) > \pi$ and $J(\theta) = (1 + \cos \theta)(K(s) - \pi)$.

(b) For $s > 0$ we may differentiate under the integral sign. With $v = s^2 + t^2$, $\partial_s \sqrt{(1 + v)/v} = -s v^{-3/2}(1 + v)^{-1/2}$, so $K'(s) = -4s \int_0^1 \sqrt{1 - t^2} v^{-3/2}(1 + v)^{-1/2} dt < 0$. Since $\int_0^\infty (s^2 + t^2)^{-3/2} dt = s^{-2}$,

$$M'(s) = K'(s) + \frac{4}{s} = 4s \left[\int_0^1 \left(1 - \sqrt{\frac{1-t^2}{1+s^2+t^2}}\right) \frac{dt}{(s^2 + t^2)^{3/2}} + \int_1^\infty \frac{dt}{(s^2 + t^2)^{3/2}} \right] > 0.$$

Write $K(s) = 4 \int_0^1 \frac{Q_s(t)-1}{\sqrt{s^2+t^2}} dt + 4 \operatorname{arsinh} \frac{1}{s}$ with $Q_s(t) = \sqrt{(1-t^2)(1+s^2+t^2)}$. Since $\operatorname{arsinh} \frac{1}{s} + \ln s = \ln(1 + \sqrt{1+s^2}) \rightarrow \ln 2$ and $|Q_s - 1| \leq |Q_s^2 - 1| = |s^2(1-t^2) - t^4| \leq s^2 + t^4$, the integrand is bounded by $s + t^3 \leq 1 + t^3$ and dominated convergence gives $M(0+) = 4 \int_0^1 \frac{\sqrt{1-t^4}-1}{t} dt + 4 \ln 2 = 2 \int_0^1 \frac{\sqrt{1-v^2}-1}{v} dv + 4 \ln 2 = 2[\sqrt{1-v^2} - \ln(1 + \sqrt{1-v^2})]_0^1 + 4 \ln 2 = 6 \ln 2 - 2$.

(c) $g'_3(s) = 4s[K(s) + \frac{s}{2}K'(s) - \pi]$ and, by (b),

$$K + \frac{s}{2}K' = 4 \int_0^1 \sqrt{1-t^2} \frac{v(1+v) - \frac{s^2}{2}}{v^{3/2}(1+v)^{1/2}} dt \geq 4 \int_0^1 \sqrt{1-t^2} \frac{v + \frac{1}{2}}{\sqrt{v(1+v)}} dt > 4 \int_0^1 \sqrt{1-t^2} dt = \pi,$$

because $s^2 \leq v$ and $(v + \frac{1}{2})^2 = v(1+v) + \frac{1}{4}$. Hence $g'_3 > 0$. Also $g_3(s) = 2s^2(M(s) - 4 \ln s - \pi) \rightarrow 0$ as $s \downarrow 0$.

(d) $J = 2(K - \pi) - 2s^2(K - \pi) = 2(M - 4 \ln s - \pi) - g_3$. The map $\theta \mapsto s$ is increasing and smooth; g_1, g_3 are continuous on $[0, \pi]$ (with the stated limits at 0), increasing, and continuously differentiable on $(0, \pi]$. A continuous increasing function g on $[0, \pi]$ that is C^1 on $(0, \pi]$ is absolutely continuous: $g(\theta) - g(\epsilon) = \int_\epsilon^\theta g'$ and monotone convergence as $\epsilon \downarrow 0$ give $g(\theta) - g(0) = \int_0^\theta g'$ with $g' \geq 0$ integrable. $\square$

Lemma 10.4 (Riemann sums). Let $\theta_k = \pi k/N$ and $\mathcal{E}_N[f] = \sum_{k=1}^{N-1} f(\theta_k) - \frac{N}{\pi} \int_0^\pi f$.

- (a) If f is increasing on $[0, \pi]$, then $-f(\pi) \leq \mathcal{E}_N[f] \leq -f(0)$.
- (b) If f is absolutely continuous on $[0, \pi]$, then $\mathcal{E}_N[f] \rightarrow -\frac{1}{2}(f(0) + f(\pi))$ as $N \rightarrow \infty$.
- (c) $\mathcal{E}_N[\ln \sin \frac{\theta}{2}] = \frac{1}{2} \ln N + \ln 2$.

Consequently

$$\sum_{k=1}^{N-1} J(\theta_k) = \frac{N}{\pi} \int_0^\pi J - 4 \ln N - 8 \ln 2 + \varepsilon_b(N), \quad \varepsilon_b(N) := \mathcal{E}_N[J_b], \quad (10.6)$$

$$-2.2062 < -2(K(1) - \pi) \leq \varepsilon_b(N) \leq 2\pi + 4 - 12 \ln 2 + 2(K(1) - \pi) < 4.1716, \quad (10.7)$$

$$\lim_{N \rightarrow \infty} \varepsilon_b(N) = \pi + 2 - 6 \ln 2.$$

Proof. (a) With $h = \pi/N$, $h \sum_{k=0}^{N-1} f(\theta_k) \leq \int_0^\pi f \leq h \sum_{k=1}^N f(\theta_k)$. (b) Let $T_N = h[\frac{1}{2}f(0) + \sum_{k=1}^{N-1} f(\theta_k) + \frac{1}{2}f(\pi)]$, so that $\mathcal{E}_N[f] = -\frac{1}{2}(f(0) + f(\pi)) + h^{-1}(T_N - \int_0^\pi f)$. Integrating by parts on each cell (valid for absolutely continuous f), $T_N - \int_0^\pi f = \int_0^\pi p_h(x) f'(x) dx$ with $p_h(x) = x - \frac{1}{2}(\theta_{k-1} + \theta_k)$ on $[\theta_{k-1}, \theta_k]$; $|p_h| \leq \frac{h}{2}$ and $\int p_h = 0$ over each cell. Given $\epsilon > 0$ choose a continuous ψ with $\|f' - \psi\|_{L^1} \leq \epsilon$; then $|\int p_h f'| \leq \frac{h}{2}\epsilon + |\int p_h \psi|$ and $|\int p_h \psi| = |\sum_k \int_{\text{cell}} p_h(x)(\psi(x) - \psi(\theta_k)) dx| \leq \frac{h}{2}\pi \omega_\psi(h)$, with ω_ψ the modulus of continuity. Hence $h^{-1}|T_N - \int f| \leq \frac{\epsilon}{2} + \frac{\pi}{2}\omega_\psi(h) \rightarrow \frac{\epsilon}{2}$. (c) By Lemma 8.2, $\sum_{k=1}^{N-1} \ln \sin \frac{\theta_k}{2} = \frac{1}{2} \ln N - (N-1) \ln 2$, and $\int_0^\pi \ln \sin \frac{\theta}{2} d\theta = -\pi \ln 2$. (10.6) is (c) and Lemma 10.3(d). By (a) applied to g_1 and g_3 , $\varepsilon_b = \mathcal{E}_N[g_1] - \mathcal{E}_N[g_3] \in [-g_1(\pi) + g_3(0), -g_1(0) + g_3(\pi)]$, which is (10.7); by (b), $\varepsilon_b \rightarrow -\frac{1}{2}(g_1(0) + g_1(\pi) - g_3(0) - g_3(\pi)) = -\frac{1}{2}g_1(0)$. $\square$

10.3 Row sums and the exponentially small parts

Lemma 10.5 (row sums). *For $\theta \in (0, \pi]$ and $N \geq 1$,*

$$\sum_{k=0}^{N-1} \varepsilon'_k H(\theta, \theta_k) = \frac{N}{\pi} J(\theta) + e_N(\theta), \quad e_N(\theta) := \frac{N}{\pi} \int_0^\pi P_N(\theta, \theta') d\theta' > 0.$$

Proof. Let $\mathcal{K}(\theta_1, \theta_2, \theta_3) = \prod_i (1 + \cos \theta_i) / (3 - \sum_i \cos \theta_i)$, which is symmetric and positive. By (8.3) (with $\cosh \varphi = 3 - \cos \theta_1 - \cos \theta_2$), $H(\theta_1, \theta_2) = \frac{1}{\pi} \int_0^\pi \mathcal{K} d\theta_3$. Fix $\theta_1 = \theta$ and let $\chi = \phi(\theta, \theta_3)$, i.e. $\cosh \chi = 3 - \cos \theta - \cos \theta_3$. Summing over $\theta_2 = \theta_k$ under the integral and using (8.1),

$$\sum_{k=0}^{N-1} \varepsilon'_k H(\theta, \theta_k) = \frac{1}{\pi} \int_0^\pi (1 + \cos \theta)(1 + \cos \theta_3) N \left[\coth \frac{\chi}{2} \coth(N\chi) - 1 \right] d\theta_3,$$

while integrating over θ_2 first (Tonelli) and using (8.3), $J(\theta) = \frac{1}{\pi} \int_0^\pi \int_0^\pi \mathcal{K} d\theta_2 d\theta_3 = \int_0^\pi (1 + \cos \theta)(1 + \cos \theta_3) [\coth \frac{\chi}{2} - 1] d\theta_3$. Subtracting $\frac{N}{\pi} J(\theta)$ leaves $\frac{N}{\pi} \int_0^\pi (1 + \cos \theta)(1 + \cos \theta_3) \coth \frac{\chi}{2} (\coth N\chi - 1) d\theta_3$. $\square$

Let $\kappa_3 = \sqrt{2} \operatorname{arsinh} \sqrt{2} = 1.62099 \dots$ and (certified values in Section 13)

$$B_D = \sum_{k \in \mathbb{Z}^2 \setminus 0} \frac{1}{|k| (e^{2\kappa_3|k|} - 1)} = 0.197812 \dots, \quad B_e = \sum_{j \geq 1} \sqrt{\frac{\pi}{4\kappa_3 j}} \frac{e^{-2\kappa_3 j}}{1 - e^{-2\kappa_3 j}} = 0.029091 \dots$$

Lemma 10.6 (exponentially small parts). *Let $\rho = \sinh \frac{\phi}{2} = \sqrt{\sin^2 \frac{\theta_1}{2} + \sin^2 \frac{\theta_2}{2}}$. Then for $(\theta_1, \theta_2) \in [0, \pi]^2 \setminus \{0\}$*

$$(1 + \cos \theta_1)(1 + \cos \theta_2) \coth \frac{\phi}{2} \leq \frac{4}{\rho}, \quad N\phi \geq \kappa_3 N\rho, \quad \rho \geq \frac{|\theta|}{\pi}, \quad 0 < P_N \leq \frac{4}{\rho} \cdot \frac{2}{e^{2\kappa_3 N\rho} - 1}. \quad (10.8)$$

Consequently, for $N \geq 2$, with $D_N := \sum' \varepsilon'_{k_1} \varepsilon'_{k_2} P_N(\theta_{k_1}, \theta_{k_2})$,

$$0 < D_N \leq 2B_D N, \quad 0 < \sum_{k=1}^{N-1} e_N(\theta_k) \leq 8B_e N, \quad \lim_{N \rightarrow \infty} \frac{1}{N} \left[D_N + \sum_{k=1}^{N-1} e_N(\theta_k) \right] = E_\infty, \quad (10.9)$$

and $\operatorname{Alt}_N := \sum' \varepsilon'_{k_1} \varepsilon'_{k_2} (-1)^{k_1+k_2} (1 + c_{k_1})(1 + c_{k_2}) \coth \frac{\phi}{2} / \sinh(N\phi)$ satisfies $\operatorname{Alt}_N / N \rightarrow \frac{2}{\pi} \sum_{k \in \mathbb{Z}^2 \setminus 0} \frac{(-1)^{k_1+k_2}}{|k| \sinh(\pi|k|)}$.

Proof. With $s_i = \sin \frac{\theta_i}{2}$: $(1 + \cos \theta_1)(1 + \cos \theta_2) = 4(1 - s_1^2)(1 - s_2^2) \leq 4(1 - \frac{\rho^2}{2})^2$ by the arithmetic–geometric mean inequality, $\coth \frac{\phi}{2} = \sqrt{1 + \rho^2}/\rho$, and $(1 - \frac{\rho^2}{2})^2 \sqrt{1 + \rho^2} \leq 1$ by Lemma 7.1(e) with $y = \rho^2/2 \in [0, 1]$. Since $\rho \leq \sqrt{2}$, Lemma 7.1(c) gives $\phi = 2 \operatorname{arsinh} \rho \geq 2\rho \operatorname{arsinh}(\sqrt{2})/\sqrt{2} = \kappa_3 \rho$, and Lemma 7.1(a) gives $s_i \geq \theta_i/\pi$. This proves (10.8) ($\coth x - 1 = 2/(e^{2x} - 1)$ is decreasing).

At a lattice point, $\rho \geq |k|/N$, and $\rho \mapsto \rho^{-1}(e^{2\kappa_3 N \rho} - 1)^{-1}$ is decreasing, so $P_N \leq 8N \tilde{F}(|k|)$ with $\tilde{F}(r) = 1/(r(e^{2\kappa_3 r} - 1))$; as $\sum_{k \in \mathbb{Z}^2 \setminus 0} f(|k|) = 4 \sum'_{k \in \mathbb{N}_0^2 \setminus 0} \varepsilon'_{k_1} \varepsilon'_{k_2} f(|k|)$, $D_N \leq 2N \sum_{\mathbb{Z}^2 \setminus 0} \tilde{F} = 2B_D N$. Substituting $\theta' = \pi v/N$, $e_N(\theta_k) = \int_0^N P_N(\theta_k, \pi v/N) dv \leq 8N \int_0^\infty \tilde{F}(\sqrt{k^2 + v^2}) dv$, and with $v = k \sinh x$ (so $dv/\sqrt{k^2 + v^2} = dx$)

$$\int_0^\infty \tilde{F}(\sqrt{k^2 + v^2}) dv = \int_0^\infty \frac{dx}{e^{2\kappa_3 k \cosh x} - 1} \leq \frac{e^{-2\kappa_3 k}}{1 - e^{-2\kappa_3 k}} \int_0^\infty e^{-\kappa_3 k x^2} dx = \sqrt{\frac{\pi}{4\kappa_3 k}} \frac{e^{-2\kappa_3 k}}{1 - e^{-2\kappa_3 k}},$$

using $\cosh x \geq 1 + \frac{x^2}{2}$. Summing over $k \geq 1$ gives the second bound in (10.9).

For the limits, fix $k = (k_1, k_2) \neq 0$. As $N \rightarrow \infty$, $\rho = \frac{\pi|k|}{2N}(1 + o(1))$, $N\phi \rightarrow \pi|k|$, $(1 + c_{k_1})(1 + c_{k_2}) \rightarrow 4$, hence $P_N/N \rightarrow \frac{16}{\pi} F_*(|k|)$, and likewise for fixed $k \geq 1$, $v \geq 0$: $P_N(\theta_k, \pi v/N)/N \rightarrow \frac{16}{\pi} F_*(\sqrt{k^2 + v^2})$. The bounds just proved, $P_N/N \leq 8\tilde{F}$, are summable (integrable) and independent of N , so dominated convergence gives $D_N/N \rightarrow \frac{16}{\pi} \sum' \varepsilon' \varepsilon' F_*(|k|) = \frac{4}{\pi} \sum_{\mathbb{Z}^2 \setminus 0} F_*(|k|)$ and $\frac{1}{N} \sum_k e_N(\theta_k) \rightarrow \frac{16}{\pi} \sum_{k \geq 1} \int_0^\infty F_*(\sqrt{k^2 + v^2}) dv$. The statement on Alt_N follows in the same way: its terms divided by N converge to $(-1)^{k_1+k_2} \frac{8}{\pi} \frac{1}{|k| \sinh \pi|k|}$ and are dominated by $\frac{4}{|k| \sinh(\kappa_3|k|)}$. $\square$

10.4 Proof of Theorem 10.1(a),(b)

(a) $C_3 = \frac{3}{\pi^2} \int_0^\pi \int_0^\pi H = \frac{3}{\pi^3} \int_{[0,\pi]^3} \mathcal{K}$ by the proof of Lemma 10.5. Writing $1/(3 - \sum_i \cos \theta_i) = \int_0^\infty e^{-s(3 - \sum_i \cos \theta_i)} ds$ and using Tonelli's theorem, $C_3 = 3 \int_0^\infty a(s)^3 ds$ with $a(s) = \frac{1}{\pi} \int_0^\pi (1 + \cos \theta) e^{-s(1 - \cos \theta)} d\theta = e^{-s}(I_0(s) + I_1(s))$ (DLMF 10.32.3). Expanding $\prod_i (1 + \cos \theta_i) = \sum_{y \in \{0,1\}^3} \prod_i \cos(y_i \theta_i)$ gives $C_3 = \sum_y u(y)$ with $u(y) = \pi^{-3} \int_{[0,\pi]^3} \prod_i \cos(y_i \theta_i) (1 - \frac{1}{3} \sum_i \cos \theta_i)^{-1} d\theta$, the Fourier representation of the Green's function of the simple random walk on $\mathbb{Z}^3$. The enclosure is proved in Lemma 10.7 below.

(b) Split the sum in (8.8) into the rows $k_1 \geq 1$ and the row $k_1 = 0$. By Lemma 10.5 and (10.6),

$$\sum_{k_1=1}^{N-1} \sum_{k_2=0}^{N-1} \varepsilon'_{k_2} H(\theta_{k_1}, \theta_{k_2}) = \frac{N}{\pi} \left[\frac{N}{\pi} \int_0^\pi J - 4 \ln N - 8 \ln 2 + \varepsilon_b(N) \right] + \sum_{k=1}^{N-1} e_N(\theta_k).$$

For the row $k_1 = 0$: $\sinh \frac{\phi}{2} = \sin \frac{\theta}{2}$, so with $x = \theta/2$, $H(0, \theta) = 4 \cos^2 x (\sqrt{1 + \sin^2 x} / \sin x - 1) = 4\Phi(x) - 4 \cos^2 x$, and by (9.7) and Lemma 8.2

$$\frac{1}{2} \sum_{k=1}^{N-1} H(0, \theta_k) = 2 \sum_{k=1}^{N-1} \Phi(x_k) - (N-1) = \frac{4N}{\pi} (\ln N + \gamma + I_\Psi) - \frac{4N}{\pi} \eta_N + 2e_1(N) - N + 1.$$

The terms $\mp \frac{4N}{\pi} \ln N$ cancel. Inserting into (8.8) and using $\frac{3N^3}{\pi^2} \int_0^\pi J = C_3 N^3$, $(N-1)(2N-1) = 2N^2 - 3N + 1$:

$$\tau_3 = C_3 N^3 + N^2 \left[A_0 + \frac{3}{\pi} \varepsilon_b(N) \right] + 3N \left[\sum_{k=1}^{N-1} e_N(\theta_k) + D_N \right] + \rho_N, \quad (10.10)$$

$$A_0 = -1 - \frac{24 \ln 2}{\pi} + \frac{12}{\pi} (\gamma + I_\Psi) = -6.4014175 \dots,$$

$$\rho_N = 1 - \frac{12N^2}{\pi} \eta_N + 6N e_1(N) \in \left(1 - \frac{2}{\pi} - \frac{\pi^2 M_2}{8}, 1 + \frac{\pi^2 M_2}{8} \right) \subset (-2.536, 3.900),$$

by Lemmas 9.4 and 9.6. With (10.7) and (10.9),

$$\begin{aligned} -8.5082 < A_0 - \frac{6}{\pi}(K(1) - \pi) &\leq \frac{\tau_3 - C_3 N^3 - \rho_N}{N^2} \\ &\leq A_0 + \frac{3}{\pi}(2\pi + 4 - 12 \ln 2 + 2(K(1) - \pi)) \\ &\quad + 3(8B_e + 2B_D) < -0.5328, \end{aligned}$$

which is the first part of (10.4). The second part is Lemma 7.2 with $d = 3$, $W \leq 6$ and $\Lambda_1 = \frac{2}{3} \sin^2 \frac{\pi}{2N} \geq \frac{2}{3N^2}$.

Parts (c) and (d) are proved in Section 10.5. $\square$

Lemma 10.7 (enclosure of C_3). *Let $a(s) = e^{-s}(I_0(s) + I_1(s))$. Then $a(s) = \frac{4}{\pi} \int_0^1 \sqrt{1-t^2} e^{-2st^2} dt$, and*

$$\sqrt{\frac{2}{\pi s}} \left(1 - \frac{1}{8s} - \frac{3}{32s^2}\right) \leq a(s) \leq \sqrt{\frac{2}{\pi s}} \quad (s > 0), \quad a(s) \leq \sqrt{\frac{2}{\pi s}} \left(1 - \frac{1}{8s}\right) + \frac{e^{-2s}}{\pi} \quad (s \geq 1).$$

Consequently, for $S \geq 200$, with $b = (2/\pi)^{3/2}$, $b(2S^{-1/2} - \frac{1}{4}S^{-3/2} - \frac{9}{80}S^{-5/2}) \leq \int_S^\infty a^3 \leq b(2S^{-1/2}(1 + 4e^{-150}) - \frac{1}{4}S^{-3/2} + \frac{3}{160}S^{-5/2})$, and $C_3 = 4.32046016937353\dots$, in particular $4.320460169373 < C_3 < 4.320460169375$.

Proof. In $a(s) = \frac{1}{\pi} \int_0^\pi (1 + \cos \theta) e^{-s(1-\cos \theta)} d\theta$ substitute $t = \sin \frac{\theta}{2}$. By Lemma 7.1(f) with $y = t^2$, $1 - \frac{t^2}{2} - \frac{t^4}{2} \leq \sqrt{1-t^2} \leq 1 - \frac{t^2}{2} \leq 1$ on $[0, 1]$. Using $\int_0^\infty t^{2j} e^{-2st^2} dt = \frac{1}{2} \sqrt{\frac{\pi}{2s}} \{1, \frac{1}{4s}, \frac{3}{16s^2}\}$ for $j = 0, 1, 2$: the bound $\sqrt{1-t^2} \leq 1$ gives $a \leq \sqrt{2/(\pi s)}$; the lower bound gives $a \geq \frac{4}{\pi} \int_0^\infty (1 - \frac{t^2}{2} - \frac{t^4}{2}) e^{-2st^2} dt$ (the integrand is negative for $t > 1$), which is the stated lower bound; and $a \leq \frac{4}{\pi} \int_0^\infty (1 - \frac{t^2}{2}) e^{-2st^2} dt + \frac{4}{\pi} \int_1^\infty \frac{t^2}{2} e^{-2st^2} dt$, where $\int_1^\infty \frac{t^2}{2} e^{-2st^2} dt \leq \int_1^\infty \frac{t^3}{2} e^{-2st^2} dt = \frac{e^{-2s}}{4} (\frac{1}{2s} + \frac{1}{4s^2}) \leq \frac{e^{-2s}}{4}$ for $s \geq 1$. For $s \geq 200$ the last term is at most $e^{-2s} \sqrt{s} \cdot \sqrt{2/(\pi s)} (1 - \frac{1}{8s})$, so $a^3 \leq b s^{-3/2} (1 - \frac{1}{8s})^3 (1 + e^{-2s} \sqrt{s})^3 \leq b s^{-3/2} (1 - \frac{3}{8s} + \frac{3}{64s^2})(1 + 4e^{-150})$, and $a^3 \geq b s^{-3/2} (1 - \frac{3}{8s} - \frac{9}{32s^2})$ by Bernoulli's inequality; integrate from S to ∞ (in the upper bound the factor $1 + 4e^{-150}$ is needed only on the leading term since the sum of the two others is negative). With $S = 10^6$ the two tail bounds differ by $3b \cdot \frac{21}{160} S^{-5/2} < 2.1 \cdot 10^{-16}$. The integral $\int_0^S a^3$ of the entire function a^3 is enclosed rigorously by the Petras algorithm of Arb (`acb.integral`; F. Johansson, *Numerical integration in arbitrary-precision ball arithmetic*, Proc. ICMS 2018, LNCS 10931, 255–263) in `06_certify_constants.py`; adding the tail gives $C_3 = 4.320460169373537 \pm 2.3 \cdot 10^{-16}$ (the same computation with $S = 200$ gives the consistent enclosure $[4.3204598, 4.3204603]$). $\square$

10.5 Second order: closed form of K , the $O(1)$ remainder and the constant term

Throughout, $\mathbf{K}(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \varphi)^{-1/2} d\varphi$ and $\mathbf{E}(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \varphi)^{1/2} d\varphi$ are the complete elliptic integrals with modulus $k \in [0, 1)$, $k' = \sqrt{1 - k^2}$. We use the classical facts (DLMF 19.4.1, 19.4.2, 19.9.1, 19.12.1)

$$\frac{d\mathbf{K}}{dk} = \frac{\mathbf{E} - k'^2 \mathbf{K}}{k k'^2}, \quad \frac{d\mathbf{E}}{dk} = \frac{\mathbf{E} - \mathbf{K}}{k}, \quad \ln 4 \leq \mathbf{K}(k) + \ln k' \leq \frac{\pi}{2}, \quad \lim_{k \rightarrow 1} (\mathbf{K}(k) + \ln k') = \ln 4, \quad (10.11)$$

and $1 \leq \mathbf{E} \leq \frac{\pi}{2}$, $\mathbf{E}(1) = 1$, $\mathbf{K} > \mathbf{E}$ for $k > 0$ (obvious from the integrals). The limit is the term $m = 0$ of the expansion DLMF 19.12.1, $\mathbf{K}(k) = \sum_{m \geq 0} \left(\frac{(1/2)_m}{m!}\right)^2 k'^{2m} (\ln \frac{1}{k'} + d(m))$ with $d(0) = \psi(1) - \psi(\frac{1}{2}) = \ln 4$.

Lemma 10.8 (closed form of K ; the functions B, b, e, ℓ). For $s \in (0, 1]$ let $k = \frac{1}{1+s^2}$, so that $k' = \frac{s\sqrt{s^2+2}}{1+s^2}$, and write $\mathbf{K} = \mathbf{K}(k)$, $\mathbf{E} = \mathbf{E}(k)$. Define

$$B(s) = 1 - (1+s^2)\mathbf{E} + \frac{s^2(s^2+2)}{1+s^2}\mathbf{K}, \quad b(s) = \frac{B(s)}{s^2} - (\mathbf{K} - \mathbf{E}),$$

$$e(s) = \frac{\mathbf{E}}{(1+s^2)(2+s^2)}, \quad \ell(s) = \mathbf{K} - \mathbf{E} + \ln s.$$

(a) $K(s) = (4+2s^2)\mathbf{K} - (2+2s^2)\mathbf{E}$.

(b) With $' = d/ds$:

$$\mathbf{E}' = \frac{2s(\mathbf{K} - \mathbf{E})}{1+s^2}, \quad (\mathbf{K} - \mathbf{E})' = -\frac{2e}{s}, \quad M' = \frac{4B}{s}, \quad B' = 2s(\mathbf{K} - \mathbf{E}),$$

$$M'' = 8(\mathbf{K} - \mathbf{E}) - \frac{4B}{s^2}, \quad \left(\frac{B}{s^2}\right)' = -\frac{2b}{s}, \quad (s^2b)' = 2se,$$

$$b' = \frac{2(e-b)}{s}, \quad \ell' = \frac{1-2e}{s}.$$

(c) $B(0+) = 0$, $B > 0$, $b > 0$; $M, B, \mathbf{E}$ and s^2b are increasing in s and $\mathbf{K} - \mathbf{E}$ is decreasing; $e(0+) = b(0+) = \frac{1}{2}$, $\ell(0+) = \frac{3}{2}\ln 2 - 1$, $M(0+) = 6\ln 2 - 2$.

(d) For $0 < s \leq 0.78$: $k' \geq s$, $0 \leq \mathbf{E} - 1 \leq k'^2\mathbf{K} \leq 2s^2(\frac{\pi}{2} + \ln \frac{1}{s})$, $0 < \frac{B}{s^2} \leq \frac{\pi}{2} - \frac{1}{2} + \ln \frac{1}{s}$, and with $w := 1 + 2b - 4e$: $|w(s)| \leq s^2(13.11 + 5\ln \frac{1}{s})$. For all $s \in (0, 1]$: $0 < b \leq \frac{\pi}{4}$, $|\ell| \leq 0.72$, $-0.983 \leq M - \pi \leq 1.104$.

Proof. (a) Fix $\theta \in (0, \pi]$, $s = \sin \frac{\theta}{2}$, $E = 3 - \cos \theta = 2 + 2s^2$, and put

$$g_{mn} = \frac{1}{\pi^2} \int_{[0, \pi]^2} \frac{\cos(m\theta_2) \cos(n\theta_3)}{E - \cos \theta_2 - \cos \theta_3} d\theta_2 d\theta_3.$$

By the proof of Lemma 10.5 and Lemma 10.3(a), $(1 + \cos \theta)(K(s) - \pi) = J(\theta) = (1 + \cos \theta)\pi(g_{00} + 2g_{10} + g_{11})$, using $g_{10} = g_{01}$. Integrating the identity $1 = (E - \cos \theta_2 - \cos \theta_3)/(E - \cos \theta_2 - \cos \theta_3)$ gives $Eg_{00} - 2g_{10} = 1$, hence

$$K(s) = \pi[(1 + E)g_{00} + g_{11}].$$

Let Φ be continuous on $\mathbb{R}^2$, 2π -periodic and even in each variable. Then $\pi^{-2} \int_{[0, \pi]^2} \Phi = (2\pi)^{-2} \int_{[-\pi, \pi]^2} \Phi(\theta_2, \theta_3) d\theta_2 d\theta_3 = (2\pi)^{-2} \int_{[-\pi, \pi]^2} \Phi(u + v, u - v) du dv$: for fixed v , periodicity in u gives $\int_{-\pi}^{\pi} \Phi(u + v, u - v) du = \int_{-\pi}^{\pi} \Phi(w, w - 2v) dw$, and for fixed w , as v runs over $[-\pi, \pi]$ the point $w - 2v$ runs twice over a period, so $\frac{1}{2\pi} \int_{-\pi}^{\pi} \Phi(w, w - 2v) dv = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Phi(w, t) dt$. Since $\cos \theta_2 + \cos \theta_3 = 2\cos u \cos v$ and $\cos \theta_2 \cos \theta_3 = \frac{1}{2}(\cos 2u + \cos 2v)$, and $\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{dv}{E - \beta \cos v} = (E^2 - \beta^2)^{-1/2}$ for $|\beta| < E$ (this is (8.4) integrated over a period), we get with $k = 2/E = 1/(1+s^2)$

$$g_{00} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{du}{\sqrt{E^2 - 4\cos^2 u}} = \frac{2}{\pi E} \mathbf{K}, \quad g_{11} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\cos 2u du}{\sqrt{E^2 - 4\cos^2 u}} = \frac{2}{\pi E} \left[\frac{2}{k^2} (\mathbf{K} - \mathbf{E}) - \mathbf{K} \right]$$

(in g_{11} the two terms $\cos 2u$ and $\cos 2v$ contribute equally by the symmetry $u \leftrightarrow v$; then $\cos 2u = 2\cos^2 u - 1$ and $\int_0^{\pi/2} \cos^2 u (1 - k^2 \cos^2 u)^{-1/2} du = (\mathbf{K} - \mathbf{E})/k^2$). Therefore $K = \frac{2}{E} [(1 + E)\mathbf{K} + \frac{E^2}{2} (\mathbf{K} - \mathbf{E}) - \mathbf{K}] = (2 + E)\mathbf{K} - E\mathbf{E}$.

(b) $dk/ds = -2sk^2$ and $k'^2 = s^2(s^2+2)/(1+s^2)^2$, so by (10.11) $\mathbf{K}' = -\frac{2(1+s^2)}{s(s^2+2)}(\mathbf{E} - k'^2\mathbf{K})$ and $\mathbf{E}' = \frac{2s(\mathbf{K} - \mathbf{E})}{1+s^2}$; moreover $\frac{d(\mathbf{K} - \mathbf{E})}{dk} = \frac{k\mathbf{E}}{k'^2}$, which gives $(\mathbf{K} - \mathbf{E})' = -\frac{2\mathbf{E}}{s(1+s^2)(s^2+2)} = -\frac{2e}{s}$. Differentiating (a): $K' = 4s(\mathbf{K} - \mathbf{E}) + (4 + 2s^2)\mathbf{K}' - (2 + 2s^2)\mathbf{E}' = -\frac{4}{s}[(1 + s^2)\mathbf{E} - \frac{s^2(s^2+2)}{1+s^2}\mathbf{K}] = -\frac{4}{s} + \frac{4B}{s}$,

i.e. $M' = K' + \frac{4}{s} = \frac{4B}{s}$. With $c(s) = \frac{s^2(s^2+2)}{1+s^2}$ we have $c\mathbf{K}' = -2s(\mathbf{E} - k'^2\mathbf{K})$, $c' = \frac{2s(2+2s^2+s^4)}{(1+s^2)^2}$ and $c' - 2s + 2sk'^2 = 2s$, hence $B' = -2s\mathbf{E} - (1+s^2)\mathbf{E}' + c'\mathbf{K} + c\mathbf{K}' = -2s\mathbf{E} + (c' - 2s + 2sk'^2)\mathbf{K} = 2s(\mathbf{K} - \mathbf{E})$. The remaining identities follow: $(B/s^2)' = \frac{2(\mathbf{K}-\mathbf{E})}{s^3} - \frac{2B}{s^3} = -\frac{2b}{s}$; $b' = -\frac{2b}{s} + \frac{2e}{s}$; $(s^2b)' = 2sb + s^2b' = 2se$; $\ell' = -\frac{2e}{s} + \frac{1}{s}$; $M'' = (4B/s)' = 8(\mathbf{K} - \mathbf{E}) - 4B/s^2$.

(c) As $s \downarrow 0$, $k \rightarrow 1$, $\mathbf{E} \rightarrow 1$ and $c\mathbf{K} \leq 3s^2(\frac{\pi}{2} + \ln \frac{1}{k'}) \rightarrow 0$, so $B(0+) = 0$ and $B(s) = \int_0^s 2\sigma(\mathbf{K} - \mathbf{E}) d\sigma > 0$; likewise $s^2b = B - s^2(\mathbf{K} - \mathbf{E}) \rightarrow 0$, so $s^2b(s) = \int_0^s 2\sigma e(\sigma) d\sigma > 0$. The monotonicity statements are the signs of the derivatives in (b) ($M' > 0$, $B' > 0$, $\mathbf{E}' > 0$, $(s^2b)' > 0$, $(\mathbf{K} - \mathbf{E})' < 0$). e is continuous on $[0, 1]$ with $e(0) = \frac{1}{2}\mathbf{E}(1) = \frac{1}{2}$, hence $b(s) = s^{-2} \int_0^s 2\sigma e \rightarrow \frac{1}{2}$. By (10.11), $\mathbf{K} + \ln s = (\mathbf{K} + \ln k') + \ln \frac{s}{k'} \rightarrow \ln 4 - \frac{1}{2} \ln 2 = \frac{3}{2} \ln 2$, so $\ell(0+) = \frac{3}{2} \ln 2 - 1$ and, by (a), $M(0+) = \lim [4(\mathbf{K} + \ln s) + 2s^2\mathbf{K} - (2 + 2s^2)\mathbf{E}] = 6 \ln 2 - 2$ (in agreement with Lemma 10.3(b)).

(d) $k' \geq s \iff \sqrt{s^2+2} \geq 1+s^2 \iff s^4 + s^2 \leq 1$, true for $s \leq 0.78$. Next $\mathbf{E} - 1 = \int_0^{\pi/2} [\sqrt{\cos^2 \varphi + k'^2 \sin^2 \varphi} - \cos \varphi] d\varphi = \int_0^{\pi/2} \frac{k'^2 \sin^2 \varphi d\varphi}{\sqrt{\cos^2 \varphi + k'^2 \sin^2 \varphi} + \cos \varphi} \leq k'^2 \mathbf{K}$, and $k'^2 \leq 2s^2$, $\mathbf{K} \leq \frac{\pi}{2} + \ln \frac{1}{k'} \leq \frac{\pi}{2} + \ln \frac{1}{s}$. Hence $B = \int_0^s 2\sigma(\mathbf{K} - \mathbf{E}) \leq \int_0^s 2\sigma(\frac{\pi}{2} - 1 + \ln \frac{1}{\sigma}) d\sigma = s^2(\frac{\pi}{2} - \frac{1}{2} + \ln \frac{1}{s})$. Write $e = \frac{1}{2} + \eta$; then $\eta(\sigma) = \frac{\mathbf{E}-1}{(1+\sigma^2)(2+\sigma^2)} - \frac{3\sigma^2+\sigma^4}{2(1+\sigma^2)(2+\sigma^2)}$, where the first term lies in $[0, \sigma^2(\frac{\pi}{2} + \ln \frac{1}{\sigma})]$ and the second in $[-\sigma^2, 0]$, so $|\eta(\sigma)| \leq \sigma^2(\frac{\pi}{2} + 1 + \ln \frac{1}{\sigma})$. Since $b - \frac{1}{2} = s^{-2} \int_0^s 2\sigma\eta(\sigma) d\sigma$, $|b - \frac{1}{2}| \leq \frac{2}{s^2} \int_0^s \sigma^3(\frac{\pi}{2} + 1 + \ln \frac{1}{\sigma}) d\sigma = s^2(\frac{\pi/2+1}{2} + \frac{1}{2} \ln \frac{1}{s} + \frac{1}{8})$. Therefore $|w| = |2(b - \frac{1}{2}) - 4\eta| \leq s^2(5(\frac{\pi}{2} + 1) + \frac{1}{4} + 5 \ln \frac{1}{s}) \leq s^2(13.11 + 5 \ln \frac{1}{s})$. For all s : $\frac{1}{6} \leq e \leq \frac{\pi}{4}$ (as $1 \leq \mathbf{E} \leq \frac{\pi}{2}$ and $2 \leq (1+s^2)(2+s^2) \leq 6$), so $0 < b = s^{-2} \int_0^s 2\sigma e \leq \frac{\pi}{4}$; $\ell = (\mathbf{K} + \ln k') - \mathbf{E} + \ln \frac{s}{k'}$ with $\ln \frac{s}{k'} = \ln \frac{1+s^2}{\sqrt{s^2+2}} \in [-\frac{1}{2} \ln 2, \ln \frac{2}{\sqrt{3}}]$, so $\ell \in [\ln 4 - \frac{\pi}{2} - \frac{1}{2} \ln 2, \frac{\pi}{2} - 1 + \ln \frac{2}{\sqrt{3}}] \subset [-0.54, 0.72]$; and M increases from $6 \ln 2 - 2$ to $K(1)$, where $6 \ln 2 - 2 - \pi = -0.9827 \dots$ and $K(1) - \pi = 1.1030 \dots$ (Lemma 10.3(d)). $\square$

Lemma 10.9 (the second derivative of J_b). *Let J_b be as in Lemma 10.3(d), i.e. $J_b(\theta) = F(s) := 2(1-s^2)(M(s) - \pi) + 8s^2 \ln s$ with $s = \sin \frac{\theta}{2}$.*

(a) $J_b \in C^1[0, \pi] \cap C^\infty(0, \pi]$, $J'_b(0) = J'_b(\pi) = 0$, and J'_b is absolutely continuous on $[0, \pi]$.

(b) For $\theta \in (0, \pi]$, $J''_b(\theta) = 2 \ln \sin \frac{\theta}{2} + g(\theta)$ with $g(\theta) = \frac{1}{4}G(\sin \frac{\theta}{2})$ and

$$G(s) = -4(1-2s^2)(M-\pi) - 40(1-s^2)B - 8(1-s^2)^2(b-\ell) - (16s^2+8s^4) \ln s + 24 - 32s^2. \quad (10.12)$$

G extends continuously to $[0, 1]$, with $G(0) = 4\pi - 12 \ln 2 + 20 = 24.2486 \dots$ and $G(1) = 4(K(1) - \pi) - 8 = -3.5877 \dots$.

(c) $G \in C^1(0, 1]$ and, with $w = 1 + 2b - 4e$,

$$\begin{aligned} G'(s) = & s \left\{ 16(M - \pi) - 96(1 - 2s^2) \frac{B}{s^2} + 80(1 - s^2)b + 32(1 - s^2)(b - \ell) \right. \\ & \left. - (32 + 32s^2) \ln s - 80 - 8s^2 \right\} + \frac{8(1 - s^2)^2 w}{s}. \end{aligned} \quad (10.13)$$

For $0 < s \leq 0.002$: $|G'(s)| \leq s(416.4 + 168.01 \ln \frac{1}{s})$.

(d) (computer-assisted) $\int_0^\pi |g'(\theta)| d\theta = \frac{1}{4} \int_0^1 |G'(s)| ds \leq V_g := 6.97$; in particular g is absolutely continuous on $[0, \pi]$. (In fact $G' < 0$ on $[0.002, 1]$, and $\frac{1}{4}(G(0) - G(1)) = 6.9590 \dots$)

Proof. (b) With $c = \cos \frac{\theta}{2}$: $ds/d\theta = \frac{c}{2}$, $d^2s/d\theta^2 = -\frac{s}{4}$, so $J'_b = \frac{c}{2}F'(s)$ and $4J''_b = (1-s^2)F''(s) - sF'(s)$. By Lemma 10.8(b), $sM' = 4B$ and $M'' = 8(\mathbf{K} - \mathbf{E}) - 4B/s^2$, so $F' = -4s(M - \pi) + \frac{8(1-s^2)B}{s} + 16s \ln s + 8s$ and $F'' = -4(M - \pi) - 32B + 2(1-s^2)[8(\mathbf{K} - \mathbf{E}) - \frac{4B}{s^2}] + 16 \ln s + 24$. Hence

$$4J''_b = -4(1-2s^2)(M-\pi) - 40(1-s^2)B - \frac{8(1-s^2)^2B}{s^2} + 16(1-s^2)^2(\mathbf{K}-\mathbf{E}) + 16(1-2s^2) \ln s + 24 - 32s^2.$$

Insert $B/s^2 = b + \ell - \ln s$ and $\mathbf{K} - \mathbf{E} = \ell - \ln s$: the terms with $\ln s$ add up to $[8(1 - s^2)^2 - 16(1 - s^2)^2 + 16(1 - 2s^2)] \ln s = 8 \ln s - (16s^2 + 8s^4) \ln s$, which gives $4J_b'' = 8 \ln s + G(s)$ with G as in (10.12). The limits at 0 follow from Lemma 10.8(c): $G(0) = -4(6 \ln 2 - 2 - \pi) - 8(\frac{1}{2} - \frac{3}{2} \ln 2 + 1) + 24$.

(a) F is smooth on $(0, 1]$. As $s \downarrow 0$, $F'(s) \rightarrow 0$ because $B/s \leq s(\frac{\pi}{2} - \frac{1}{2} + \ln \frac{1}{s})$; hence $J_b'(\theta) \rightarrow 0$ as $\theta \downarrow 0$, and since J_b is continuous on $[0, \pi]$ (Lemma 10.3) the mean value theorem gives $J_b \in C^1[0, \pi]$ with $J_b'(0) = 0$. $J_b'(\pi) = 0$ because $c = 0$ at $\theta = \pi$ and $F'(1)$ is finite. By (b), J_b'' is integrable on $(0, \pi)$, so J_b' is absolutely continuous.

(c) Differentiate (10.12) term by term with Lemma 10.8(b): $\frac{d}{ds}[-4(1 - 2s^2)(M - \pi)] = 16s(M - \pi) - 16(1 - 2s^2)\frac{B}{s}$; $\frac{d}{ds}[-40(1 - s^2)B] = 80sB - 80s(1 - s^2)(\mathbf{K} - \mathbf{E}) = -80s(1 - 2s^2)\frac{B}{s^2} + 80s(1 - s^2)b$ (use $\mathbf{K} - \mathbf{E} = B/s^2 - b$); $\frac{d}{ds}[-8(1 - s^2)^2(b - \ell)] = 32s(1 - s^2)(b - \ell) - \frac{8(1 - s^2)^2}{s}(2(e - b) - (1 - 2e)) = 32s(1 - s^2)(b - \ell) + \frac{8(1 - s^2)^2 w}{s}$; $\frac{d}{ds}[-(16s^2 + 8s^4) \ln s + 24 - 32s^2] = -(32s + 32s^3) \ln s - 16s - 8s^3 - 64s$. The sum is (10.13). For $s \leq 0.002$, by Lemma 10.8(d) (with $L = \ln \frac{1}{s}$): $|16(M - \pi)| \leq 17.67$; $96B/s^2 \leq 96(1.0708 + L)$; $80b \leq 62.84$; $32|b - \ell| \leq 32(\frac{\pi}{4} + 0.72) \leq 48.18$; $(32 + 32s^2)|\ln s| \leq 32.01L$; $80 + 8s^2 \leq 80.01$; and $8|w|/s \leq s(104.9 + 40L)$. Adding: $|G'| \leq s(416.4 + 168.01L)$.

(d) Since $\theta \mapsto s$ is increasing and smooth, $\int_0^\pi |g'(\theta)| d\theta = \frac{1}{4} \int_0^1 |G'(s)| ds$. By (c), $\int_0^{s_1} |G'| \leq s_1^2 [\frac{416.4}{2} + 168.01(\frac{1}{2} \ln \frac{1}{s_1} + \frac{1}{4})] < 0.0031$ for $s_1 = 0.002$. On $[s_1, 1]$ the script `16_certify_d3_remainder.py` uses 90 108 cells $[s_i, s_{i+1}]$. On each cell the monotone functions M , B , $\mathbf{E}$ (increasing) and $\mathbf{K} - \mathbf{E}$ (decreasing) are enclosed by the hull of their end-point values (computed from Lemma 10.8(a) with Arb's complete elliptic integrals at 256 bits), s and $\ln s$ by the hull of the end points, and w by $w(s_i) \pm (s_{i+1} - s_i) \sup |w'|$ with $w' = \frac{4}{s}(e - b) - 4e'$, $e' = \frac{2s(\mathbf{K} - \mathbf{E})}{(1 + s^2)^2(2 + s^2)} - \frac{\mathbf{E}(6s + 4s^3)}{(1 + s^2)(2 + s^2)^2}$ evaluated on the cell; then (10.13) is evaluated in ball arithmetic. The sum of (cell width) $\times$ (upper bound of $|G'|$), plus the bound for $[0, s_1]$, divided by 4, is < 6.962 ; the same computation certifies $G' < 0$ on every cell. $\square$

Lemma 10.10 (trapezoidal sums for a logarithmically singular second derivative). *Let $f \in C^1[0, \pi]$ with f' absolutely continuous, $f'(0) = f'(\pi) = 0$, and suppose $f''(\theta) = 2 \ln \sin \frac{\theta}{2} + g(\theta)$ for a.e. θ , where g is absolutely continuous on $[0, \pi]$. Then for every $N \geq 1$, in the notation of Lemma 10.4,*

$$\mathcal{E}_N[f] = -\frac{f(0) + f(\pi)}{2} + \frac{\zeta(3)}{4N^2} + r_N, \quad |r_N| \leq \frac{\sqrt{3}\pi^2}{216N^2} \int_0^\pi |g'|, \quad N^2 r_N \rightarrow 0 \quad (N \rightarrow \infty).$$

Proof. Let $h = \pi/N$, $T_N = h[\frac{1}{2}f(0) + \sum_{k=1}^{N-1} f(\theta_k) + \frac{1}{2}f(\pi)]$ and, on each cell $[\theta_{k-1}, \theta_k]$, $q_h(\theta) = \frac{1}{2}(\theta - \theta_{k-1})(\theta_k - \theta)$. Two integrations by parts on each cell give $T_N - \int_0^\pi f = \int_0^\pi q_h f''$. Put $\varkappa_h = q_h - \frac{h^2}{12}$; since $\int_0^\pi f'' = f'(\pi) - f'(0) = 0$, $T_N - \int_0^\pi f = \int_0^\pi \varkappa_h f''$. With $t = \{\theta/h\}$, $\varkappa_h = -\frac{h^2}{2} B_2(t)$, $B_2(t) = t^2 - t + \frac{1}{6}$, and (DLMF 24.8.1) $B_2(t) = \frac{1}{\pi^2} \sum_{j \geq 1} \frac{\cos 2\pi j t}{j^2}$, i.e. $\varkappa_h(\theta) = -\frac{h^2}{2\pi^2} \sum_{j \geq 1} \frac{\cos(2Nj\theta)}{j^2}$, uniformly.

The logarithmic part. In $L^2(0, \pi)$, $\ln \sin \frac{\theta}{2} = -\ln 2 - \sum_{m \geq 1} \frac{\cos m\theta}{m}$ (for $0 < \theta < 2\pi$, Abel's theorem applied to $-\ln(1 - z) = \sum_{m \geq 1} z^m/m$ at $z = e^{i\theta}$ gives, on taking real parts, $-\ln|1 - e^{i\theta}| = -\ln(2 \sin \frac{\theta}{2}) = \sum_{m \geq 1} \frac{\cos m\theta}{m}$; the functions $\cos m\theta$, $m \geq 0$, are orthogonal in $L^2(0, \pi)$ and the coefficients are square summable, so the series also converges in $L^2(0, \pi)$ to the same function). As $\int_0^\pi \varkappa_h = 0$ and $\int_0^\pi \cos(2Nj\theta) \cos(m\theta) d\theta = \frac{\pi}{2} \delta_{m, 2Nj}$ for $m, j \geq 1$,

$$\int_0^\pi \varkappa_h(\theta) \ln \sin \frac{\theta}{2} d\theta = \sum_{j \geq 1} \frac{1}{2Nj} \cdot \frac{h^2}{2\pi^2 j^2} \cdot \frac{\pi}{2} = \frac{h^2 \zeta(3)}{8\pi N} = \frac{\pi \zeta(3)}{8N^3}.$$

The regular part. Let $\mathcal{K}_h(\theta) = \int_0^\theta \varkappa_h$. On a cell, with $t = (\theta - \theta_{k-1})/h$, $\mathcal{K}_h = h^3 \mathcal{P}(t)$, $\mathcal{P}(t) = -\frac{1}{12}t(1 - t)(1 - 2t)$; so $\mathcal{K}_h$ vanishes at all nodes, $|\mathcal{K}_h| \leq h^3 \max |\mathcal{P}| = \frac{\sqrt{3}}{216} h^3$, and $\int_0^1 \mathcal{P} = 0$. Integrating by parts, $\int_0^\pi \varkappa_h g = -\int_0^\pi \mathcal{K}_h g'$, hence $|\int_0^\pi \varkappa_h g| \leq \frac{\sqrt{3}}{216} h^3 \int_0^\pi |g'|$; moreover $h^{-3} \int_0^\pi \mathcal{K}_h g' = \int_0^\pi \mathcal{P}(\{N\theta/\pi\}) g'(\theta) d\theta \rightarrow \int_0^1 \mathcal{P} \cdot \int_0^1 g' = 0$ by the Riemann–Lebesgue (Fejér) lemma for $g' \in L^1$. Finally $\mathcal{E}_N[f] = -\frac{1}{2}(f(0) + f(\pi)) + h^{-1}(T_N - \int_0^\pi f)$ and $h^{-1} \cdot 2 \cdot \frac{\pi \zeta(3)}{8N^3} = \frac{\zeta(3)}{4N^2}$; $r_N = h^{-1} \int_0^\pi \varkappa_h g$. $\square$

Lemma 10.11 (quantitative comparison with the continuum sums). *Let $N \geq 2$, $\kappa \in [0, N]^2$ with $r = |\kappa| \geq 1$, $\theta_i = \pi\kappa_i/N$, $x = \frac{\pi\kappa}{2N}$, and ρ, ϕ as in Lemma 10.6; put $\mathcal{A} = (1 + \cos \theta_1)(1 + \cos \theta_2) \coth \frac{\phi}{2}$, $y = N\phi$, $\mathcal{A}_\infty = \frac{4}{|x|} = \frac{8N}{\pi r}$, $P_\infty(\kappa) = \mathcal{A}_\infty(\coth(\pi r) - 1) = \frac{16N}{\pi} F_*(r)$. Then*

$$|\mathcal{A} - \mathcal{A}_\infty| \leq 4|x| = \frac{2\pi r}{N}, \quad \kappa_3 r \leq y \leq \pi r, \quad 0 \leq \pi r - y \leq \frac{\pi r |x|^2}{3} = \frac{\pi^3 r^3}{12N^2}, \quad (10.14)$$

$$\begin{aligned} |P_N(\kappa) - P_\infty(\kappa)| &\leq \frac{Q(r)}{N}, \quad Q(r) = 2\pi r(\coth(\kappa_3 r) - 1) + \frac{2\pi^2 r^2}{3 \sinh^2(\kappa_3 r)}, \\ \left| \frac{\mathcal{A}}{\sinh y} - \frac{\mathcal{A}_\infty}{\sinh \pi r} \right| &\leq \frac{Q_a(r)}{N}, \quad Q_a(r) = \frac{2\pi r}{\sinh(\kappa_3 r)} + \frac{2\pi^2 r^2 \cosh(\kappa_3 r)}{3 \sinh^2(\kappa_3 r)}, \end{aligned} \quad (10.15)$$

and for fixed κ , $N(P_N(\kappa) - P_\infty(\kappa)) \rightarrow p(\kappa)$ and $N(\frac{\mathcal{A}}{\sinh y} - \frac{\mathcal{A}_\infty}{\sinh \pi r}) \rightarrow p_a(\kappa)$ as $N \rightarrow \infty$, with p, p_a from (10.3); in particular $|p| \leq Q$, $|p_a| \leq Q_a$. Consequently, with D_N , e_N , Alt_N , E_∞ from Lemma 10.6 and $A_\infty := \frac{2}{\pi} \sum_{k \in \mathbb{Z}^2 \setminus 0} \frac{(-1)^{k_1+k_2}}{|k| \sinh(\pi|k|)} = \frac{1-c_{m\tau 3}}{3}$,

$$\begin{aligned} \delta_N &:= D_N + \sum_{k=1}^{N-1} e_N(\theta_k) - E_\infty N, \quad |\delta_N| \leq \frac{c_\delta}{N}, \quad N\delta_N \rightarrow \delta_\infty; \\ |\text{Alt}_N - A_\infty N| &\leq \frac{c_a}{N}, \quad N(\text{Alt}_N - A_\infty N) \rightarrow \alpha_\infty, \end{aligned} \quad (10.16)$$

where (certified values) $c_\delta = \frac{1}{4} \sum_{k \in \mathbb{Z}^2 \setminus 0} Q(|k|) + \sum_{j \geq 1} \int_0^\infty Q(\sqrt{j^2 + v^2}) dv + 2 \cdot 10^{-4} < 4.864$ and $c_a = \frac{1}{4} \sum_{k \in \mathbb{Z}^2 \setminus 0} Q_a(|k|) + 0.08 < 27.13$.

Proof. With $s_i = \sin x_i$, $x_i \in [0, \frac{\pi}{2}]$: $x_i^2(1 - \frac{x_i^2}{3}) \leq s_i^2 \leq x_i^2$ (as in Lemma 7.1(a); $\sin v \geq v - \frac{v^3}{6} \geq 0$ on $[0, \frac{\pi}{2}]$), hence $|x|^2(1 - \frac{|x|^2}{3}) \leq \rho^2 \leq |x|^2$, and $\rho \geq \frac{2}{\pi}|x| = \frac{r}{N}$. $\mathcal{A}$: $\mathcal{A} = 4(1 - s_1^2)(1 - s_2^2)\sqrt{1 + \rho^2}/\rho \leq 4/\rho$ (Lemma 10.6), and $\frac{1}{\rho} - \frac{1}{|x|} = \frac{|x|^2 - \rho^2}{\rho|x|(|x| + \rho)} \leq \frac{|x|^2}{3\rho} \leq \frac{\pi|x|}{6}$, so $\mathcal{A} - \mathcal{A}_\infty \leq \frac{2\pi}{3}|x|$. Conversely $(1 - s_1^2)(1 - s_2^2) \geq 1 - \rho^2$, so $\mathcal{A} \geq \frac{4(1 - \rho^2)}{\rho}$ (trivially if $\rho > 1$), and $\frac{4}{\rho} - 4\rho \geq \frac{4}{|x|} - 4|x|$. Thus $|\mathcal{A} - \mathcal{A}_\infty| \leq 4|x|$. y : $y = 2N \operatorname{arsinh} \rho \leq 2N|x| = \pi r$, and $y \geq \kappa_3 N \rho \geq \kappa_3 r$ by (10.8). If $|x|^2 \geq 3$ the last bound in (10.14) is trivial. Otherwise $\operatorname{arsinh} \rho \geq \rho(1 - \frac{\rho^2}{6})$ (compare derivatives: $(1 + \rho^2)^{-1/2} \geq 1 - \frac{\rho^2}{2}$), the function $\rho \mapsto \rho(1 - \frac{\rho^2}{6})$ is increasing on $[0, \sqrt{2}]$, and $\rho \geq \rho_- := |x|(1 - \frac{|x|^2}{3})^{1/2}$ with $1 - \frac{\rho_-^2}{6} \geq 1 - \frac{|x|^2}{6} \geq (1 - \frac{|x|^2}{3})^{1/2}$; so $\operatorname{arsinh} \rho \geq |x|(1 - \frac{|x|^2}{3})$, i.e. $\pi r - y \leq \pi r|x|^2/3$. (10.15): With $F_c(z) = \coth z - 1$ (positive, decreasing, $|F'_c| = \sinh^{-2}$ decreasing), $P_N - P_\infty = (\mathcal{A} - \mathcal{A}_\infty)F_c(y) + \mathcal{A}_\infty(F_c(y) - F_c(\pi r))$, so by (10.14) and the mean value theorem $|P_N - P_\infty| \leq \frac{2\pi r}{N} F_c(\kappa_3 r) + \frac{8N}{\pi r} \cdot \frac{\pi^3 r^3}{12N^2} \sinh^{-2}(\kappa_3 r) = Q(r)/N$. The second line is the same computation with $1/\sinh z$ in place of F_c ($|\frac{d}{dz} \frac{1}{\sinh z}| = \frac{\cosh z}{\sinh^2 z}$, decreasing). *Pointwise limits:* for fixed κ , as $N \rightarrow \infty$ ($x \rightarrow 0$), with $q_4 = x_1^4 + x_2^4$, $\rho^2 = |x|^2 - \frac{q_4}{3} + O(|x|^6)$, $(1 - s_1^2)(1 - s_2^2) = 1 - |x|^2 + O(|x|^4)$, $\sqrt{1 + \rho^2} = 1 + \frac{|x|^2}{2} + O(|x|^4)$, hence $\mathcal{A} - \mathcal{A}_\infty = \frac{4}{|x|}(\frac{q_4}{6|x|^2} - \frac{|x|^2}{2}) + O(|x|^3) = \frac{2\pi}{Nr} \chi_-(\kappa) + O(N^{-3})$ and $\pi r - y = \pi r(\frac{q_4}{6|x|^2} + \frac{|x|^2}{6}) + O(N|x|^5) = \frac{\pi^3 r}{4N^2} \chi_+(\kappa) + O(N^{-4})$. Therefore $N(P_N - P_\infty) \rightarrow \frac{2\pi}{r} \chi_- F_c(\pi r) + \frac{8}{\pi r} \cdot \frac{\pi^3 r}{4} \chi_+ \sinh^{-2}(\pi r) = p(\kappa)$, and likewise for p_a . (10.16): Recall $D_N = \sum' \varepsilon'_{k_1} \varepsilon'_{k_2} P_N(k)$ and $e_N(\theta_k) = \int_0^N P_N(k, v) dv$ (proof of Lemma 10.6), and $E_\infty N = \sum'_{k \in \mathbb{N}_0^2 \setminus 0} \varepsilon' \varepsilon' P_\infty(k) + \sum_{j \geq 1} \int_0^\infty P_\infty(j, v) dv$. Hence

$$|\delta_N| \leq \frac{1}{N} \left[\frac{1}{4} \sum_{k \in \mathbb{Z}^2 \setminus 0} Q(|k|) + \sum_{j \geq 1} \int_0^\infty Q(\sqrt{j^2 + v^2}) dv \right] + T_1 + T_2,$$

where

$$\begin{aligned} T_1 &= \frac{4N}{\pi} \sum_{|k|_\infty \geq N} F_*(|k|) \leq \frac{4N}{\pi} \sum_{j \geq N} \frac{8}{e^{2\pi j} - 1} \leq \frac{32N e^{-2\pi N}}{\pi(1 - e^{-2\pi})(1 - e^{-4\pi})}, \\ T_2 &= \frac{16N}{\pi} \left[\sum_{j \geq N} \int_0^\infty F_*(\sqrt{j^2 + v^2}) dv + \sum_{j=1}^{N-1} \int_N^\infty F_*(\sqrt{j^2 + v^2}) dv \right] \\ &\leq \frac{16N}{\pi} e^{-2\pi N} \left[\frac{1}{2\sqrt{N}(1 - e^{-2\pi})(1 - e^{-4\pi})} + \frac{1}{2\pi(1 - e^{-4\pi})} \right] \end{aligned}$$

(for the first sum use the computation of Lemma 10.6 with π in place of κ_3 ; for the second $F_*(\sqrt{j^2 + v^2}) \leq F_*(v) \leq \frac{e^{-2\pi v}}{N(1 - e^{-4\pi})}$). As $N^2 e^{-2\pi N}$ is decreasing, $N(T_1 + T_2) \leq 1.8 \cdot 10^{-4}$ for $N \geq 2$. The limit $N\delta_N \rightarrow \delta_\infty$ follows by dominated convergence (dominating functions $Q(|k|)$ and $Q(\sqrt{j^2 + v^2})$, which do not depend on N ; the tails $N(T_1 + T_2) \rightarrow 0$). The statements on Alt_N are proved in the same way; its tail is $\frac{2N}{\pi} \sum_{|k|_\infty \geq N} \frac{1}{|k| \sinh \pi |k|} \leq \frac{32N e^{-\pi N}}{\pi(1 - e^{-\pi})(1 - e^{-4\pi})}$, at most $0.08/N$ for $N \geq 2$. $A_\infty = (1 - c_{m\tau_3})/3$ is the definition (10.2) of $c_{m\tau_3}$. The numerical values: Q and Q_a are summed over $|k|_\infty \leq 30$ in ball arithmetic; for $r \geq 1$, $Q(r) \leq e^{-2\kappa_3 r}(a_1 r + a_2 r^2)$ and $Q_a(r) \leq e^{-\kappa_3 r}(a_3 r + a_4 r^2)$ with $a_1 = a_3 = \frac{4\pi}{1-q}$, $a_2 = \frac{8\pi^2}{3(1-q)^2}$, $a_4 = \frac{4\pi^2(1+q)}{3(1-q)^2}$, $q = e^{-2\kappa_3}$, which bounds the tails, and with $v = j \sinh t$, $\int_0^\infty e^{-2\kappa_3 \sqrt{j^2 + v^2}}(a_1 r + a_2 r^2) dv = a_1 j^2 \frac{K_0(z) + K_2(z)}{2} + a_2 j^3 \frac{3K_1(z) + K_3(z)}{4}$, $z = 2\kappa_3 j$ (K_ν the modified Bessel functions, $K_\nu(z) = \int_0^\infty e^{-z \cosh t} \cosh \nu t dt$, DLMF 10.32.9). $\square$

10.6 Proof of Theorem 10.1(c),(d)

By (10.10), (10.16) and the algebra $A_0 + \frac{3}{\pi}(\pi + 2 - 6 \ln 2) + 3E_\infty = 2 + \frac{12}{\pi}(\gamma - \ln \pi - 2 \ln 2) + 3E_\infty = c_{\tau_3}$ (insert $I_\Psi = -\frac{1}{2} + \frac{3}{2} \ln 2 - \ln \pi$),

$$R_3(N) = \tau_3 - C_3 N^3 - c_{\tau_3} N^2 = \frac{3N^2}{\pi} (\varepsilon_b(N) - \varepsilon_b(\infty)) + 3N\delta_N + \rho_N, \quad \varepsilon_b(\infty) := \pi + 2 - 6 \ln 2. \quad (10.17)$$

By Lemma 10.9, $f = J_b$ satisfies the hypotheses of Lemma 10.10 with $\int |g'| \leq V_g$, and $-\frac{1}{2}(J_b(0) + J_b(\pi)) = -\frac{1}{2}g_1(0) = \varepsilon_b(\infty)$; so

$$\varepsilon_b(N) = \varepsilon_b(\infty) + \frac{\zeta(3)}{4N^2} + r_N, \quad \frac{3N^2}{\pi} |r_N| \leq \frac{\sqrt{3}\pi}{72} V_g < 0.527.$$

With $|3N\delta_N| \leq 3c_\delta < 14.60$ and $\rho_N \in (-2.536, 3.900)$ this gives $R_3 \geq \frac{3\zeta(3)}{4\pi} - 0.527 - 14.60 - 2.536 > -17.4$ and $R_3 \leq \frac{3\zeta(3)}{4\pi} + 0.527 + 14.60 + 3.900 < 19.4$ ($\frac{3\zeta(3)}{4\pi} = 0.28696\dots$). By (8.9), $m_3 - \tau_3 = N^2 - 1 - 3N\text{Alt}_N = (1 - 3A_\infty)N^2 - 1 - 3N(\text{Alt}_N - A_\infty N)$, so $R'_3 = -1 - 3N(\text{Alt}_N - A_\infty N)$ and $|R'_3 + 1| \leq 3c_a < 81.4$. This proves (c).

For (d): $\frac{3N^2}{\pi} (\varepsilon_b(N) - \varepsilon_b(\infty)) \rightarrow \frac{3\zeta(3)}{4\pi}$ (Lemma 10.10), $3N\delta_N \rightarrow 3\delta_\infty$ and $R'_3 \rightarrow -1 - 3\alpha_\infty$ (Lemma 10.11). It remains to show $\rho_N = 1 - \frac{12N^2}{\pi} \eta_N + 6Ne_1(N) \rightarrow 1 + \frac{\pi}{12}$. By Lemma 9.4, $N^2 \eta_N \rightarrow \frac{1}{12}$. For $e_1(N)$ (Lemma 9.6; step $h = \frac{\pi}{2N}$ on $[0, \frac{\pi}{2}]$) the cell formula of Lemma 10.10 gives $h e_1(N) = \int_0^{\pi/2} q_h \Psi'' = \frac{h^2}{12} (\Psi'(\frac{\pi}{2}) - \Psi'(0)) + \int_0^{\pi/2} \varkappa_h \Psi''$, and $|\int \varkappa_h \Psi''| = |\sum_{\text{cells}} \int \varkappa_h (\Psi'' - \Psi''(\text{mid}))| \leq \frac{h^2}{12} \cdot \frac{\pi}{2} \cdot \omega_{\Psi''}(h) = o(h^2)$ because Ψ'' is continuous (Lemma 9.5(a)). By Lemma 9.5(a), $\Psi'(x) = c'(x) + Q'(\sin x) \cos x$ with $c'(0) = \frac{1}{6}$, $Q'(0) = -\frac{1}{2}$, $c'(\frac{\pi}{2}) = \frac{4}{\pi^2}$, so $\Psi'(\frac{\pi}{2}) - \Psi'(0) = \frac{4}{\pi^2} + \frac{1}{3}$ and $6Ne_1(N) \rightarrow 6 \cdot \frac{\pi}{24} (\frac{4}{\pi^2} + \frac{1}{3}) = \frac{1}{\pi} + \frac{\pi}{12}$. Hence $\rho_N \rightarrow 1 - \frac{1}{\pi} + \frac{1}{\pi} + \frac{\pi}{12}$, and $R_3(N) \rightarrow \frac{3\zeta(3)}{4\pi} + 3\delta_\infty + 1 + \frac{\pi}{12}$. The values of $\delta_\infty, \alpha_\infty$ are certified in `16_certify_d3_remainder.py` (lattice sums over $|k|_\infty \leq 12$, rigorous integration of $p(j, \cdot)$, $j \leq 12$, on $[0, 14]$, and tails bounded through $|p(\kappa)| \leq \pi r (\coth \pi r - 1) + \frac{2\pi^2 r^2}{3 \sinh^2 \pi r}$, $|p_a(\kappa)| \leq \frac{\pi r}{\sinh \pi r} + \frac{2\pi^2 r^2 \cosh \pi r}{3 \sinh^2 \pi r}$, which follow from $|\chi_-| \leq \frac{r^2}{2}$, $0 \leq \chi_+ \leq \frac{r^2}{3}$). $\square$

Remark 10.12 (how sharp is (c)?). The true remainders are much smaller than the bounds in (c): $R_3(N)$ increases from 1.6117 ($N = 2$) to its limit 1.75740 and $R'_3(N)$ from -0.6263 to -0.29892 (script 17; in agreement with (d) to all digits shown). The loss is in the crude comparison constants c_δ , c_a of Lemma 10.11 (the true values of $N|\delta_N|$ and $N|\text{Alt}_N - A_\infty N|$ are below 0.07 and 0.24), which use the worst-case exponent $\kappa_3 = 1.62$ instead of π , and in the bound for ρ_N . The limits in (d) are proved without a rate.

11 Explicit consequences in two and three dimensions

Theorem 7.10 expresses every bound through the single number $\bar{X} = \Lambda_1 \tau$ (the ratio of the stationary-start hitting time to the relaxation time of the visible sector). Theorems 9.1 and 10.1 locate $\bar{X}$.

Corollary 11.1 ($\bar{X}$ and $\Lambda_1(m - \tau)$ in $d = 2$ and $d = 3$). *For every $N \geq 2$, $X_- \leq \bar{X} \leq X_+$ and $M_- \leq \Lambda_1(m - \tau) \leq M_+$, where, with $a_N = 1 - \frac{\pi^2}{12N^2}$, $\mu_- = \mu_-(N)$ and $\ell_W = 1 + \ln W$ from Theorem 7.10:*

$$\begin{aligned} d = 2 : \quad X_- &= a_N \left(2\pi \ln N + \frac{\pi^2}{4} c_\tau - \frac{\pi^2}{4} \frac{2.14}{N^2} \right), & X_+ &= 2\pi \ln N + \frac{\pi^2}{4} c_\tau + \frac{\pi^2}{4} \frac{3.53}{N^2}, \\ M_- &= \max \left\{ \mu_-, a_N \left(\pi \ln 2 - \frac{\pi^2}{4} \frac{9.06}{N^2} \right) \right\}, & M_+ &= \min \left\{ \ell_W, \pi \ln 2 + \frac{\pi^2}{4} \frac{7.72}{N^2} \right\}, \\ d = 3 : \quad X_- &= a_N \frac{\pi^2}{6} \left(C_3 N + c_{\tau 3} - \frac{17.4}{N^2} \right), & X_+ &= \frac{\pi^2}{6} \left(C_3 N + c_{\tau 3} + \frac{19.4}{N^2} \right), \\ M_- &= \max \left\{ \mu_-, a_N \frac{\pi^2}{6} \left(c_{m\tau 3} - \frac{82.4}{N^2} \right) \right\}, & M_+ &= \min \left\{ \ell_W, \frac{\pi^2}{6} \left(c_{m\tau 3} + \frac{80.4}{N^2} \right) \right\}. \end{aligned}$$

In particular, with the exact constants $\frac{\pi^2}{4} c_\tau = -1.7960326\dots$, $\frac{\pi^2}{6} C_3 = 7.1068721\dots$, $\frac{\pi^2}{6} c_{\tau 3} = -8.9136329\dots$, $\frac{\pi^2}{6} c_{m\tau 3} = 2.5193561\dots$,

$$\begin{aligned} d = 2 : \quad \left| \bar{X} - \left(2\pi \ln N + \frac{\pi^2}{4} c_\tau \right) \right| &\leq \frac{8.8 + 5.2 \ln N}{N^2}, & \left| \Lambda_1(m_2 - \tau_2) - \pi \ln 2 \right| &\leq \frac{24.2}{N^2}; \\ d = 3 : \quad -\frac{5.85}{N} - \frac{21.3}{N^2} &\leq \bar{X} - \frac{\pi^2}{6} (C_3 N + c_{\tau 3}) \leq \frac{31.92}{N^2}, & \left| \Lambda_1(m_3 - \tau_3) - \frac{\pi^2}{6} c_{m\tau 3} \right| &\leq \frac{137.7}{N^2}. \end{aligned}$$

(The exact constants must not be replaced by truncated decimals if these bounds are used for very large N . For $d = 3$ and small N the elementary bounds (10.4) may be sharper than $X_\pm$.)

Proof. $\Lambda_1 = \frac{2}{d} \sin^2 u$ with $u = \frac{\pi}{2N} \leq \frac{\pi}{4}$, and $u^2(1 - \frac{u^2}{3}) \leq \sin^2 u \leq u^2$ (Lemma 7.1(a)), i.e. $a_N \frac{\pi^2}{2dN^2} \leq \Lambda_1 \leq \frac{\pi^2}{2dN^2}$; multiply by the bounds (9.3), (9.4), (10.5) for τ and $m - \tau$ (as $\tau > 0$ and $m - \tau > 0$, the lower bound for Λ_1 may be multiplied by a lower bound for τ or $m - \tau$ even if the latter is negative), and use $\frac{\pi^2}{4} c_{m\tau} = \pi \ln 2$ (Lemma 9.2) and (7.5), (7.4). The simplified forms follow from $(1 - \frac{a}{N^2})(B - \frac{b}{N^2}) \geq B - \frac{b+a \max(B,0)}{N^2}$ for $a, b \geq 0$, with $a = \frac{\pi^2}{12} = 0.8225$: for $d = 2$, $B = 2\pi \ln N + \frac{\pi^2}{4} c_\tau \leq 2\pi \ln N$, $b = \frac{\pi^2}{4} \cdot 2.14 < 5.281$, so $b + aB \leq 3.81 + 5.17 \ln N$, and $\frac{\pi^2}{4} \cdot 3.53 < 8.711$; for $\Lambda_1(m_2 - \tau_2)$: $B = \pi \ln 2$, $b = \frac{\pi^2}{4} \cdot 9.06 < 22.36$, $b + aB < 24.15$, and $\frac{\pi^2}{4} \cdot 7.72 < 19.05$; for $d = 3$, $B = \frac{\pi^2}{6} (C_3 N + c_{\tau 3}) > 0$, $b = \frac{\pi^2}{6} \cdot 17.4 < 28.63$, $\frac{aB}{N^2} \leq \frac{5.8452}{N} - \frac{7.331}{N^2}$, and $\frac{\pi^2}{6} \cdot 19.4 < 31.92$; for $\Lambda_1(m_3 - \tau_3)$: $B = \frac{\pi^2}{6} c_{m\tau 3}$, $b = \frac{\pi^2}{6} \cdot 82.4 < 135.55$, $b + aB < 137.62$, and $\frac{\pi^2}{6} \cdot 80.4 < 132.26$ (all products certified in scripts 09, 16). $\square$

Corollary 11.2 (explicit bounds in terms of N). *Let $d \in \{2, 3\}$, let Z be any upper bound for Z_d (for instance 6.02682 resp. 16.53232 by (7.7), or the sharper certified values (7.11), which are used in Tables 1–3), and $X_\pm$, $M_\pm$ as in Corollary 11.1; assume $X_- > 1$ and put $\hat{\beta} = X_-/(X_- - 1)$ (the hypothesis $X_- > 1$ holds for every $N \geq 3$, since both factors of X_- are positive and increasing in N for $N \geq 3$ and $X_-(3) = 4.10\dots$ resp. $8.38\dots$; for $N = 2$ it fails in both dimensions, $X_- = 0.984\dots$ for $d = 2$ and $X_- < 0$ for $d = 3$, although $\bar{X} = \frac{5}{2}$ resp. $\frac{29}{6}$, so the corollary is void for $N = 2$,*

where Theorem 7.10 with the exact $\bar{X}$ applies). Then all statements of Theorem 7.10 hold with $\bar{X}$, μ_- , ℓ_W replaced as follows:

- (i) $t_- := \frac{2}{1 + \sqrt{1 + 4\hat{\beta}Z/X_-^2}} \leq \sigma_0\tau \leq t_+ := \frac{2}{1 + \sqrt{1 + 4W/X_+^2}};$
- (ii) $\frac{M_-}{X_+} \leq \mu \leq \frac{M_+}{X_-}$ and $(1 + \frac{M_-}{X_+})t_- \leq \sigma_0 m \leq (1 + \frac{M_+}{X_-})t_+;$
- (iii) $\delta \leq \delta_+ := W/(X_- - W - 1 + \vartheta_0)$ if the denominator is positive; if $\delta_+ < 1$, $\omega := 2(1 + \delta_+)/(1 - \delta_+)$ and $\delta \geq \delta_- := W/(X_+ - W + \omega(Z - W) - \frac{1}{1 + \delta_+});$
- (iv) $\Lambda_1(1 + \delta_- - \frac{1}{X_-}) \leq \sigma_1 - \sigma_0 \leq \Lambda_1(1 + \delta_+ - \frac{1}{1 + X_+});$
- (v) $(1 + \frac{M_-}{X_+})(1 - \frac{\hat{\beta}(1 + \hat{\beta})Z}{X_-^2}) \leq r_0 \leq 1 + \frac{M_+}{X_-} + \frac{\hat{\beta}Z}{X_-^2}$ (the lower bound if $\hat{\beta}(1 + \hat{\beta})Z \leq X_-^2$);
- (vi) the bounds of Theorem 7.10(vi) for r_1 , computed with these $\delta_{\pm}$, ω and with $y_- = W - M_+ - \omega(Z - W) - 1$, $y_+ = W - M_- + \omega(Z - W) - \frac{1}{1 + \delta_+}.$

Proof. Each bound of Theorem 7.10 is monotone in $\bar{X}$: the lower bounds in (i), (v) increase with $\bar{X}$ (β decreases), the upper bound in (i) increases, and $\mu = \Lambda_1(m - \tau)/\bar{X} \in [M_-/X_+, M_+/X_-]$, so that $\sigma_0 m = (1 + \mu)\sigma_0\tau$ and (6.9) give (ii) and (v) exactly as in the proof of Theorem 7.10 (there $\mu_- \leq \Lambda_1(m - \tau) \leq \ell_W$ was used; here $M_- \leq \Lambda_1(m - \tau) \leq M_+$, in particular $Y_{x_0}(0) = W - \Lambda_1(m - \tau) \in [W - M_+, W - M_-]$); δ_+ decreases with $\bar{X}$, and by Remark 6.9(b),(c) the lower bound for δ and the bounds for r_1 remain valid when δ_+ (hence ω) is replaced by a larger number and $\bar{X}$ in δ_- by a larger number. $\square$

Theorem 11.3 (first-order pole structure; $d = 2, 3$). *For the corner-to-corner geometry in $d \in \{2, 3\}$ let*

$$Z_d = \sum_{k \in \mathbb{Z}^d \setminus 0} \frac{1}{|k|^4}, \quad Z'_d = \sum_{k \in \mathbb{Z}^d, |k|^2 \geq 2} \frac{1}{|k|^2(|k|^2 - 1)}, \quad \mu_\infty = \begin{cases} \pi \ln 2 = 2.17758 \dots & (d = 2), \\ \frac{\pi^2}{6} c_{m\tau 3} = 2.51935 \dots & (d = 3). \end{cases}$$

($Z_2 = 6.02681 \dots$, $Z_3 = 16.53231 \dots$, $Z'_2 = 3.22591 \dots$, $Z'_3 = 14.70229 \dots$, certified in Lemma 7.6.) Then, as $N \rightarrow \infty$ (so that $\bar{X} \rightarrow \infty$ by Corollary 11.1):

- (a) $\Lambda_1^2 K_2 \rightarrow Z_d$ and $\sigma_0\tau = 1 - \frac{Z_d}{\bar{X}^2} + o(\bar{X}^{-2}).$
- (b) $\Lambda_1(m - \tau) \rightarrow \mu_\infty$, so that $X := \Lambda_1 m = \bar{X} + \mu_\infty + o(1)$ and $\sigma_0 m = 1 + \frac{\mu_\infty}{\bar{X}} + o(\bar{X}^{-1}).$ In both dimensions $\Lambda_1(m - \tau) = \mu_\infty + O(N^{-2})$ with the explicit constants of Corollary 11.1; for $d = 2$, $\sigma_0 m_2 - 1 = \frac{\ln 2}{2 \ln N} (1 + O(1/\ln N))$; for $d = 3$, $\bar{X} = \frac{\pi^2}{6} (C_3 N + c_{\tau 3}) + O(N^{-1})$ and $\sigma_0 m_3 - 1 = \frac{\mu_\infty}{\frac{\pi^2}{6} C_3 N} (1 + O(N^{-1})) \approx \frac{0.3545}{N} (1 + O(N^{-1})).$
- (c) $\frac{\sigma_1}{\Lambda_1} = 1 + \delta$ with $\frac{W}{\delta} = \bar{X} - (2d + 1 - Z'_d) + o(1)$; in particular $\delta = \frac{2d}{\bar{X}} (1 + O(\bar{X}^{-1})).$
- (d) $\frac{\sigma_1 - \sigma_0}{\Lambda_1} = 1 + \frac{2d - 1}{\bar{X}} + O(\bar{X}^{-2}).$
- (e) $r_0 = 1 + \frac{\mu_\infty}{\bar{X}} + o(\bar{X}^{-1}), \quad r_1 = -\frac{2d}{\bar{X}} (1 + O(\bar{X}^{-1})), \quad \frac{r_1 \sigma_1}{r_0 \sigma_0} \rightarrow -2d.$

All $O(\cdot)$ terms are explicit through Theorem 7.10 and Corollary 11.2; the $o(\cdot)$ terms come from dominated convergence.

Proof. (a) $\Lambda_1^2 K_2 = \sum_{k \neq 0} w_k (\Lambda_1/\lambda_k)^2$, where for fixed k , $w_k \rightarrow \prod_i \varepsilon_{k_i}$ and $\lambda_k/\Lambda_1 = \sum_i \sin^2(k_i u)/\sin^2 u \rightarrow |k|^2$ as $N \rightarrow \infty$, and each term is bounded by $\prod_i \varepsilon_{k_i} |k|^{-4}$ (proof of Lemma 7.4), which is summable for $d \leq 3$. So $\kappa \bar{X}^2 \rightarrow Z_d$; now use (6.6), $1 - \beta\kappa \leq \sigma_0\tau \leq 1 - \kappa + 2\kappa^2$, with $\beta = 1 + O(\bar{X}^{-1})$. (b) Corollary 11.1 (which rests on (9.4), Lemma 9.2 and Theorem 10.1(c)). Then $\sigma_0 m = (1 + \mu)\sigma_0\tau$ with $\mu = \Lambda_1(m - \tau)/\bar{X}$, and (a). For $d = 2$, $\bar{X} = 2\pi \ln N + O(1)$. (c) By Theorem 7.10(iii), $\delta \leq \delta_+ \rightarrow 0$. By (6.8), $W/\delta = Y(\sigma_1) - \frac{1}{1+\delta}$, $Y(\sigma_1) = Y_0 + [Y(\sigma_1) - Y_0]$, $Y_0 = \bar{X} - W$, $W \rightarrow 2d$, and

$$Y(\sigma_1) - Y_0 = \sum_{k: |k|^2 \geq 2} w_k \frac{\Lambda_1 \sigma_1}{\lambda_k (\lambda_k - \sigma_1)}.$$

For fixed k the summand tends to $\prod_i \varepsilon_{k_i} / (|k|^2 (|k|^2 - 1))$. Once $\delta_+ \leq \frac{1}{2}$ we have $\sigma_1 \leq \frac{3}{2}\Lambda_1 \leq \frac{3}{4}\lambda_k$ for these k (as $\lambda_k \geq 2\Lambda_1$), so the summand is at most $6 w_k (\Lambda_1/\lambda_k)^2 \leq 6 \prod_i \varepsilon_{k_i} |k|^{-4}$, and dominated convergence gives $Y(\sigma_1) - Y_0 \rightarrow Z'_d$. (d) $\sigma_0/\Lambda_1 = \sigma_0\tau/\bar{X} = \bar{X}^{-1} + O(\bar{X}^{-3})$ by (a), and (c). (e) Theorem 7.10(v) gives $r_0 = 1 + \mu + O(\bar{X}^{-2})$. In (6.11) with $A_{x_0} = W$, the quantities $Y_{x_0}(\sigma_1) - \varsigma$ and $\varsigma + (1 + \delta)n\Lambda_1^2 h'(\sigma_1)$ are bounded (Theorem 6.8 and Remark 6.9), so $-r_1 = \delta(1 + O(\delta))$. Finally $\frac{r_1 \sigma_1}{r_0 \sigma_0} = \frac{r_1}{r_0} (1 + \delta) \frac{\bar{X}}{\sigma_0 \tau} \rightarrow -2d$. $\square$

Remark 11.4 (comparison with the formal perturbation theory of the computations of Sections 5–6 of the article). In the notation of the computations of Sections 5–6 of the article $\nu_j = \sigma_j$, $a_j = r_j \sigma_j$, $\mu_1 = \Lambda_1$, $X = \Lambda_1 \cdot \text{MFPT} = \bar{X} + \mu_\infty + o(1)$, $A = W$. Theorem 11.3 proves the four relations “ $\nu_0 = \text{MFPT}^{-1}(1 + O(1/X))$ ”, “ $\nu_1 = \mu_1(1 + W/X + O(X^{-2}))$ ”, “ $a_0 \approx \nu_0$ ”, “ $a_1 \approx -A\nu_0$ ” and identifies the constants: $\nu_0 \cdot \text{MFPT} = 1 + \mu_\infty/X + \dots$ with $\mu_\infty = \pi \ln 2$ ($d = 2$) or 2.519 ($d = 3$), and $(\nu_1 - \nu_0)/\mu_1 = 1 + (2d - 1)/X + \dots$, which is the rate “ $\mu_1(X + W - 1)/X$ ” underlying the closed form mode $\approx \text{MFPT} \ln(2dX)/(X + 2d - 1)$. The numerical values of Sections 5–6 of the article “ $\nu_0 \cdot \text{MFPT} = 1.164 \rightarrow 1.022$ (2D)”, “ $1.076 \rightarrow 1.0003$ (3D)” and “ $a_1/a_0 \rightarrow -4.09, -5.998$ ” agree with (b) and (e). What Theorem 11.3 does *not* give is a theorem about the mode itself; see Section 14.

Proposition 11.5 (the constants of Corollary 11.1 may not be truncated). *If the “in particular” part of Corollary 11.1 is stated with truncated decimal constants, the two statements become false:*

(a) ($d = 2$) The inequality $|\bar{X} - (2\pi \ln N - 1.796034)| \leq \frac{8.8 + 5.2 \ln N}{N^2}$ is violated for every $N \geq 8872$; more precisely $\bar{X} > 2\pi \ln N - 1.796034 + \frac{8.8 + 5.2 \ln N}{N^2}$ for all $N \geq 8872$. By direct evaluation it holds at $N = 8539$ (and at $N = 100, 1000, 7000, 8000$) and is violated at $N = 8540, 10^4, 2 \cdot 10^4, 10^5$.

(b) ($d = 3$) The inequality $7.1069 N - 14.0 - \frac{5.85}{N} - \frac{4.18}{N^2} \leq \bar{X}$ is violated for every $N \geq 182420$.

The admissible statements are those of Corollary 11.1 (exact constants $\frac{\pi^2}{4}c_\tau$, $\frac{\pi^2}{6}C_3$, $\frac{\pi^2}{6}c_{\tau 3}$); they hold for every $N \geq 2$.

Proof. (Computer-assisted through Corollary 11.1 and the certified constants; script 20.) (a) Put $\beta = \frac{\pi^2}{4}c_\tau = -1.7960326424\dots$, $\Delta = 1.796034 + \beta \in (1.3574 \cdot 10^{-6}, 1.3576 \cdot 10^{-6})$, $b = \frac{\pi^2}{4} \cdot 2.14$ and $B = 2\pi \ln N + \beta > 0$. By Corollary 11.1, $\bar{X} \geq X_- = (1 - \frac{\pi^2}{12N^2})(B - \frac{b}{N^2}) \geq B - \frac{b + \frac{\pi^2}{12}B}{N^2}$. Hence

$$N^2 \left[\bar{X} - \left(2\pi \ln N - 1.796034 + \frac{8.8 + 5.2 \ln N}{N^2} \right) \right] \geq h_0(N) := \Delta N^2 - b - \frac{\pi^2}{12} \beta - \left(\frac{\pi^3}{6} + 5.2 \right) \ln N - 8.8.$$

$h'_0(N) = 2\Delta N - (\frac{\pi^3}{6} + 5.2)/N > 0$ for $N \geq 1955$, and $h_0(8872) > 0$ (certified; $h_0(8871) < 0$). So $h_0(N) > 0$ for all $N \geq 8872$. The direct evaluations use the single sum (8.6) in ball arithmetic; at each of these N the inequality of Corollary 11.1 is verified as well. (b) By Corollary 11.1, $\bar{X} \leq X_+ = \frac{\pi^2}{6}(C_3 N + c_{\tau 3} + \frac{19.4}{N^2})$, so the lower bound with truncated constants minus $\bar{X}$ is at least

$$g(N) := \left(7.1069 - \frac{\pi^2}{6}C_3\right)N - 14.0 - \frac{\pi^2}{6}c_{\tau 3} - \frac{5.85}{N} - \frac{4.18 + \frac{\pi^2}{6}19.4}{N^2},$$

which is increasing in N because $7.1069 - \frac{\pi^2}{6}C_3 = 2.788 \dots 10^{-5} > 0$; and $g(182420) > 0$ (certified; $g(182419) < 0$). $\square$

Remark 11.6. In $d = 2$ the statement (a) is confirmed by explicit counter-examples that do not use Corollary 11.1 (e.g. $N = 10^4$: $|\bar{X} - (2\pi \ln N - 1.796034)| > 9.18 \cdot 10^{-7}$, while $(8.8 + 5.2 \ln N)/N^2 < 5.67 \cdot 10^{-7}$). In $d = 3$ no direct counter-example was computed (at $N \approx 1.8 \cdot 10^5$ the double sum (8.8) has $3 \cdot 10^{10}$ terms); (b) is a consequence of the upper bound $\bar{X} \leq X_+$, which is proved for all N but was verified by direct evaluation only for $N \leq 800$ (script 17; separately written code of a check, within the same project, reached $N = 1600$). The lesson is the parenthetical warning in Corollary 11.1: an $O(N^{-2})$ error term cannot absorb a constant that is wrong in the sixth decimal.

Part IV. Neglected poles, certification, open points

12 The neglected poles: a rigorous tail bound

A mode theorem based on the two slowest poles needs a bound on the contribution of all the others. The following bound is valid for every finite N .

Theorem 12.1 (tail of the spectral expansion). *Assume (1.1), $x \in \Omega'$, and let $1 \leq J \leq p - 1$. Write $S_x(t) = \mathbb{P}_x(T > t)$, $f_x = -S'_x$ for the unit-rate walk in continuous time, $p_t(x, y)$ for the transition kernel without target, and ϱ_j as in (5.2). Then for all $t_0 > 0$:*

$$\left| S_x(t) - \sum_{j < J} r_j(x) e^{-\sigma_j t} \right| \leq \left(n p_{t_0}(x, x) \sum_{j \geq J} \varrho_j \right)^{1/2} e^{-\sigma_J(t-t_0/2)} \quad (t \geq \frac{t_0}{2}), \quad (12.1)$$

$$\left| f_x(t) - \sum_{j < J} r_j(x) \sigma_j e^{-\sigma_j t} \right| \leq \left(n p_{t_0}(x, x) \sum_{j \geq J} \varrho_j \right)^{1/2} \sigma_J e^{-\sigma_J(t-t_0/2)} \quad (t \geq \frac{t_0}{2} + \frac{1}{\sigma_J}). \quad (12.2)$$

Here $\sum_{j \geq J} \varrho_j \leq \sum_{j \geq 1} \varrho_j \leq \beta^2 \kappa$ if $\bar{X} > 1$ (Theorem 6.7), and σ_J may be replaced by any lower bound $\Lambda \leq \sigma_J$ (in (12.2): for $t \geq \frac{t_0}{2} + \frac{1}{\Lambda}$), e.g. $\Lambda = \Lambda_J$. In discrete time with activity q , for integers $0 \leq t_1 \leq t$,

$$\left| \mathbb{P}_x(T > t) - \sum_{j < J} r_j(x) \nu_j^t \right| \leq \left(n P^{2t_1}(x, x) \sum_{j \geq J} \varrho_j \right)^{1/2} \nu_*^{t-t_1}, \quad \nu_* = \max_{j \geq J} |\nu_j|. \quad (12.3)$$

For activity q in continuous time replace t, t_0 by qt, qt_0 .

Proof. Let $\Pi = \sum_{j \geq J} \psi_j \psi_j^\top / \|\psi_j\|^2$, the orthogonal projector onto $\text{span}\{\psi_j : j \geq J\} \subset K_0$; it commutes with L_a . By the expansion in the proof of Theorem 3.1, $\sum_{j \geq J} r_j(x) e^{-\sigma_j t} = \langle e_x, e^{-tL_a} \Pi \mathbf{1}_{\Omega'} \rangle = \langle e_x, e^{-\frac{t_0}{2}L_a} e_x, e^{-(t-\frac{t_0}{2})L_a} \Pi \mathbf{1}_{\Omega'} \rangle$. Now $\|e^{-\frac{t_0}{2}L_a} e_x\|^2 = \langle e_x, e^{-t_0L_a} e_x \rangle = \mathbb{P}_x(X_{t_0} = x, T > t_0) \leq p_{t_0}(x, x)$; the operator e^{-sL_a} restricted to the range of Π has norm $e^{-\sigma_J s}$; and $\|\Pi \mathbf{1}_{\Omega'}\|^2 = \sum_{j \geq J} \langle \psi_j, \mathbf{1} \rangle^2 / \|\psi_j\|^2 = n \sum_{j \geq J} \varrho_j$ by Proposition 5.4. Cauchy–Schwarz gives (12.1).

For (12.2) replace e^{-sL_a} by $L_a e^{-sL_a}$, whose norm on the range of Π is $\max_{j \geq J} \sigma_j e^{-\sigma_j s} = \sigma_J e^{-\sigma_J s}$ when $s \geq 1/\sigma_J$ (the function $\sigma \mapsto \sigma e^{-\sigma s}$ decreases on $[1/s, \infty)$); the same monotonicity allows replacing σ_J by $\Lambda \leq \sigma_J$ if $s \geq 1/\Lambda$. In discrete time, $\sum_{j \geq J} r_j \nu_j^t = \langle Q^{t_1} e_x, Q^{t-t_1} \Pi \mathbf{1}_{\Omega'} \rangle$, $\|Q^{t_1} e_x\|^2 = Q^{2t_1}(x, x) \leq P^{2t_1}(x, x)$, and $\|Q^{t-t_1} \Pi\| = \nu_*^{t-t_1}$. $\square$

Corollary 12.2 (corner to corner, $d \geq 2$). *For the box, CC, $d \geq 2$, $N \geq 2$, $\bar{X} > 1$, with $Z \geq \Lambda_1^2 K_2$ as in Theorem 7.10 and $c_* = 1 + \frac{\pi^{3/2}}{2} = 3.7841 \dots$:*

(a) $n p_t(x_0, x_0) \leq (1 + N \sqrt{\pi d / (2t)})^d$ for $t > 0$; in particular $n p_{1/\Lambda_1}(x_0, x_0) \leq c_*^d$.

(b) (continuous time, unit rate) for $t \geq \frac{1}{2\Lambda_1}$:

$$|S_{x_0}(t) - r_0 e^{-\sigma_0 t}| \leq \sqrt{e} c_*^{d/2} \frac{\beta \sqrt{Z}}{\bar{X}} e^{-\Lambda_1 t}, \quad |S_{x_0}(t) - r_0 e^{-\sigma_0 t} - r_1 e^{-\sigma_1 t}| \leq e c_*^{d/2} \frac{\beta \sqrt{Z}}{\bar{X}} e^{-2\Lambda_1 t},$$

$$\text{and for } t \geq \frac{1}{\Lambda_1}: |f_{x_0}(t) - r_0 \sigma_0 e^{-\sigma_0 t} - r_1 \sigma_1 e^{-\sigma_1 t}| \leq 2e c_*^{d/2} \frac{\beta \sqrt{Z}}{\bar{X}} \Lambda_1 e^{-2\Lambda_1 t}.$$

(c) (discrete time) For every $q \in (0, 1]$ and $j \geq 2$: $|\nu_j| < 1 - 2q\Lambda_1$. If $q \leq \frac{1}{2}$ and $t_1 = \lceil \frac{1}{2q\Lambda_1} \rceil$, then for $t \geq t_1$

$$|\mathbb{P}_{x_0}(T > t) - r_0 \nu_0^t - r_1 \nu_1^t| \leq c_*^{d/2} \frac{\beta \sqrt{Z}}{\bar{X}} (1 - 2q\Lambda_1)^{t-t_1}.$$

For $\frac{1}{2} < q \leq 1$ the same holds with c_*^d replaced by $n P^{2t_1}(x_0, x_0) = \sum_k w_k (1 - q\lambda_k)^{2t_1}$.

Proof. (a) As in Lemma 7.2, $n p_t(x_0, x_0) = a_N(t/d)^d$ with $a_N(s) = \sum_{k=0}^{N-1} \varepsilon_k \cos^2(ku) e^{-(1-c_k)s} \leq 1 + 2 \sum_{k \geq 1} e^{-2k^2 s / N^2} \leq 1 + 2 \int_0^\infty e^{-2x^2 s / N^2} dx = 1 + N \sqrt{\pi / (2s)}$, using $1 - c_k = 2 \sin^2(ku) \geq 2k^2 / N^2$. For $t = 1/\Lambda_1$, $s = 1/(d\Lambda_1) = 1/(2 \sin^2 u) \geq 2N^2/\pi^2$, so $N \sqrt{\pi / (2s)} \leq \pi^{3/2}/2$. (b) Theorem 12.1 with $t_0 = 1/\Lambda_1$, $J = 1$ ($\sigma_1 > \Lambda_1$) and $J = 2$ ($\sigma_2 > \Lambda_2 = 2\Lambda_1$, Corollary 4.2(c); if $p = 2$ the left sides vanish), together with $\sum_{j \geq 1} \varrho_j \leq \beta^2 \kappa \leq \beta^2 Z / \bar{X}^2$. (c) For $j \geq 2$, $\nu_j = 1 - q\sigma_j < 1 - 2q\Lambda_1$, and $\nu_j > 1 - q\Lambda_p$ where $\Lambda_p = \lambda_{(N-1, \dots, N-1)} = 1 + \cos \frac{\pi}{N} = 2 - d\Lambda_1$ is the largest eigenvalue of L ; so $\nu_j > -(1 - 2q\Lambda_1) + [2(1 - q) + q(d - 2)\Lambda_1] \geq -(1 - 2q\Lambda_1)$. If $q \leq \frac{1}{2}$ then $0 \leq 1 - q\lambda_k \leq e^{-q\lambda_k}$, hence $n P^{2t_1}(x_0, x_0) \leq n p_{2qt_1}(x_0, x_0) \leq c_*^d$ by (a) and monotonicity in t , since $2qt_1 \geq 1/\Lambda_1$. Apply (12.3) with $J = 2$. $\square$

Remark 12.3 (what the tail bound says near the mode). At times t with $e^{-\Lambda_1 t} \asymp 1/(2d\bar{X})$, where the two slowest terms of f_{x_0} are of the same order $\Lambda_1 \bar{X}^{-1}$ (Theorem 11.3), the neglected part is at most $2e c_*^{d/2} \beta \sqrt{Z} \Lambda_1 \bar{X}^{-1} e^{-2\Lambda_1 t} = O(\Lambda_1 \bar{X}^{-3})$, i.e. smaller by a factor $O(\bar{X}^{-2})$. For $t \geq 1/\Lambda_1$ this bound is absolute: it is a relative error $O(\bar{X}^{-2})$ only where $e^{-\Lambda_1 t} \asymp 1/\bar{X}$, that is near the mode, whereas at $t = 1/\Lambda_1$ it is of the same order $\Lambda_1 \bar{X}^{-1}$ as the two slowest terms themselves (for large $\bar{X}$ their sum is even negative there, since $a_1/a_0 \rightarrow -2d$ and $1 - 2d/e < 0$); so the density is a two-exponential function up to a small relative error near the mode, not on the whole window $t \geq 1/\Lambda_1$ ($\bar{X} \sim 2\pi \ln N$ in $d = 2$, $\bar{X} \sim 7.1069N$ in $d = 3$). This, with Theorems 7.10 and 11.3, is the complete list of spectral inputs for a mode theorem.

13 Computer-assisted certification

All certified computations use Arb ball arithmetic through `python-flint` 0.9.0 (F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. 66 (2017) 1281–1292) at 200–256 bits. A comparison of two balls is accepted only if it holds for all points of the balls. Scripts are in `scripts/`, their outputs in `data/` and `logs/`.

Constants (scripts `06_certify_constants.py` and `09_certify_part3.py`). $\sup |\Psi''| \leq 2.35$ (Lemma 9.5(d)); Λ_*, Σ_* (series with explicit geometric tail bounds; the closed forms of Lemma 9.2 overlap); $c_\tau, c_{m\tau}, C'_2; S_1, \dots, S_4$; elementary enclosures of $Z_2, Z_3; C_3$ (Lemma 10.7, split point $S = 10^6$, cross-checked with $S = 200$); $K(1)$ (rigorous integration of the analytic integrand $4 \cos^2 \varphi \sqrt{(2 + \sin^2 \varphi)/(1 + \sin^2 \varphi)}$ on $[0, \frac{\pi}{2}]$); B_D (partial sum over $|k|_\infty \leq 30$ plus the tail bound $8e^{-62\kappa_3}/(1 - e^{-2\kappa_3})^2$); $B_e; E_\infty, c_{\tau 3}, c_{m\tau 3}$ (lattice sums with exponential tail bounds and rigorous integration). The script asserts that the rounded constants quoted in Theorems 9.1 and 10.1(b) and in Corollary 11.1 are implied by the enclosures. Balls passed from one script to the next as decimal strings are re-inflated to cover the rounding of the strings.

Epstein sums (`14_certify_epstein.py`). Z_2, Z_3, Z'_2, Z'_3 by Lemma 7.6 (300 bits), with the three cross-checks described there.

Second order in $d = 3$ (`16_certify_d3_remainder.py`). $K(1) = 6\mathbf{K}(\frac{1}{2}) - 4\mathbf{E}(\frac{1}{2})$ (agrees with the quadrature value); V_g (Lemma 10.9(d)); c_δ, c_a (Lemma 10.11); $\delta_\infty, \alpha_\infty, c_{\tau 3}^{(0)}, c_{m\tau 3}^{(0)}, C_3'''$ (Theorem 10.1(d)); the rounded constants of Theorem 10.1(c) and of Corollary 11.1. As a cross-check the script also bounds $\int_0^\pi |J_b''| \leq 2.651$, which gives the weaker first-order estimate $|\varepsilon_b(N) - \varepsilon_b(\infty)| \leq 0.694/N$.

Poles and residues (`07_certify_poles.py, 09_certify_part3.py`). For CC, $d = 2$ ($N \leq 300$) and $d = 3$ ($N \leq 50$; τ, m also for $N = 70, 100$), the functionals τ, m, K_2 are evaluated as exact spectral sums (7.1) in ball arithmetic. A pole is certified as follows: by Theorem 2.2(i), G is strictly increasing on $(\Lambda_j, \Lambda_{j+1})$ and has exactly one zero there; if $\Lambda_j < s_- < s_+ < \Lambda_{j+1}$ and $G(s_-) < 0 < G(s_+)$ are verified with balls, then $\sigma_j \in (s_-, s_+)$. The brackets have relative width 2^{-149} . The residues $r_j = -G_{x_0}(\sigma_j)/(\sigma_j G'(\sigma_j))$ are then evaluated on the bracket. For each N the scripts verify with certified comparisons: the enclosures of Theorems 9.1, 10.1(b),(c) and both forms of Corollary 11.1; Lemmas 7.2, 7.4, 7.7; $r_0 > 1, \sigma_0 m > 1, r_1 < 0$ (Theorem 7.9, Corollary 5.7); every inequality of Theorems 6.2–6.8 with the exact functionals; and every inequality of Theorem 7.10 and Corollary 11.2 whose hypothesis holds (10 to 38 certified comparisons per case, 37 cases, all passed). Tables 1–3 show a selection.

Signs of the residues (`13_certify_residue_signs.py`). Proposition 14.1; exact grouping of the eigenvalues modulo the cyclotomic polynomial, 300-bit balls; as a consistency check the script also asserts the statements of Theorem 5.10 (which is proved without computation) for each of its 20 cases, including the exact purity and parity of the two eigenvalues used in part (b).

Limits in $d = 2$ (`19_certify_d2_limits.py`). $S_5, \dots, S_8, c_\tau^{(0)}, c_{m\tau}^{(0)}, C_2^{(0)}$ (Theorem 9.8); comparison with the closed forms of Remark 9.9 (agreement to 10^{-50}); certified values of $R_\tau(N), R_{m\tau}(N)$ for 22 values of $N \leq 10^6$ (inside the proved windows, increasing, below the limits) and certified strict monotonicity for all consecutive $N \leq 300$; floating-point checks of every step of the proof (Lemma 9.6, the majorants, the pointwise limits, the four sums).

Truncated constants (`20_certify_rounded_constants.py`). Proposition 11.5: the thresholds 8872 and 182 420, the monotonicity conditions, and the direct evaluations in $d = 2$.

Printed decimals (`21_check_printed_decimals.py`). Every decimal string of the text that ends in dots is compared with a 40-digit value of the constant and must be its truncation.

Sanity checks of every lemma (floating point). 01: Theorems 2.2–5.1 against brute-force diagonalisation of L_a for 29 geometries in $d = 1, 2, 3$ (corner, centre, generic and degenerate targets), including the time-domain formulas (3.2), (3.3). 03: Section 6. 02, 04, 05: Propositions 8.3, 8.5, Lemmas 8.1, 8.2, 9.5, 9.6 and the remainder pieces of Theorem 9.1. 08: Proposition 5.4, Lemmas 7.1–7.7, Corollary 7.11 ($d = 3, 4, 5$), Lemmas 9.4, 10.3–10.6, 10.7, the enclosures of Theorems 9.1, 10.1 up to $N = 5000$ ($d = 2$) and $N = 200$ ($d = 3$), and Theorem 12.1/Corollary 12.2 against brute force (the ratio remainder/bound is between 0.01 and 0.15; discrete time checked for $q = 0.3, 0.5, 0.8, 1$). 10: the limits of Theorem 11.3. 12: Theorem 5.6 and Remark 5.8 (formula (5.4), $\eta_1 \leq \Lambda_1$, $G^+(\eta_1) = 0$, and the sign rule for every pole). 15: Proposition 5.5, the lower bound of Lemma 7.2, Lemma 7.8 (continuous and discrete time, $d = 1, \dots, 4$), Theorem 7.9, and the sharpened Theorem 7.10(ii),(v),(vi) up to $N = 640$ ($d = 2$), $N = 64$ ($d = 3$). 17: Lemmas 10.8–10.11 (closed form of K against quadrature, the formulas for M' , M'' , J_b'' , the comparison inequalities at 24000 random points) and Theorem 10.1(c),(d) for $N \leq 800$. 18: Proposition 5.9 (3282 pure eigenvalues of 352 random chains with an involution and 2986 of boxes with arbitrary symmetric pairs), Remark 5.11 (198 random start–target pairs: degree and leading coefficient of $\mathcal{N}_x$, both sign relations, the three counting consequences), and every spectral claim in the proof of Theorem 5.10 (part (b) for 104 pairs (d, N) with $d \geq 2$, $N \geq 3$, part (c) for 91 of them) together with floating-point sign patterns for 49 pairs (the statistics of Observation 14.2 are asserted there).

N	$\bar{X}$	$[X_-, X_+]$	$\sigma_0 \tau$	bounds (i)	$\sigma_0 m$	bounds (ii)
5	8.1304	[7.839, 8.665]	0.928025	[0.907421, 0.955961]	1.16428	[1.06743, 1.23475]
10	12.5893	[12.515, 12.759]	0.964332	[0.961349, 0.977113]	1.12909	[1.10737, 1.16149]
20	16.9972	[16.979, 17.048]	0.979370	[0.978720, 0.986684]	1.10446	[1.10027, 1.11600]
50	22.7773	[22.774, 22.787]	0.988292	[0.988133, 0.992421]	1.08273	[1.08214, 1.08764]
100	27.1371	[27.136, 27.140]	0.991723	[0.991644, 0.994629]	1.07129	[1.07112, 1.07451]
200	31.4937	[31.493, 31.494]	0.993849	[0.993802, 0.996000]	1.06256	[1.06250, 1.06488]
300	34.0416	[34.042, 34.042]	0.994735	[0.994698, 0.996572]	1.05837	[1.05832, 1.06033]

N	$\delta = \sigma_1/\Lambda_1 - 1$	bounds (iii) $[\delta_-, \delta_+]$	r_0	bounds (v)	r_1	bounds (vi)
5	0.40891	[0.03923, 0.89586]	1.18370	[0.89247, 1.40406]	-0.29622	[-19.90502, -]
10	0.30187	[0.19978, 0.45562]	1.14042	[1.05136, 1.23052]	-0.25470	[-0.92575, -]
20	0.23498	[0.19848, 0.30601]	1.11172	[1.07268, 1.15328]	-0.21108	[-0.42176, -0.04962]
50	0.17929	[0.16496, 0.21282]	1.08723	[1.06791, 1.10810]	-0.16718	[-0.25043, -0.08339]
100	0.15128	[0.14262, 0.17284]	1.07461	[1.06144, 1.08881]	-0.14318	[-0.19350, -0.08739]
200	0.13057	[0.12477, 0.14548]	1.06511	[1.05548, 1.07543]	-0.12477	[-0.15828, -0.08525]
300	0.12082	[0.11608, 0.13315]	1.06058	[1.05239, 1.06933]	-0.11594	[-0.14321, -0.08288]

Table 1: $d = 2$, corner to corner: certified values (ball arithmetic, all digits shown are correct up to rounding) and the bounds of Corollary 11.2, which depend on N only ($X_\pm$, $M_\pm$ from Corollary 11.1; Z is the certified upper end 6.0268120396919401... of Z_2 from Lemma 7.6, not the rounded value 6.02682, with which the bounds change in the last digit shown at most, e.g. the lower bound for r_1 at $N = 5$ becomes -19.90509). A dash means that the hypothesis of the corresponding bound is not satisfied.

14 Numerical observations and what is not proved

Proposition 14.1 (certified sign patterns; the residues do not alternate for $d \geq 2$, $N \geq 3$). *For the corner-to-corner geometry the signs of $(r_0, r_1, \dots, r_{p-1})$ are (computer-assisted, ball arithmetic; S = number of sign changes, $D - 1 = d(N - 1) - 1$ the lower bound of Theorem 5.1(d), $p - 1$ the number needed for alternation):*

N	$\bar{X}$	$[X_-, X_+]$	$\sigma_0\tau$	bounds (i)	$\sigma_0 m$	bounds (ii)
4	18.7009	[16.814, 21.508]	0.973816	[0.944530, 0.989168]	1.09643	[1.00884, 1.14409]
6	33.0422	[32.180, 34.614]	0.989375	[0.984045, 0.995371]	1.06265	[1.02848, 1.07958]
10	61.6742	[61.360, 62.474]	0.996421	[0.995576, 0.998505]	1.03671	[1.02131, 1.04353]
20	132.9573	[132.878, 133.304]	0.999146	[0.999058, 0.999665]	1.01803	[1.01537, 1.02062]
30	204.1091	[204.074, 204.328]	0.999626	[0.999601, 0.999857]	1.01195	[1.01118, 1.01292]
50	346.3172	[346.305, 346.443]	0.999867	[0.999862, 0.999950]	1.00714	[1.00697, 1.00738]

N	$\delta = \sigma_1/\Lambda_1 - 1$	bounds (iii) $[\delta_-, \delta_+]$	r_0	bounds (v)	r_1	bounds (vi)
4	0.22154	[0.07462, 0.39767]	1.10263	[0.93107, 1.21880]	-0.20422	[-1.97022, -]
6	0.14461	[0.09194, 0.19856]	1.06516	[1.01016, 1.10108]	-0.14124	[-0.41132, -]
10	0.08574	[0.07144, 0.10204]	1.03755	[1.01662, 1.04956]	-0.08567	[-0.14382, -0.03428]
20	0.04255	[0.03987, 0.04627]	1.01823	[1.01440, 1.02191]	-0.04273	[-0.05342, -0.03136]
30	0.02829	[0.02723, 0.02991]	1.01204	[1.01077, 1.01346]	-0.02840	[-0.03274, -0.02362]
50	0.01694	[0.01659, 0.01751]	1.00717	[1.00683, 1.00757]	-0.01699	[-0.01845, -0.01535]

Table 2: $d = 3$, corner to corner: certified values and the bounds of Corollary 11.2 (functions of N only), with $X_{\pm}$, $M_{\pm}$ from Corollary 11.1, i.e. from Theorem 10.1(c), and Z the certified upper end of Z_3 from Lemma 7.6. A dash means that the hypothesis of the corresponding bound is not satisfied.

N	$\bar{X}$	$1 - \sigma_0\tau$	bounds (i) on $1 - \sigma_0\tau$	δ	$[\delta_-, \delta_+]$	r_1 and bounds (vi)
4	18.701	2.618e-02	[1.423e-02, 4.550e-02]	0.22154	[0.08550, 0.34685]	-0.20422 [-1.39532, -]
6	33.042	1.063e-02	[5.076e-03, 1.515e-02]	0.14461	[0.09502, 0.19267]	-0.14124 [-0.38882, -]
10	61.674	3.579e-03	[1.534e-03, 4.379e-03]	0.08574	[0.07218, 0.10148]	-0.08567 [-0.14262, -0.03485]
20	132.957	8.544e-04	[3.371e-04, 9.405e-04]	0.04255	[0.03996, 0.04624]	-0.04273 [-0.05357, -0.03144]
30	204.109	3.742e-04	[1.436e-04, 3.985e-04]	0.02829	[0.02725, 0.02990]	-0.02840 [-0.03284, -0.02363]
50	346.317	1.332e-04	[4.997e-05, 1.382e-04]	0.01694	[0.01660, 0.01751]	-0.01699 [-0.01849, -0.01534]

Table 3: $d = 3$: bounds of Theorem 7.10 evaluated with the exact (certified) $\bar{X}$ and with Z the certified upper end of Z_3 (Lemma 7.6).

(d, N)	$signs\ of\ r_0, \dots, r_{p-1}$	p	S	$D - 1$	$negative$
(2, 2)	$+-$	2	1	1	1
(2, 3)	$+-+ -$	5	3	3	3
(2, 4)	$+-+ + - + -$	8	5	5	3
(2, 5)	$+-+ + - + + - + -$	13	7	7	7
(2, 6)	$+-+ + - + + + - + -$	16	9	9	8
(2, 7)	$+-+ + - + + + - + + - + -$	25	17	11	13
(2, 8)	$+-+ + - + + + - + + + - + + - + -$	32	21	13	16
(2, 10)	$+-+ + - + + + - + + + - + + + - + + + - + + - + -$	50	29	17	25
(2, 11)	$+-+ + - + + + - + + + - + + + - + + + - + + + - + + - + -$	61	35	19	34
(2, 12)	$+-+ + - + + + - + + + - + + + - + + + - + + + - + + + - + + - + -$	64	37	21	31
(3, 2)	$+-+$	3	2	2	1
(3, 3)	$+-+ + -$	8	5	5	4
(3, 4)	$+-+ + - + + - + -$	15	12	8	7
(3, 5)	$+-+ + - + + + - + + + - + + -$	29	17	11	14
(3, 6)	$+-+ + - + + + - + + + - + + + - + + -$	33	16	14	16
(4, 2)	$+-+ -$	4	3	3	2
(4, 3)	$+-+ + - + -$	11	7	7	6
(4, 4)	$+-+ + - + + - + - + + - + -$	24	19	11	11
(5, 2)	$+-+ +$	5	4	4	2
(5, 3)	$+-+ + - + -$	14	9	9	8

In particular all residues are non-zero in these cases; for $N = 2$ the sequence alternates and for $N \geq 3$ it does not ($S < p - 1$), in agreement with Theorem 5.10 (which does not depend on this proposition). In every row with $N \geq 3$ the three signs ($\text{sign } r_{i-1}, \text{sign } r_i, \text{sign } r_{i+1}$) named in Theorem 5.10(b) are $-+$ when Λ_i, Λ_{i+1} are purely even and $+-$ when they are purely odd. The lower bound $D - 1$ of Theorem 5.1(d) is attained in the rows with $N \leq 3$, for $d = 2$, $N \leq 6$, but

not in general; about one half of the residues are negative.

Proof. Script `13_certify_residue_signs.py`. The distinct eigenvalues are determined exactly: with $\zeta = e^{i\pi/N}$, $2\sum_i \cos(\pi k_i/N) = \sum_i (\zeta^{k_i} + \zeta^{2N-k_i})$, and two such sums are equal if and only if the corresponding integer polynomials agree modulo the cyclotomic polynomial Φ_{2N} (exact arithmetic). The distinct values are then separated by ball arithmetic, every zero σ_j of G is bracketed by a certified sign change of G (Theorem 2.2(i)), and the sign of $r_j = -G_{x_0}(\sigma_j)/(\sigma_j G'(\sigma_j))$ is certified on the bracket. The script also asserts, for every row with $N \geq 3$, the purity and parity of Λ_i, Λ_{i+1} (exact) and the monotonicity of the three signs of Theorem 5.10(b), and for every row covered by Theorem 5.10(c) that the pair named there does not have the excluded signs. $\square$

Numerical observation 14.2 (signs of the residues, larger N). In floating-point arithmetic (script 18; 49 pairs (d, N) with $p \leq 140$, among them $d = 2$, $N \leq 16$ and $d = 3$, $N \leq 9$) the number of negative residues stays between $0.33p$ and $0.6p$, the number S of sign changes stays strictly between $D - 1$ and $p - 1$ for $7 \leq N \leq 16$ ($d = 2$) and $4 \leq N \leq 9$ ($d = 3$), and for $N = 3$ the lower bound is attained, $S = D - 1 = 2d - 1$, in every dimension examined ($2 \leq d \leq 11$). That the residues do not alternate for *every* $d \geq 2$, $N \geq 3$ is Theorem 5.10; the exact pattern for a given (d, N) is given by Remark 5.8 and Proposition 5.9, and no closed description of it is known to us.

What is not proved in this note.

- (G1) *The mode.* No statement about the location of the maximum of f or about unimodality is proved here. The spectral inputs are complete (Theorems 7.10, 11.3, 12.1): for $t \geq 1/\Lambda_1$ the density is an explicit two-exponential function up to an absolute error $O(\Lambda_1 \bar{X}^{-1} e^{-2\Lambda_1 t})$, a relative error $O(\bar{X}^{-2})$ near the mode (Remark 12.3). What is missing for a mode theorem is (i) a quantitative analysis of the critical point of the two-exponential function under this perturbation, and (ii) control of f on $[0, 1/\Lambda_1]$, where all poles contribute (Theorem 12.1 does not apply there, and the elementary bound $f(t) \leq \frac{1}{n}(\frac{1}{2}n p_t(x_0, a) + \partial_t n p_t(x_0, a))$ is too weak by a factor of order $\ln N$ in $d = 2$). In $d = 2$ every rate of convergence is only logarithmic in N ($\bar{X} \sim 2\pi \ln N$), so explicit thresholds for N would be astronomically large; this is intrinsic, not an artefact.
- (G2) *Sharpness in $d = 3$.* For every $N \geq 2$, $R_3(N) \in [-17.4, 19.4]$ and $R'_3(N) \in [-82.4, 80.4]$ are proved, and the limits 1.7574 and -0.2989 are proved; the true remainders stay within $[1.61, 1.76]$ and $[-0.63, -0.29]$ (Remark 10.12). The loss is in the comparison constants c_δ , c_a (worst-case exponent κ_3) and in the range of ρ_N ; no rate is proved for the convergence to the limits (this would need a third-order analysis of $P_N - P_\infty$ and of g').
- (G3) *Sharpness in $d = 2$.* $R_\tau \in [-2.14, 3.53]$ is proved for every N and the limit $R_\tau \rightarrow 0.8830352 \dots$ is proved (Theorem 9.8); $R_\tau(N) \in [0.8512, 0.8831]$ for all N , the monotonicity in N and the rate $O(N^{-2})$ are only observed. An explicit rate would follow from the same proof with second-order remainders (a certified bound for Ψ''' and second-order versions of (9.6)); it was not carried out.
- (G4) *Computer assistance.* The numerical constants listed in Section 13 are certified by interval arithmetic, not by hand. The structure theorems (Sections 2–6), Lemmas 7.2–7.4, 7.7, 7.8, Theorem 7.9, Theorem 7.10 with $Z = Z_d(N)$, the limit statements of Theorem 11.3 and Theorem 12.1 do not depend on any computation.
- (G5) *Other geometries.* Sections 2–6 and Theorem 12.1 hold for every start and target (and every finite symmetric chain). The lattice-sum evaluations of Sections 7–11 are carried out for opposite corners only. For a centre target the first visible eigenvalue is $\Lambda_1 = \frac{1}{d}(1 - \cos \frac{2\pi}{N})$ and for a centre start with corner target $A_{x_0} = 0$ and the residue r_1 changes sign (Supplementary Section S7 of the article); the general theorems apply but the functionals were not evaluated.

- (G6) *Quality of the lower bound for δ and of the bounds for r_1 .* They contain the factor $\omega = 2(1 + \delta_+)/ (1 - \delta_+)$ and are informative only when δ_+ is small ($N \geq 20$ in $d = 2$, $N \geq 10$ in $d = 3$; see the tables).
- (G7) *Discrete time, $q > \frac{1}{2}$.* Corollary 12.2(c) is explicit for $q \leq \frac{1}{2}$; for $\frac{1}{2} < q \leq 1$ the prefactor $n P^{2t_1}(x_0, x_0)$ is left as a finite sum (near-bipartite oscillations of the non-lazy walk).
- (G8) *Classical identities.* The closed forms of Lemma 9.2 rest on the cited DLMF formulas (theta-function and elliptic-integral identities at the lemniscatic point); the bounds of Theorem 9.1 do not. Theorem 10.1(c),(d) rests on the classical derivative formulas and the inequality $\ln 4 \leq \mathbf{K} + \ln k' \leq \frac{\pi}{2}$ for complete elliptic integrals (10.11), on the Fourier series of B_2 and of $\ln \sin \frac{\theta}{2}$, and on Arb's evaluation of $\mathbf{K}$, $\mathbf{E}$, K_ν . The identification of C_3 with lattice Green's function values is elementary, but the closed form of Watson's integral $u(0)$ is quoted from the literature and not used. The values of Z_3 , Z'_2 , Z'_3 rest on Jacobi's theta transformation (cited) and on Arb's incomplete gamma function.

References

- [1] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions*, National Bureau of Standards, 1964 (formula 4.3.68).
- [2] D. J. Aldous and M. Brown, Inequalities for rare events in time-reversible Markov chains I, in: *Stochastic Inequalities*, IMS Lecture Notes–Monograph Series 22 (1992), 1–16.
- [3] *NIST Digital Library of Mathematical Functions*, <https://dlmf.nist.gov>, formulas 10.32.3, 10.32.9, 19.4.1, 19.4.2, 19.9.1, 19.12.1, 20.4.6, 20.5.3, 20.7.5, 20.7.32, 20.9.1, 20.9.2, 23.5.5, 23.15.9, 23.17.8, 24.8.1 (all checked against the online edition on 1 October 2026).
- [4] J. W. Essam and F. Y. Wu, The exact evaluation of the corner-to-corner resistance of an $M \times N$ resistor network: asymptotic expansion, *J. Phys. A* 42 (2009) 025205.
- [5] M. L. Glasser and I. J. Zucker, Extended Watson integrals for the cubic lattices, *Proc. Natl. Acad. Sci. USA* 74 (1977) 1800–1801.
- [6] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013 (Theorems 4.3.17, 8.3.1, 8.4.4).
- [7] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* 66 (2017) 1281–1292.
- [8] F. Johansson, Numerical integration in arbitrary-precision ball arithmetic, in: *Mathematical Software – ICMS 2018*, LNCS 10931, 255–263.
- [9] G. N. Watson, Three triple integrals, *Quart. J. Math. Oxford* 10 (1939) 266–276.
- [10] I. J. Zucker and M. M. Robertson, Exact values of some two-dimensional lattice sums, *J. Phys. A* 8 (1975) 874–881.